THE CENTRAL CONCEPTS OF AUTOMATA THEORY

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* Alphabets

- A language consists of various symbols from which the words, statements etc., can be obtained. These symbols are called alphabets.
- An alphabet is defined as a finite non-empty set of symbols. (or characters)
- The symbol Σ denotes the set of alphabets of a language.
- Example : ① The alphabets of C language has the letters from A to Z, a to z, digits from 0 to 9, symbols such as +, -, *, /, (,), {, }, etc. & is denoted by

$\Sigma = \{ a, b, \ldots z, A, B, \ldots Z, 0, 1, \ldots 9, \#, (,), \{, \}, <, >, !, [,], \ldots \}$

② The machine language is made up of only 0's & 1's & so, the alphabets of machine language is denoted by

$\Sigma = \{0, 1\}$

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* Strings

- The sequence of symbols obtained from the alphabets of a language is called a string.
- A string is defined as a finite sequence of symbols from the alphabet Σ.
- Example : $\Sigma = \{0, 1\}$ is set of alphabets.

The various strings that can be obtained from Σ are $\{\varepsilon, 0, 1, 00, 01, 10, 11, 1010, 010101, \ldots\ldots\}$

✗ **NOTE** : • An empty string is denoted by the symbol ε (string with zero occurrences of symbols)

• $\varepsilon \notin \Sigma$ i.e., ε is not part of Σ.

- An infinite number of strings can be generated from Σ and once the string is generated, it has finite number of symbols in it and has a definite sequence.
- The lowercase letters a, b, c etc along with the symbols such as +, -, (,), { and so on are used to denote the symbols in Σ
- The lowercase letters such as u, v, w, x, y, z are normally used to indicate the strings.
- Example : $w = 010101$

  The symbols 0 and 1 are in Σ and the letter w denotes the string with a specific value.

* What is concatenation of two strings?

- The concatenation of two strings u and v is the string obtained by writing the letters of string u followed by the letters of string v.
- Example : ① $u = a_1 a_2 a_3 \dots a_n$

  $v = b_1 b_2 b_3 \dots b_m$

  $uv = a_1 a_2 a_3 \dots a_n b_1 b_2 b_3 \dots b_m$

  written as uv or $u \cdot v$

  ② u : Computer

  v : Science

  uv : Computer Science

  $|uv| = |u| + |v|$

String replication

→ for each string w and each natural number i, the string w^i is defined as

$w^0 = \varepsilon$

$w^{i+1} = w^i w$

Ex $a^3 = aaa$

$(comp)^2 = comp\,comp$ (No space)

$a^0 b^3 = bbb$

NOTE : Concatenation is not commutative. $uv \neq vu$

Concatenation on strings is associative, $\forall s, t, w \; ((st)w = s(tw))$

* What is sub string?

- Let w is a string obtained from the symbols in Σ. The string w if it can be decomposed into three strings x, y and z such that $w = xyz$ then x is a sub string, y is a sub string and z is a sub string of string w.

  A string s is a substring of a string t iff s occurs contiguously as part of t. Ex : aaa - s = aaabbbaaa

  [and z is a sub string → aaaaaa NOT aaabbbaa

  Proper Substring : A string s is a proper substring of a string t iff s is a substring of t & $s \neq t$.

* What is prefix?

- A prefix is string of any number of leading symbols.
- Example : • $w = xyz$

  The string w has prefix ε, x, xy and xyz

- In the string "Rama", the various prefixes are ε, "R", "Ra", "Ram" and "Rama"

* What is suffix ?

- The suffix is a string of any number of trailing symbols.
- Example : • $\omega = xyz$

  The string ω has suffix ε, z, yz and xyz

  • In the string "Rama", the various suffixes are ε, "a", "ma", "ama" and "Rama".

* What is reversal of a string ?

- The reversal of a string is obtained by writing the symbols in reverse order.
- Example : If $u = a_1 a_2 a_3 \cdots a_n$

  then the reverse of u is denoted by u^R

  $u^R = a_n a_{n-1} a_{n-2} \cdots a_3 a_2 a_1$

$$\omega^R = \begin{cases} \varepsilon & \text{if } \omega = \varepsilon \\ a & \text{if } \omega = a \\ (xa)^R = ax^R & \text{if } \omega = xa \end{cases} \qquad a \in \Sigma, \ \omega, x \in \Sigma^*$$

NOTE : $u = \varepsilon$, $u^R = \varepsilon = \varepsilon^R$; $u = a$, $u^R = a = a^R$

* What is the length of a string ?

- The length of a string u is the number of symbols (positions) in u & is denoted by $|u|$
- Example : • $u = a_1 a_2 a_3 \cdots a_n$, $|u| = n$

  • The length of an empty string ε is 0

  $|\varepsilon| = 0$

$\varepsilon\omega = \omega\varepsilon = \omega$

for any symbol c and string s, define the function $\#_c(s)$ to be the no of times that the symbol c occurs in s

Ex. $\#_a(abbaaa) = 4$

* What is powers of an alphabet ?

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- The power of an alphabet denoted by Σ^i is the set of words of length i.

- Example : $\Sigma = \{0,1\}$ then

$\Sigma^0 = \{\varepsilon\}$ denote the set of words of length 0.

$\Sigma^1 = \{0,1\}$ " " " " " " " " 1.

$\Sigma^2 = \{00,01,10,11\}$ " " " " " 2.

$\Sigma^3 = \{000,001,010,100,011,101,110,111\}$ " 3.

⋮

and so on.

* What are the two variations of powers of an alphabet?

Variations of power of an alphabet → Kleene closure / Kleene star
Variations of power of an alphabet → Kleene plus

* What is Kleene closure / Kleene star / star operator ?

- The Kleene closure is defined as

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \ldots\ldots$$

which is the set of words of any length including null string. Each string is made up of symbols only from Σ.

- Example : ① $\Sigma = \{0,1\}$

$\Sigma^* = \{\varepsilon, 0, 1, 00, 01, 10, 11, 000, 001, \ldots\ldots\}$

which is the set of strings of 0's & 1's of any length.

② $\Sigma = \{"a", "bc"\}$

$\Sigma^* = \{\varepsilon, "a", "bc", "abc", "bca", "aa", "bcbc", \ldots\ldots\}$

which is the set of strings of a's & bc's of any length.

* What is Kleene plus ?

- The Kleene plus denoted by Σ^+ is defined as

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots\ldots$$

which is the set of words of any length except the null string.

- ε (Σ^0) is not part of Σ^+ & hence $\varepsilon \notin \Sigma^+$

- Example : $\Sigma = \{0,1\}$

$\Sigma^+ = \{0,1,00,01,10,11,000,001,\ldots\ldots\}$ is the set of strings of 0's & 1's of any length except the null string.

$\Sigma^* = \Sigma^+ + \varepsilon$

$\Sigma^+ = \Sigma^* - \varepsilon$

* Recursive definition for length of the string

If a is the symbol and ω is the string derived from the alphabet Σ, the length of a string can be defined as

$$|\omega| = \begin{cases} 0 & \text{if } \omega = \varepsilon \\ 1 & \text{if } \omega = a \\ |ua| = |u|+1 & \text{if } \omega = ua \text{ / otherwise} \end{cases}$$

for each $a \in \Sigma$ and each $u, \omega \in \Sigma^*$

* Recursive definition for reverse of a string

If a is the symbol and ω is the string derived from the alphabet Σ, the reverse of a string can be defined as

$$\omega^R = \begin{cases} \varepsilon & \text{if } \omega = \varepsilon \\ a & \text{if } \omega = a \\ (xa)^R = ax^R & \text{if } \omega = xa \text{ / otherwise} \end{cases}$$

for each $a \in \Sigma$ and each $\omega, x \in \Sigma^*$

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* Languages

- A language can be defined as a set of strings obtained from Σ^* where Σ is set of alphabets of a particular language.
- A language is subset of Σ^*, denoted by $L \subseteq \Sigma^*$
- Example : ① A language of strings consisting of equal number of 0's & 1's can be represented as

$\Sigma = \{0,1\}$ $\{\varepsilon, 01, 10, 0011, 1010, 0101, 0011, \ldots\}$

② The language of strings consisting of n number of 0's followed by n number of 1's can be represented as

$\{\varepsilon, 01, 0011, 000111, \ldots\}$

③ A language containing empty string ε is denoted by $\{\varepsilon\}$

④ An empty language is denoted by ϕ.

NOTE: $\phi \neq \{\varepsilon\}$

↓ No strings $\{\}$; ↓ one string

* Sentence

- A string that belongs to a language is called word or sentence of that language.
- Example: A language of strings consisting of equal number of 0's and 1's can be represented as

  $\{\varepsilon, 01, 10, 0011, 1010, 0101, 1100, \ldots\}$

  So, 01 is a sentence

  10 " " "

  0011 " " "

* Different types of languages

Type 0	Language	-	Phrase structured language
Type 1	"	-	Context sensitive "
Type 2	"	-	Context free "
Type 3	"	-	Regular "

* Regular Languages

- The regular languages are defined as the languages accepted by finite Automata
  - DFA's - Deterministic finite Automata
  - NFA's - Nondeterministic " "
  - ε-NFA's - " " " with ε-moves
- They are also the languages defined by Regular expressions.

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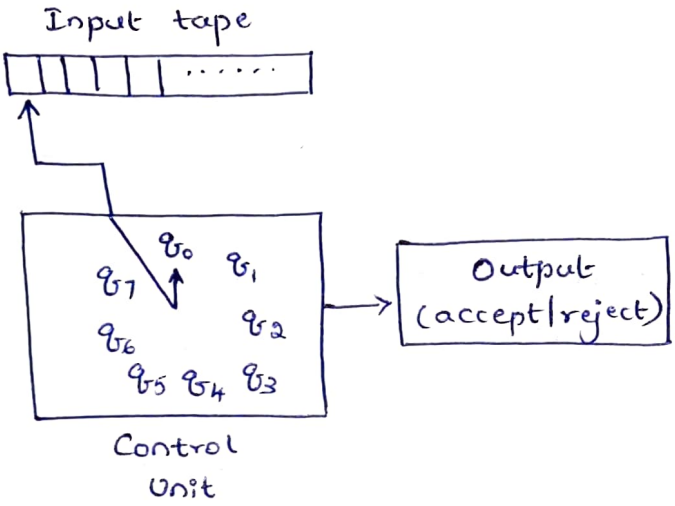
* Abstract Machine

- An Abstract machine / Abstract compuleī is a conceptual or theoretical model of a compuleī hardware or software system which really does not exisīs.
- These machines are not actual machines and hence, they are also called hypothetical compulers.
- These machines have commonly encounlered hardware features and conceplīs and avoids most of the details that are often found in real machines.
- The various types of abstract machines are
  - finīlē Aulomata
  - Linear bounded Aulomata
  - Push down Aulomata
  - Turing machine

* Finīlē Aulomata

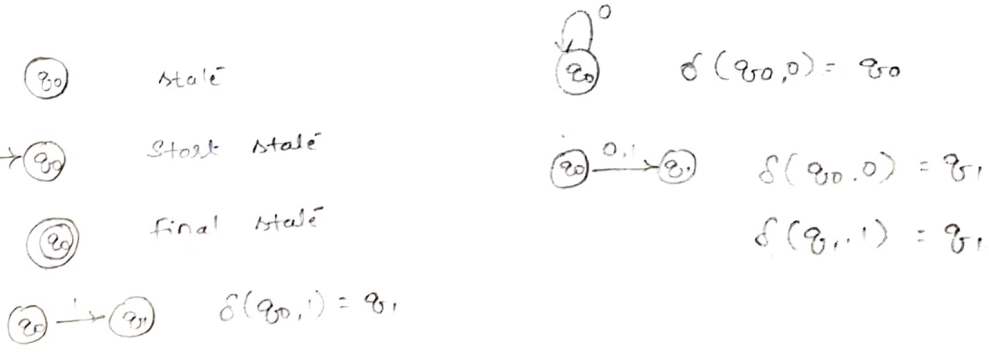
- A finīlē aulomata is a mathematical model which is used to study the abstract machines or abstract computing devices with the inpulīs chosen from Σ.
- Σ stands for set of alphabelīs using which any string can be obtained.
- On reading the string, the machine may accept the string or reject the string.
- Using this abstract model, the behavior of the actual syslēm can be understood and build to perform various activities.
- Finīlē aulomaton is an abstract model of a digital compuleī which has three componenīs • Input file • Output • Control Unit

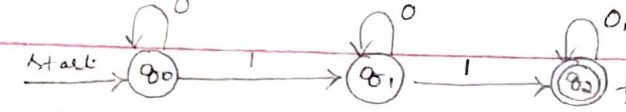


- **Input tape** — The input tape is divided into cells, each of which can hold one symbol.
  The string to be processed is stored in these cells.
- **Control Unit**
  - The machine has some states one of which is the start state designated as q_0 and at least one final state.
  - Apart from these, it has some finite states designated by q_1, q_2 and so on.
  - Based on the current input symbol, the state of the machine can change.
- **Output**
  - Output may be accept or reject.
  - When end of input is encountered, the control unit may be in accept or reject state.

* **The different types of finite automata !**

- Deterministic finite Automata (DFA)
- Non-Deterministic finite Automata (NFA)
- Non-Deterministic finite Automata with ε-moves (ε-NFA)





$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{0, 1\}$
$M = (Q, \Sigma, \delta, q_0, F)$
$\delta : (Q \times \Sigma) \to Q$
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Deterministic finite Automata

- The Deterministic finite Automata is 5-tuple or quintuple indicating five components

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where,

M is the name of the machine. It can also be called by any name.

Q is non-empty, finite set of states

Σ is non-empty, finite set of input alphabets

$\delta : Q \times \Sigma$ to Q

δ is transition function which is a mapping from $Q \times \Sigma$ to Q. Based on the current state and input symbol, the machine enters into another state.

$q_0 \in Q$ is a start state

$F \subseteq Q$ is set of accepting or final states.

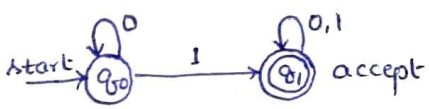
Thus, name DFA emerges from the following facts:

- Deterministic — There is exactly one transition for every input symbol from the state. So, it is possible to determine exactly to which state the machine enters into after consuming the input symbol. So, the machine is deterministic.
- finite — Has finite number of states and arcs. So, it is deterministic and finite
- Automaton — Automaton is a machine which may accept the string or reject the string. So, it is DFA.

Simple Notations for DFA's

- The two notations using which the DFA's can be easily represented are:
  - (1) Transition diagram / Transition graph
  - (2) Transition table

- The transition diagram for DFA $M = (Q, \Sigma, \delta, q_0, F)$ is defined as a graph with circles, arrow and arcs with labels, two circles etc.
- It is formally defined as
  - ✓ Each state of Q corresponds to one node or vertex represented using a circle or two circles.
  - ✓ Alphabets in Σ are represented as labels along with the edges.
  - ✓ The transition from one state to another state is indicated by the directed edge)
    Let $\delta(q_i, a) = q_j$. This indicates that there is a directed edge from q_i to q_j and the edge is labeled a.
  - The start state is a state which has an arrow not originating from any node and entering into the state. This is labeled with start.
  - The final states or accepting states which are in F are represented by double circles. The states which are not in F are represented by a single circle.
- Example:



  - It has two nodes corresponding to the two states of machine q_0 and q_1.
  - q_0 has a start arrow (incoming arrow not originating from any node) and hence it is the start state.
  - q_1 is the accepting state and hence it is denoted by two circles.
  - There are 3 arcs: the first one with label 0 from q_0 to q_0,
    the second " " " 1 " q_0 " q_1,
    & " third " " " 0,1 both from q_1 to q_1.

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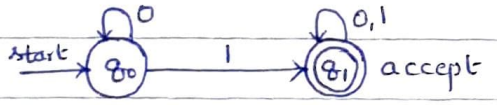
- The transition table for DFA $M = (Q, \Sigma, \delta, q_0, F)$ is defined as a conventional, tabular representation of a transition function such as δ which takes two arguments and returns a value with :
  - The rows of the table correspond to the states of DFA obtained from Q.
  - The columns correspond to the input symbols obtained from Σ. (There is one row for each state, and one column for each input.
  - If q_i is the current state of DFA and a is the current input symbol, the value returned from $\delta(q_i, a)$ represent the next state of DFA and is entered in row q_i and column a.
  - The start state is marked with an arrow
  - The final state is marked with a star.

- Example :

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_0	q_1
$*q_1$	q_1	q_1

* The transition diagram of DFA has two states q_0 and q_1
* There are two input symbols 0 and 1
* Start state is represented using an arrow mark not originating from any state and labeled start.

* Represented using two rows q_0 and q_1
* The two input symbols 0 and 1 correspond to two columns.
* The start state is identified by putting an arrow with direction towards right.

* The final states are represented by two circles.
* The transition from state q_s on input symbol a to state p is represented by a directed edge originating from state q_s and ending at state p with label a.
* The final states are represented by putting stars by the side of states.
* The equivalent transition is represented by writing p in row q_s and column a.

How a DFA processes strings

- The string w is processed by the DFA as

  * Initialization

    Let q_0 is the start of DFA and let w is the string to be processed. Initially, the input pointer points to the left most symbol in string w.

  * Processing

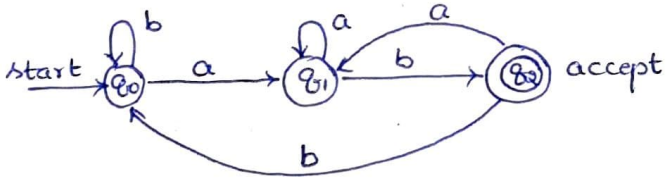
    The DFA reads one symbol from the input string at a time. The machine in state q_0 on reading the input symbol, consult the transition function δ.

    If there is a transition $\delta(q_0, a) = q_1$ then the machine changes its state to q_1 and the input pointer points to the next symbol. Now, the machine in state q_1 after reading the next input symbol may change its state q_2 and so on.

  * Accept and reject

    When the input points to the end of the string, if the DFA is in final state, the string w is accepted by DFA. After the end of the input, if the DFA is not in a final state, then the string w is rejected by the DFA.

- Example :



  * The moves made by the DFA for the input string "abaab" is,

$$q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2$$

    At the end of the string abaab, the DFA will be in state q_2 which is the final state. So, the string "abaab" is accepted by the machine.

* The moves made by the DFA for the input string "abb" is,

$$q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{b} q_0$$

At the end of the string abb, the DFA will be in state q_0 which is not the final state. So, the string abb is rejected by the machine.

Extended transition function of DFA to strings

- Let $M = (Q, \Sigma, \delta, q_0, F)$ be a FA. The extended transition function $\delta^* : Q \times \Sigma^* \to Q$ is defined recursively as

  - Basis

    $\delta^*(q, \varepsilon) = q$

    This indicates that if the machine is in state q and read no input, then the machine is still in state q.

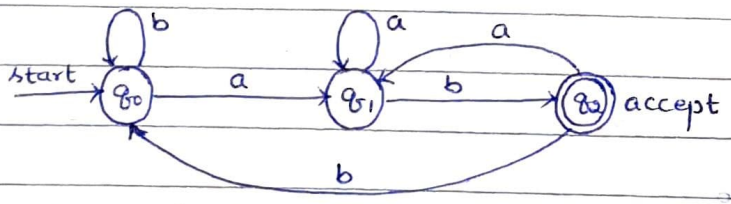
  - Induction

    Let $w = xa$ where a is the last symbol of w and x is the remaining string of w. Let q is the current state and w is the string to be processed and after consuming the string w, let the state of the machine is p. Then,

    $$\delta^*(q, w) = \delta^*(q, xa) = \delta(\delta^*(q, x), a) = p$$

NOTE : We can also use $\hat{\delta}$ in place of δ^*.

- Example :



The moves made by the DFA for the input string "abaab" using δ^* is obtained starting from ε and taking prefix of abaab in increasing size as

- for prefix ε : $\delta^*(q_0, \varepsilon) = q_0$
- for prefix a : $\delta^*(q_0, a) = \delta(\delta^*(q_0, \varepsilon), a) = \delta(q_0, a) = q_1$

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- for prefix ab : $\delta^*(q_0, ab) = \delta(\delta^*(q_0, a), b)$
  $= \delta(q_1, b)$
  $= q_2$
- for prefix aba : $\delta^*(q_0, aba) = \delta(\delta^*(q_0, ab), a)$
  $= \delta(q_2, a)$
  $= q_1$
- for prefix abaa : $\delta^*(q_0, abaa) = \delta(\delta^*(q_0, aba), a)$
  $= \delta(q_1, a)$
  $= q_1$
- for prefix abaab : $\delta^*(q_0, abaab) = \delta(\delta^*(q_0, abaa), b)$
  $= \delta(q_1, b)$
  $= q_2$

- The sequence of states the DFA is in during the processing of string abaab is :

$$q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2$$

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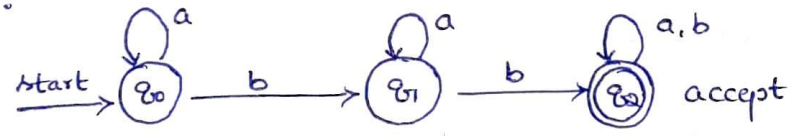
Language accepted by a DFA

- Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA. A string ω is accepted by the machine M, if it takes the initial state q_0 to final state i.e., $\delta^*(q_0, \omega)$ is in F. Thus, the language accepted by DFA represented as L(M) can be formally written as :

$$L(M) = \{\omega \mid \omega \in \Sigma^* \text{ and } \delta^*(q_0, \omega) \text{ is in } F\}$$

- Example :



The moves made by the DFA for the input string "abab" is

$$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2$$

$\delta(q_0, a) = q_0$
$\delta(q_0, b) = q_1$
$\delta(q_1, a) = q_1$
$\delta(q_1, b) = q_2$

At the end of the string abab, the DFA will be in state q_2 which is the final state. So, the string abab is accepted by the machine.

Name of Experiment..... Date.....

Experiment No..... Experiment Result.....

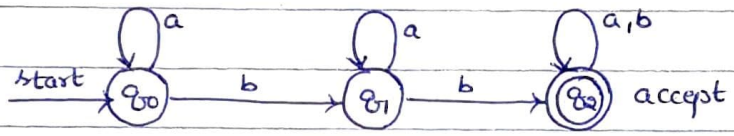
Language rejected by DFA

- Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA. The non-acceptance of a language indicates that after the string ω is processed, the DFA will not be in the final state.

Thus, the non-acceptance of the string ω by an FA or DFA can be defined in formal notation as

$$L(M) = \{ \omega \mid \omega \in \Sigma^* \text{ and } \delta^*(q_0, \omega) \text{ is not in } F \}$$

- Example :



The moves made by the DFA for the input string "abaa" is

$$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{a} q_1 \xrightarrow{a} q_1$$

$\delta(q_0, a) = q_0$

$\delta(q_0, b) = q_1$

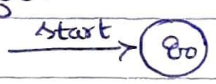
$\delta(q_1, a) = q_1$

$\delta(q_1, a) = q_1$

At the end of the string abaa, the DFA will be in state q_1 which is not the final state. So, the string abaa is rejected by the machine.

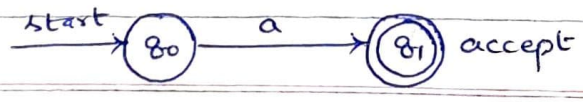
NOTE : 1) DFA to accept empty language $L = \{\phi\}$

- If there is no final state in DFA, the DFA accepts empty language.

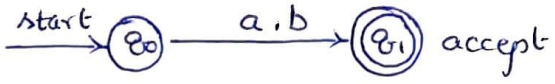


2) DFA to accept an empty string $L = \{\varepsilon\}$

3) DFA to accept exactly one a.



4) DFA to accept one a or one b.



5) DFA to accept zero or more a's.

DFA Design Techniques

- There are three types of problems for which we can construct a DFA:
  - Pattern recognition problems
  - Divisible by k "
  - Modulo-k-counter "

(1) PATTERN RECOGNITION PROBLEMS

STEP 1 : Identify the minimum string

STEP 2 : Identify the alphabets

STEP 3 : Construct a skeleton DFA

STEP 4 : Identify other transitions not defined in step 3.

STEP 5 : Construct the DFA using transitions in steps and step4.

DFSM

Q1. Draw a DFA to accept string of a's having atleast one a.

The DFA is defined as : $M = (Q, \Sigma, \delta, q_0, F)$

$M = (K, \Sigma, \delta, A, F)$

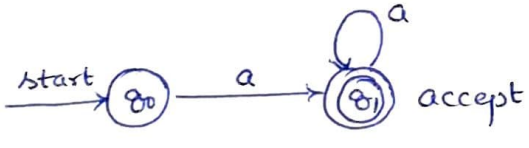
$Q = \{q_0, q_1\}$

$\Sigma = \{a\}$

q_0 is the start state

$F = \{q_1\}$

Transition diagram



Transition table

δ	a
$\rightarrow q_0$	q_1
$*q_1$	q_1

NOTE: The language accepted by DFA is

$L = \{a^n \mid n \geq 1\}$

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Q2. Draw a DFA to accept strings of a's and b's having at least one a.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

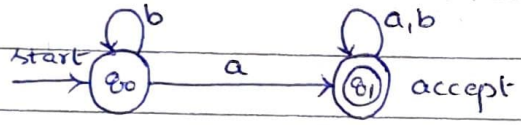
$Q = \{q_0, q_1\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_1\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
$*q_1$	q_1	q_1

$\delta(q_0, b) = q_0$

$\delta(q_0, a) = q_1$

$\delta(q_1, b) = q_1$

NOTE: The language accepted by DFA is

$L = \{\omega : n_a(\omega) \geq 1, \omega \in \{a,b\}^*\}$

Once the machine enters into state q_1, irrespective of any number of inputs of a's and b's, the machine remains in same state q_1 and can not come out of the state. So, the machine is trapped in state q_1 and hence it is called trap state. Since, q_1 is also a final state, the state q_1 is called trapped final state.

Q3. Draw a DFA to accept strings of a's and b's having exactly one a.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

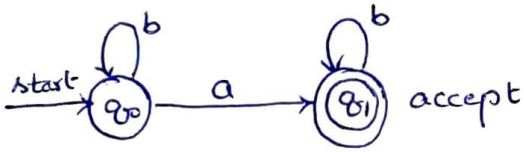
$Q = \{q_0, q_1\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_1\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
$*q_1$	-	q_1

$\delta(q_0, b) = q_0$

$\delta(q_1, a) = -$

$\delta(q_1, b) = q_1$

NOTE : * The Language accepted by DFA is

$L = \{ \omega : n_a(\omega) = 1 , \omega \in \{a, b\}^* \}$

*

accept Trap stalē / Dead Stalē

* A stalē for which there exists transitions to itself for all the input symbols chosen from Σ is called a trap stalē.

* The non final trap stalē is also called dead stalē.

* A final stalē can also be a trap stalē but it is not a dead stalē since the string is accepted at this stalē.

Q4. Obtain a DFA to accept strings of a's and b's starting with string ab.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

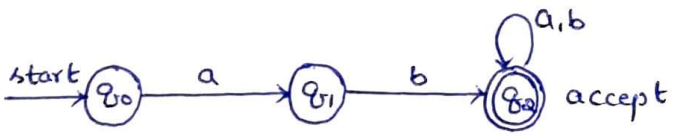
$Q = \{ q_0, q_1, q_2 \}$

$\Sigma = \{ a, b \}$

q_0 is the start stalē

$F = \{ q_2 \}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	-
q_1	-	q_2
$*q_2$	q_2	q_2

$\delta(q_0, b) = -$

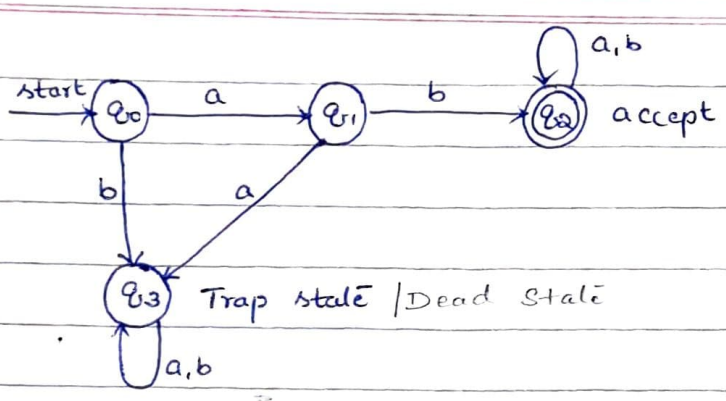
$\delta(q_1, a) = -$

$\delta(q_2, a) = q_2$

$\delta(q_2, b) = q_2$

$L = \{ ab (a+b)^n : n \geq 0 \}$

NOTE :



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Q5. Draw a DFA to accept string of a's and b's ending with the string abb.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

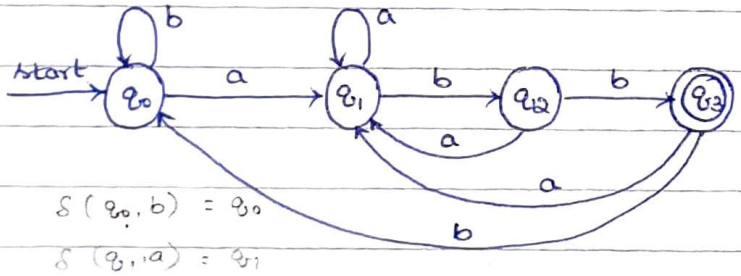
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_3\}$

Transition diagram



$\delta(q_0, b) = q_0$
$\delta(q_1, a) = q_1$
$\delta(q_0, a) = q_1$
$\delta(q_2, a) = q_1$
$\delta(q_3, b) = q_0$

Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_1	q_2
q_2	q_1	q_3
$*q_3$	q_1	q_0

NOTE : The language accepted by DFA is

$$L = \{(a+b)^n abb : n \geq 0\}$$

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Q6. Draw a DFA to accept string of a's and b's which do not end with the string abb.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

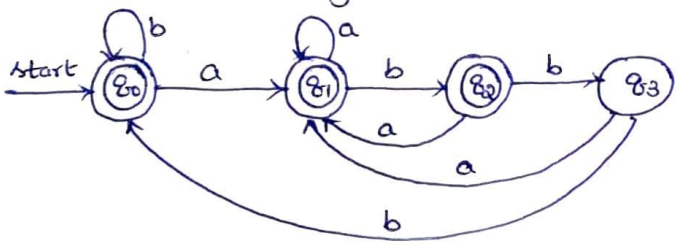
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_0, q_1, q_2\}$

Transition diagram



$L = \{ w \mid w \neq (a+b)^n abb, n \geq 0 \}$

Transition table

δ	a	b
$\rightarrow *q_0$	q_1	q_0
$*q_1$	q_1	q_2
$*q_2$	q_1	q_3
q_3	q_1	q_0

Q7. Draw a DFA to accept string of a's and b's having a substring aab.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

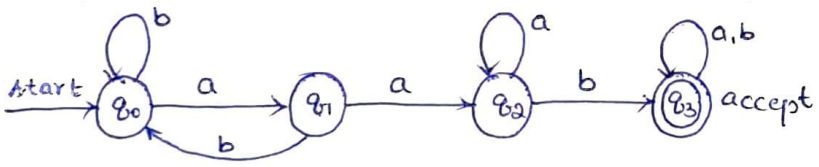
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_3\}$

Transition diagram



$\delta(q_0, b) = q_0$

$\delta(q_1, b) = q_0$

$\delta(q_0, a) = q_2$

$\delta(q_2, a) = q_3$

$\delta(q_3, b) = q_3$

Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
$*q_3$	q_3	q_3

$L = \{ (a+b)^n aab (a+b)^n : n \geq 0 \}$

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Q8. Draw a DFA to accept strings of a's and b's except those having the substring aab.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

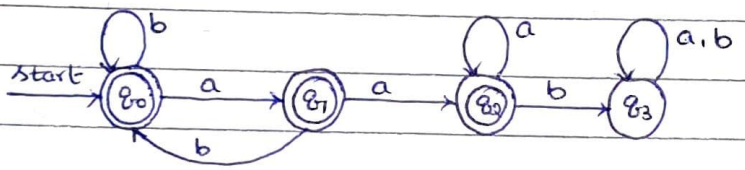
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_0, q_1, q_2\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow *q_0$	q_1	q_0
$*q_1$	q_2	q_0
$*q_2$	q_2	q_3
q_3	q_3	q_3

$L = \{ w \mid w \notin (a+b)^* aab (a+b)^*, n \geq 0 \}$

Q9. Draw a DFA to accept strings of 0's and 1's having three consecutive 0's.

OR

Draw a DFA to accept strings of 0's and 1's having a substring 000.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

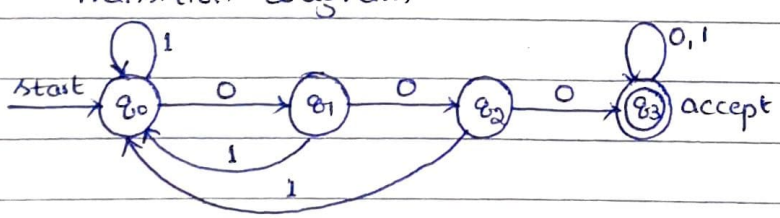
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

q_0 is the start state

$F = \{q_3\}$

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_3	q_0
$*q_3$	q_3	q_3

$L = \{ (1+0)^* 000 (1+0)^* : n \geq 0 \}$

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Q10. Draw a DFA to accept strings of a's and b's such that

$L = \{ awa \mid w \in (a+b)^n \text{ where } n \geq 0 \}$

OR

Draw a DFA to accept strings of a's and b's starting with one a and ending with one a with minimum string aa.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2 \}$

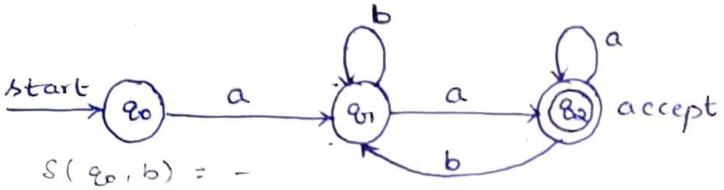
minimum string is "aa"

$\Sigma = \{ a, b \}$

q_0 is the start state

$F = \{ q_2 \}$

Transition diagram



$\delta(q_0, b) = -$

$\delta(q_1, b) = q_1$

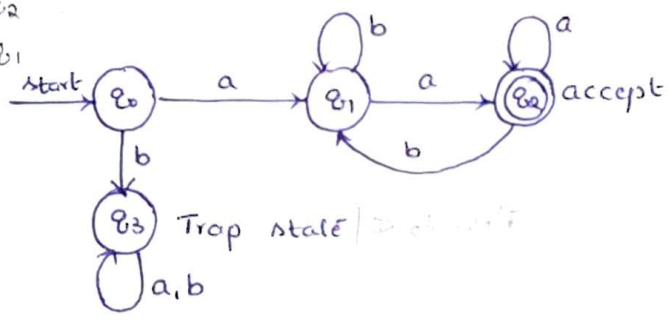
$\delta(q_2, a) = q_2$

$\delta(q_2, b) = q_1$

Transition table

δ	a	b
$\rightarrow q_0$	q_1	-
q_1	q_2	q_1
$* q_2$	q_2	q_1

NOTE :



Q11. Draw a DFA to accept strings of a's and b's ending with ab or ba.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4 \}$

$\Sigma = \{ a, b \}$

q_0 is the start state

$F = \{ q_2, q_4 \}$

$L = \{ w(ab + ba) \mid w \in \{a, b\}^* \}$

$L = \{ (a+b)^n (ab+ba) : n \geq 0 \}$

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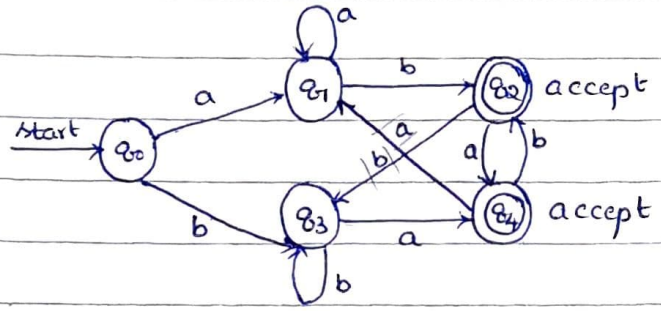
Transition diagram



$\delta(q_0, b) = q_3$
$\delta(q_1, a) = q_1$
$\delta(q_2, a) = q_4$
$\delta(q_3, b) = q_3$
$\delta(q_4, a) = q_1$
$\delta(q_4, b) = q_2$

{aba
bab} Should accept

{abba
baab} Should accept

Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_3
q_1	q_1	q_2
$*q_2$	q_4	q_3
q_3	q_4	q_3
$*q_4$	q_1	q_2

Q12. Obtain a DFA to accept strings of a's and b's having four a's where $\Sigma = \{a, b\}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

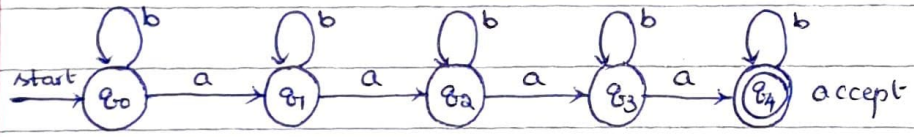
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_4\}$

Transition diagram



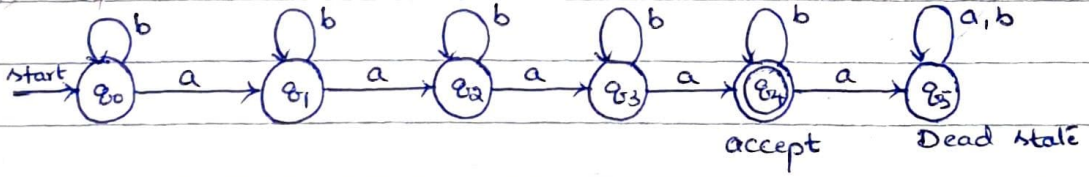
$\delta(q_0, b) = q_0$
$\delta(q_1, b) = q_1$
$\delta(q_2, b) = q_2$
$\delta(q_3, b) = q_3$
$\delta(q_4, a) =$ -
$\delta(q_4, b) = q_4$

$L = \{w \mid n_a(w) = 4, w \in \{a,b\}^*\}$

Transition Table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_1
q_2	q_3	q_2
q_3	q_4	q_3
$*q_4$	-	q_4

NOTE:



L =

Q13. Obtain a DFA to accept strings of 0's, 1's and 2's beginning with a '0' followed by odd number of 1's and ending with a '2'.

The DFA is defined as $M=(Q, \Sigma, \delta, q_0, F)$

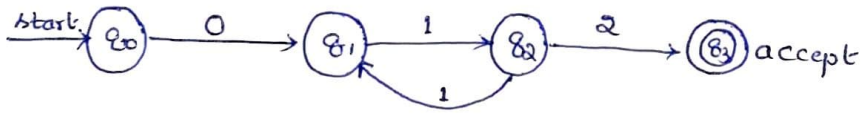
$Q = \{q_0, q_1, q_2, q_3\}$ minimum string 012

$\Sigma = \{0, 1, 2\}$

q_0 is the start state

$F = \{q_3\}$ $L = \{0 1^n 2 , n \text{ is odd}\}$

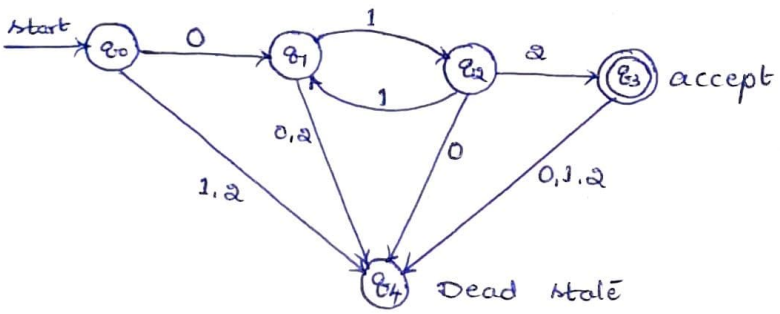
Transition diagram



Transition table

δ	0	1	2
→ q_0	q_1	-	-
q_1	-	q_2	-
q_2	-	q_1	q_3
* q_3	-	-	-

NOTE :



Q14. Obtain a DFA to accept strings of a's and b's with at most (max) two consecutive b's.

The DFA is defined as $M=(Q, \Sigma, \delta, q_0, F)$

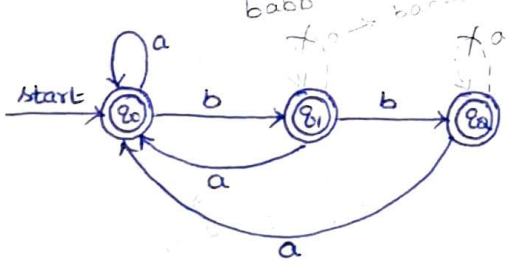
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$ ϵ, b, bb

q_0 is the start state

$F = \{q_0, q_1, q_2\}$

Transition diagram $L = \{w \in \{a,b\}^* \text{ where } b = \epsilon | b | bb\}$



bbabb

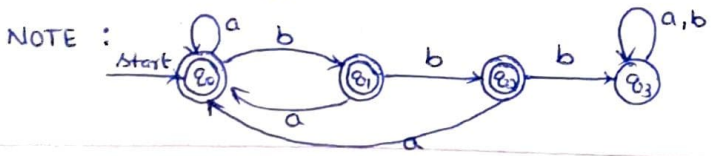
bbabba

babb

Transition table

δ	a	b
→* q_0	q_0	q_1
* q_1	q_0	q_2
* q_2	q_0	-

NOTE :



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Q15. Obtain a DFA to accept strings of 0's and 1's starting with at least two 0's and ending with at least two 1's.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

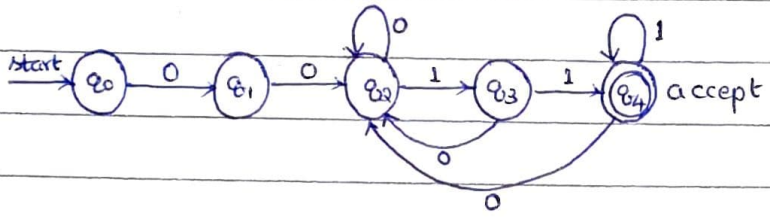
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

q_0 is the start state

$F = \{q_4\}$

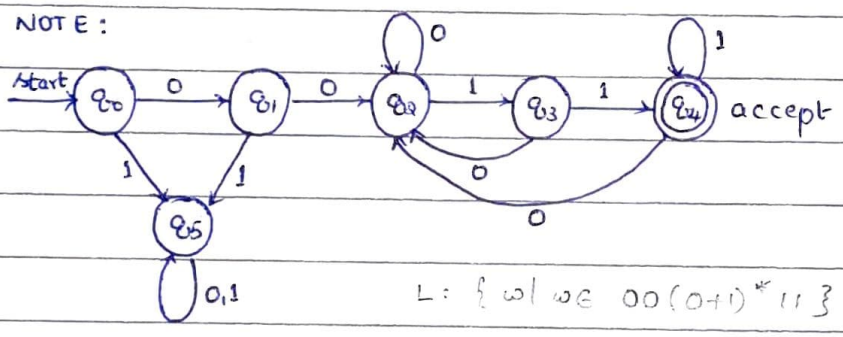
Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	-
q_1	q_2	-
q_2	q_2	q_3
q_3	q_2	q_4
$*q_4$	q_2	q_4

NOTE:



$L = \{w \mid w \in 00(0+1)^*11\}$

Q16. Obtain a DFA to accept the language $L = \{wbab \mid w \in \{a,b\}^*\}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

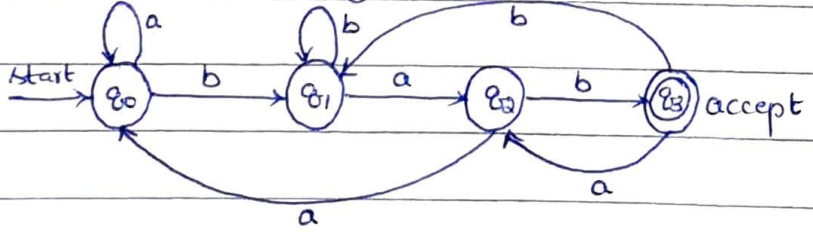
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_3\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_0	q_1
q_1	q_2	q_1
q_2	q_0	q_3
$*q_3$	q_2	q_1

Q17. Draw a DFA to accept strings of a's and b's having not more than three a's.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

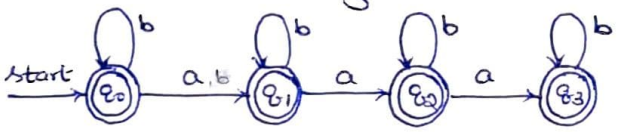
$F = \{q_0, q_1, q_2, q_3\}$

ε, a, aa, aaa

but not continuous

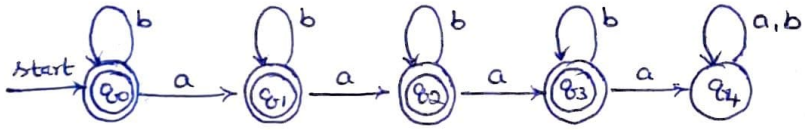
b

Transition diagram



baaa ✗

NOTE:



Transition table

δ	a	b
$\rightarrow *q_0$	q_1	q_0
$*q_1$	q_2	q_1
$*q_2$	q_3	q_2
$*q_3$	-	q_3

$L = \{\omega : n_a(\omega) \leq 3,\ \omega \in \{a,b\}^*\}$

Dec08 Jun09 Q18. Draw a DFA to accept set of all strings on the alphabet $\Sigma = \{0,1\}$ that either begins or ends or both with substring 01.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

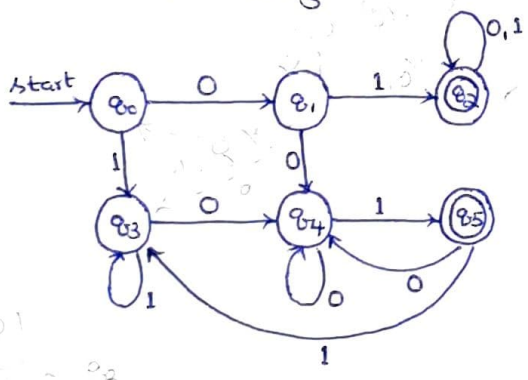
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{0, 1\}$

q_0 is the start state

$F = \{q_2, q_5\}$

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	q_3
q_1	q_4	q_2
$*q_2$	q_2	q_2
q_3	q_4	q_3
q_4	q_4	q_5
$*q_5$	q_4	q_3

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Q19. Draw a DFA to accept the language $L = \{ \omega : n_a(\omega) \geq 1, n_b(\omega) = 2 \}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

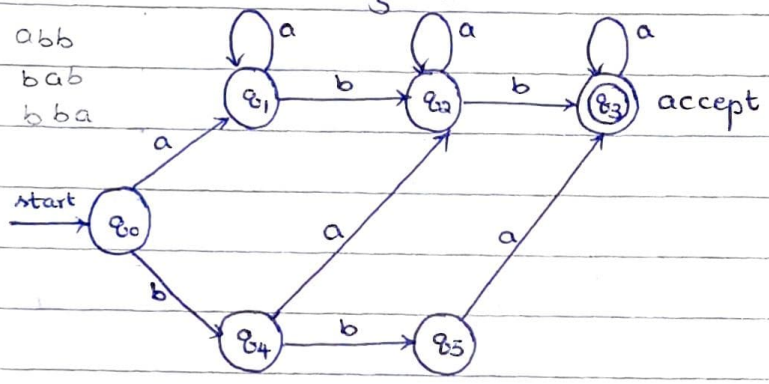
$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \}$

$\Sigma = \{ a, b \}$

q_0 is the start state

$F = \{ q_3 \}$

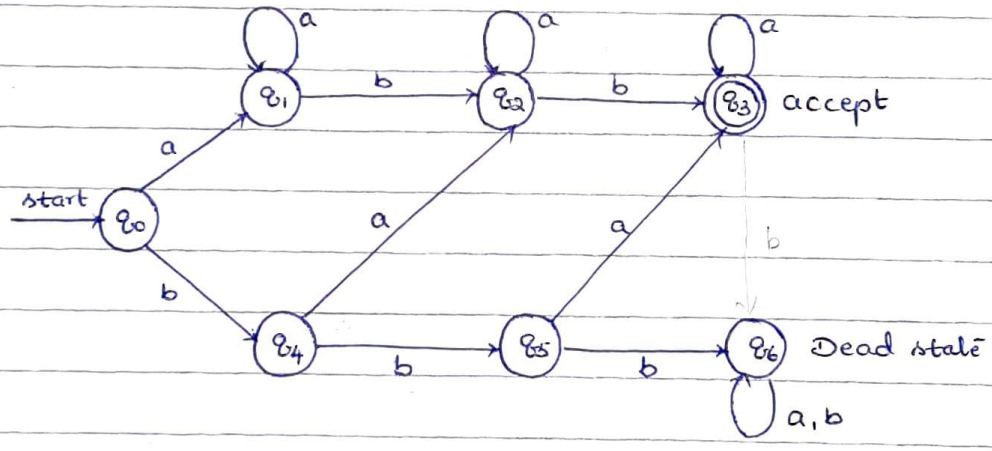
Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_4
q_1	q_1	q_2
q_2	q_2	q_3
$*q_3$	q_3	-
q_4	q_2	q_5
q_5	q_3	-

NOTE :



Q20. Draw a DFA to accept the language

$L = \{ \omega : n_a(\omega) = 2, n_b(\omega) \geq 3 \}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11} \}$

$\Sigma = \{ a, b \}$

q_0 is the start state

$F = \{q_5\}$

① aabbb ② baabb bbaab bbbaa

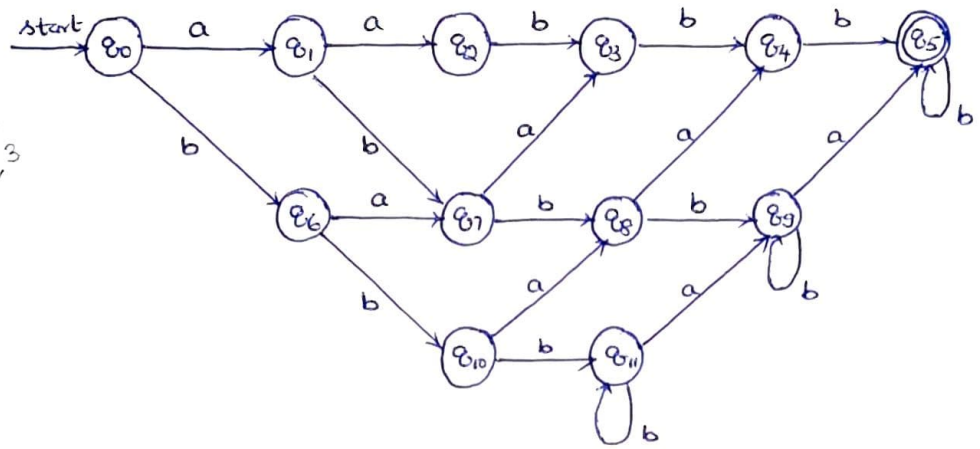
③ ababb babab bbabb

abbab babba

abbba

Transition diagram

(10)



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_6
q_1	q_2	q_7
q_2	-	q_3
q_3	-	q_4
q_4	-	q_5
* q_5	-	q_5
q_6	q_7	q_{10}
q_7	-	q_8
q_8	q_4	q_9
q_9	q_5	q_9
q_{10}	q_8	q_{11}
q_{11}	q_9	q_{11}

NOTE : Run in a string

A run in a string is a substring whose length is at least two but consisting of same symbol.

Ex : abbbbaaab

run of b's of length 4

run of a's of length 3

Q21. Draw a DFA to accept the language $L = \{w : w$ contains no runs of length less than four $\}$ where $\Sigma = \{a, b\}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_4, q_8\}$

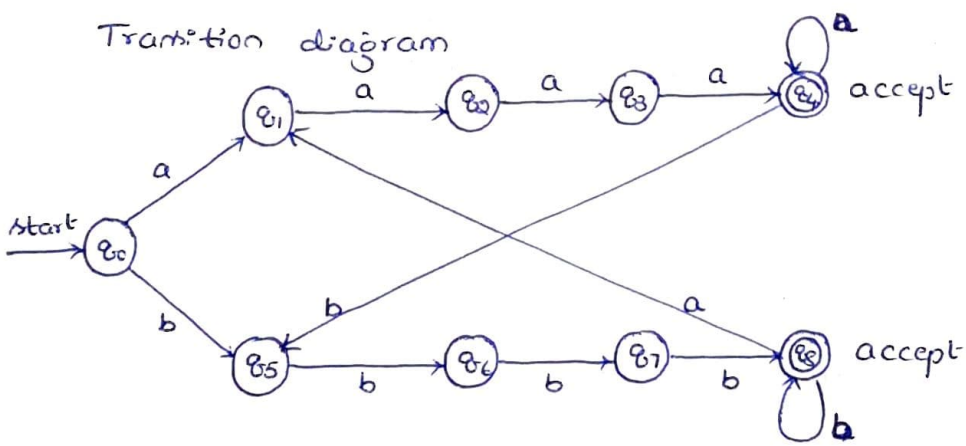
aaaa

bbbb

aaaabbbb.....

bbbbaaaa.....

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_5
q_1	q_2	-
q_2	q_3	-
q_3	q_4	-
* q_4	q_4	q_5
q_5	-	q_6
q_6	-	q_7
q_7	-	q_8
* q_8	q_1	q_8

Name of Experiment..... Date.....

Experiment No..... Experiment Result.....

NTCSE BIT

Q22. Draw a DFA to accept the language

$L = \{w : \text{every run of a's has length either two or three}\}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

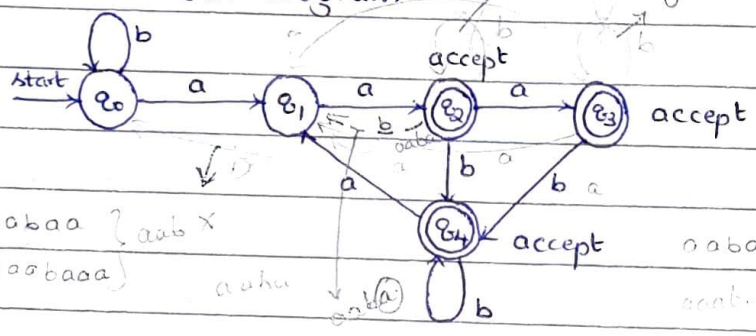
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_2, q_3, q_4\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_2	–
$*q_2$	q_3	q_4
$*q_3$	–	q_4
$*q_4$	q_1	q_4

(@) Divisible by k problems

- for these types of problems, the transitions can be obtained using the following relation

$$\delta(q_i, a) = q_j \quad \text{where} \quad j = (r * i + d) \bmod k$$

Where

r is the radix of input. for binary $r = 2$

i is the remainder obtained after dividing by k

d represents digits. for binary $d = \{0, 1\}$

k is divisor

- STEP 1 : Identify the radix, input alphabets and the divider k

STEP 2 : Compute the possible remainders. These remainders represent the states of DFA.

STEP 3 : find the transitions using $\delta(q_i, a) = q_j$ where $j = (r * i + d) \bmod k$

STEP 4 : Construct the DFA using the transitions obtained in step3.

Q1. Construct a DFA which accepts strings of 0's and 1's where the value of each string is represented as a binary number. Only the strings representing zero modulo five should be accepted. for example, 0000, 0101, 1010, 1111 etc. should be accepted.
0 5 10 15

Dec 09
Dec 11

STEP 1 : $r = 2$ Radix

$\Sigma = \{0, 1\} = d$ alphabets / binary digits

$k = 5$ divisor

STEP 2 : $i = 0, 1, 2, 3, 4$ Remainders

STEP 3 : $\delta(q_i, a) = q_j$ where $j = (r * i + d) \bmod k$

remainder	d	$(2*i+d) \bmod 5 = j$	$\delta(q_i, d) = q_j$
$i=0$	0	$(2*0+0) \bmod 5 = 0$	$\delta(q_0, 0) = q_0$
	1	$(2*0+1) \bmod 5 = 1$	$\delta(q_0, 1) = q_1$
$i=1$	0	$(2*1+0) \bmod 5 = 2$	$\delta(q_1, 0) = q_2$
	1	$(2*1+1) \bmod 5 = 3$	$\delta(q_1, 1) = q_3$
$i=2$	0	$(2*2+0) \bmod 5 = 4$	$\delta(q_2, 0) = q_4$
	1	$(2*2+1) \bmod 5 = 0$	$\delta(q_2, 1) = q_0$
$i=3$	0	$(2*3+0) \bmod 5 = 1$	$\delta(q_3, 0) = q_1$
	1	$(2*3+1) \bmod 5 = 2$	$\delta(q_3, 1) = q_2$
$i=4$	0	$(2*4+0) \bmod 5 = 3$	$\delta(q_4, 0) = q_3$
	1	$(2*4+1) \bmod 5 = 4$	$\delta(q_4, 1) = q_4$

STEP 4 : The DFA can be defined as

$M = (Q, \Sigma, \delta, q_0, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

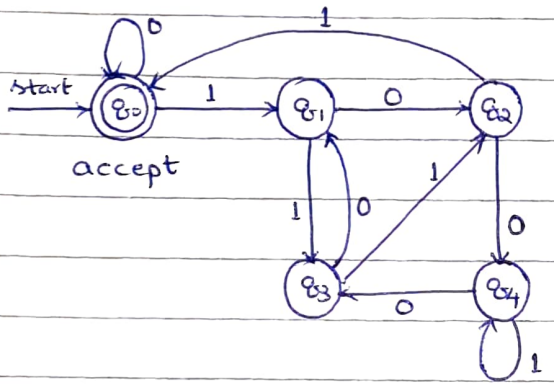
q_0 is the start state

$F = \{q_0\}$

Name of Experiment..... Date..... Page No. 118

Experiment No..... Experiment Result.....

Transition diagram



Transition table

δ	0	1
$\rightarrow *q_0$	q_0	q_1
q_1	q_2	q_3
q_2	q_4	q_0
q_3	q_1	q_2
q_4	q_3	q_4

NOTE : In general, for modulo k, number of states of DFA will be k.

Dec 13 Q2. Obtain a DFA which accepts the set of all strings beginning with a 1 that when interpreted as a binary integer, is a multiple of 5. for example, 101, 1010, 1111 etc., are multiples of 5. Note that 0101 is not beginning with 1 and it should not be accepted.

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

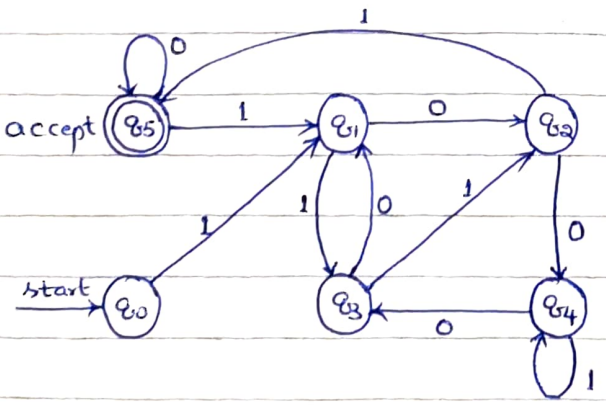
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{0, 1\}$

q_0 is the start state

$F = \{q_5\}$

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	-	q_1
q_1	q_2	q_3
q_2	q_4	q_5
q_3	q_1	q_2
q_4	q_3	q_4
$*q_5$	q_5	q_1

Q3. Obtain a DFA that accepts set of all strings that, when interpreted in reverse as a binary integer, is divisible by 5. Examples of strings in the language are 0, 10011, 1001100 and 0101.

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

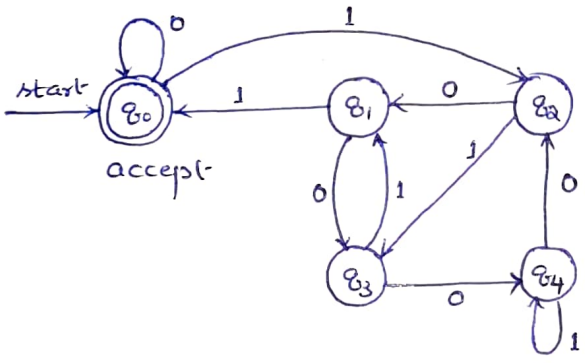
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

q_0 is the start state

$F = \{q_0\}$

Transition diagram



Transition table

δ	0	1
$\rightarrow *q_0$	q_0	q_2
q_1	q_3	q_0
q_2	q_1	q_3
q_3	q_4	q_1
q_4	q_2	q_4

Q4. Draw a DFA to accept decimal strings divisible by 3.

STEP 1 : $r = 10$

$d = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$k = 3$

STEP 2 : $i = 0, 1, 2$

$\{0, 3, 6, 9\}$ with 0 as the remainder $\Rightarrow q_0$

$\{1, 4, 7\}$ " 1 " " " $\Rightarrow q_1$

$\{2, 5, 8\}$ " 2 " " " $\Rightarrow q_2$

STEP 3 :

remainder	d	$(10 * i + d) \bmod 3 = j$	$\delta(q_i, d) = q_j$	
$i = 0$	0	$(10*0+0) \bmod 3 = 0$	$\delta(q_0, 0) = q_0$	$\Rightarrow \delta(q_0, \{0,3,6,9\}) = q_0$
	1	$(10*0+1) \bmod 3 = 1$	$\delta(q_0, 1) = q_1$	$\Rightarrow \delta(q_0, \{1,4,7\}) = q_1$
	2	$(10*0+2) \bmod 3 = 2$	$\delta(q_0, 2) = q_2$	$\Rightarrow \delta(q_0, \{2,5,8\}) = q_2$

i = 1	0	(10 * 1 + 0) mod 3 = 1	$\delta(q_1, 0) = q_1$	$\Rightarrow \delta(q_1, \{0,3,6,9\}) = q_1$
	1	(10 * 1 + 1) mod 3 = 2	$\delta(q_1, 1) = q_2$	$\Rightarrow \delta(q_1, \{1,4,7\}) = q_2$
	2	(10 * 1 + 2) mod 3 = 0	$\delta(q_1, 2) = q_0$	$\Rightarrow \delta(q_1, \{2,5,8\}) = q_0$
i = 2	0	(10 * 2 + 0) mod 3 = 2	$\delta(q_2, 0) = q_2$	$\Rightarrow \delta(q_2, \{0,3,6,9\}) = q_2$
	1	(10 * 2 + 1) mod 3 = 0	$\delta(q_2, 1) = q_0$	$\Rightarrow \delta(q_2, \{1,4,7\}) = q_0$
	2	(10 * 2 + 2) mod 3 = 1	$\delta(q_2, 2) = q_1$	$\Rightarrow \delta(q_2, \{2,5,8\}) = q_1$

STEP 4 : The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

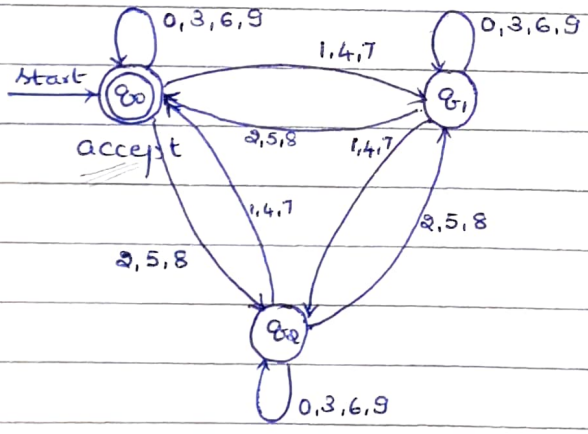
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

q_0 is the start state

$F = \{q_0\}$

Transition diagram



Transition table

δ	0,3,6,9	1,4,7	2,5,8
$\rightarrow * q_0$	q_0	q_1	q_2
q_1	q_1	q_2	q_0
q_2	q_2	q_0	q_1

(3) Modulo k counter problems

June 09 Q1. Obtain a DFA to accept strings of even number of a's.

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

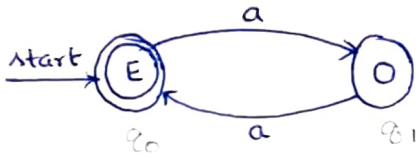
$Q = \{E, O\}$ $L = \{w \mid w$ is a even number$\}$

$\Sigma = \{a\}$

$q_0 = E$ is the start state

F: {E}

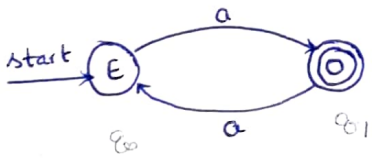
Transition diagram



Transition table

δ	a
→*E	O
O	E

NOTE : DFA is to accept strings of odd number of a's



δ	a
→E	O
*O	E

Q2. Obtain a DFA to accept the language $L = \{\omega : |\omega| \bmod 3 = 0\}$ where $\Sigma = \{a\}$

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

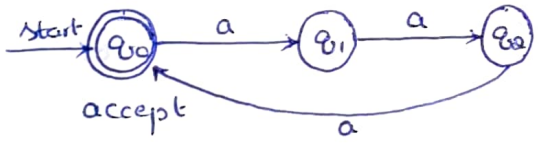
$Q = \{q_0, q_1, q_2\}$ $i = 0, 1, 2$

$\Sigma = \{a\}$

q_0 is the start state

$F = \{q_0\}$

Transition diagram



Transition table

δ	a
→*q_0	q_1
q_1	q_2
q_2	q_0

NOTE : The given language accepted by above DFA can also be represented as

$L = \{\omega : |\omega| \bmod 3 = 0\}$ where $\Sigma = \{a\}$

OR

$L = \{\omega : n_a(\omega)$ are divisible by $3\}$ where $\Sigma = \{a\}$

OR

$L = \{a^{3n} : n \geq 0\}$

Dec10

Q3. Obtain a DFA to accept the language $L = \{w : |w| \bmod 3 = 0\}$ on $\Sigma = \{a, b\}$

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

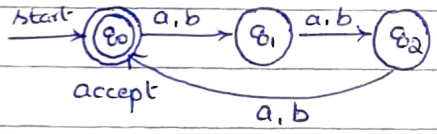
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_0\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow *q_0$	q_1	q_1
q_1	q_2	q_2
q_2	q_0	q_0

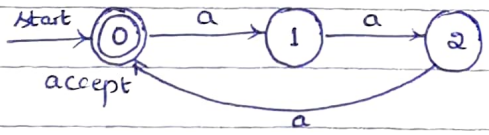
Q4. Obtain a DFA to accept the following language

(a) $L = \{w : |w| \bmod 3 \geq |w| \bmod 2\}$ where $w \in \Sigma^*$ and $\Sigma = \{a\}$

(b) $L = \{w : |w| \bmod 3 \neq |w| \bmod 2\}$ where $w \in \Sigma^*$ and $\Sigma = \{a\}$

The DFA is to accept a string w such that $|w| \bmod 3 = 0$

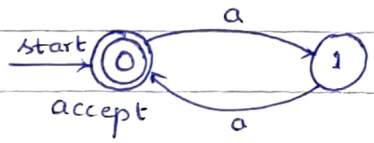
M1



$Q1 = \{0, 1, 2\}$

The DFA is to accept a string w such that $|w| \bmod 2 = 0$

M2



$Q2 = \{0, 1\}$

$Q1 \times Q2 = \{(0,0)\ (0,1)\ (1,0)\ (1,1)\ (2,0)\ (2,1)\}$

$\delta((0,0), a) = (\overset{M1}{\delta(0,a)}, \overset{M2}{\delta(0,a)}) = (1,1)$

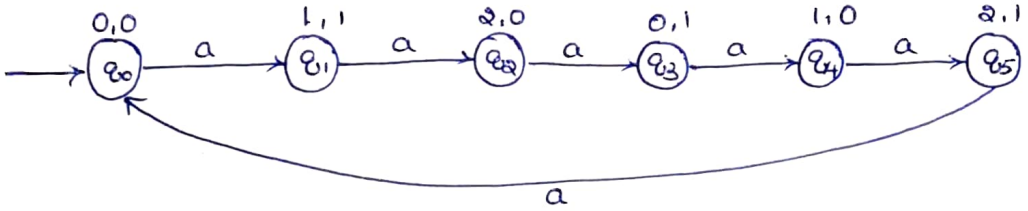
$\delta((1,1), a) = (\delta(1,a), \delta(1,a)) = (2,0)$

$\delta((2,0), a) = (\delta(2,a), \delta(0,a)) = (0,1)$

$\delta((0,1), a) = (\delta(0,a), \delta(1,a)) = (1,0)$

$\delta((1,0),a) = (\delta(1,a), \delta(0,a)) = (2,1)$

$\delta((2,1),a) = (\delta(2,a), \delta(1,a)) = (0,0)$



(a) The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

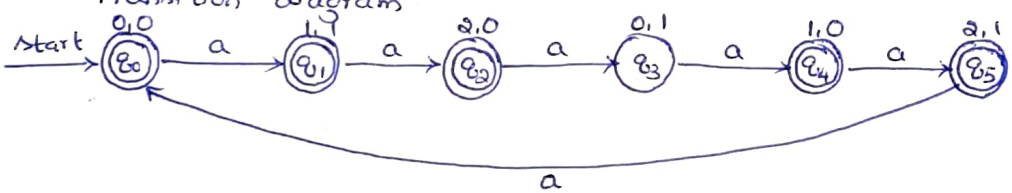
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{a\}$

q_0 is the start state

$F = \{q_0, q_1, q_2, q_4, q_5\}$ the pairs (x,y) such that $x \geq y$ are final states.

Transition diagram



Transition table

δ	a
$\rightarrow * q_0$	q_1
$* q_1$	q_2
$* q_2$	q_3
q_3	q_4
$* q_4$	q_5
$* q_5$	q_0

(b) The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

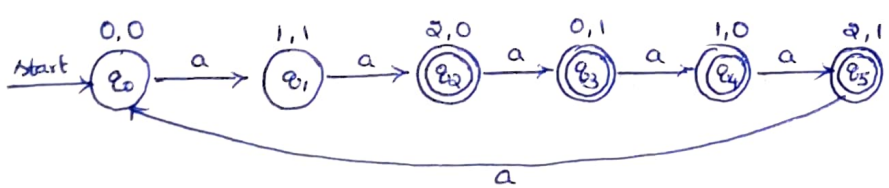
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{a\}$

q_0 is the start state

$F = \{q_3, q_4, q_5, q_2\}$ the pairs (x,y) such that $x \neq y$ are final states.

Transition diagram



Transition table

δ	a
$\rightarrow q_0$	q_1
q_1	q_2
$* q_2$	q_3
$* q_3$	q_4
$* q_4$	q_5
$* q_5$	q_0

Q5. Obtain a DFA to accept the following language

(a) $L = \{\omega : |\omega| \bmod 3 \geq |\omega| \bmod 2\}$ where $\omega \in \Sigma^*$ and $\Sigma = \{a, b\}$

(b) $L = \{\omega : |\omega| \bmod 3 \neq |\omega| \bmod 2\}$ where $\omega \in \Sigma^*$ and $\Sigma = \{a, b\}$

(a) The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

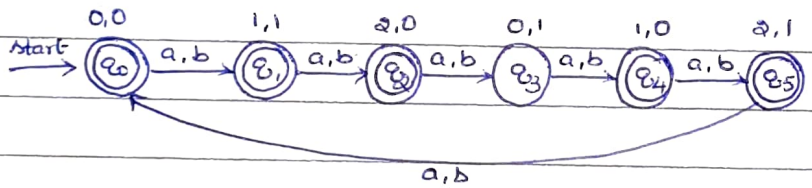
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_0, q_1, q_2, q_4, q_5\}$ $x \geq y$

Transition diagram



Transition table

δ	a	b
$\rightarrow *q_0$	q_1	q_1
$*q_1$	q_2	q_2
$*q_2$	q_3	q_3
q_3	q_4	q_4
$*q_4$	q_5	q_5
$*q_5$	q_0	q_0

(b) The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

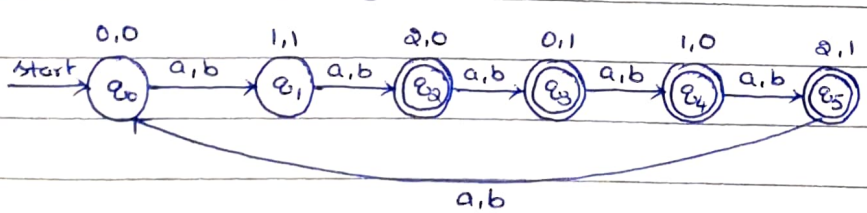
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_2, q_3, q_4, q_5\}$ $x \neq y$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_1
q_1	q_2	q_2
$*q_2$	q_3	q_3
$*q_3$	q_4	q_4
$*q_4$	q_5	q_5
$*q_5$	q_0	q_0

Q6. Obtain a DFA to accept the language $L = \{\omega : |\omega| \bmod 5 \neq 0\}$ where $\Sigma = \{a\}$

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The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

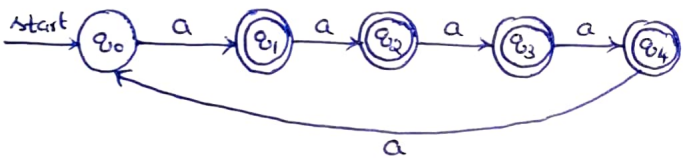
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{a\}$

q_0 is the start state

$F = \{q_1, q_2, q_3, q_4\}$

Transition diagram



Transition table

δ	a
$\rightarrow q_0$	q_1
$*q_1$	q_2
$*q_2$	q_3
$*q_3$	q_4
$*q_4$	q_0

Q7. Obtain a DFA to accept the language $L = \{\omega : |\omega| \bmod 5 \neq 0\}$ on $\Sigma = \{a, b\}$

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

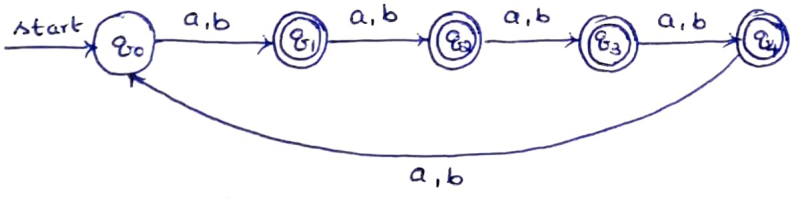
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_1, q_2, q_3, q_4\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_1
$*q_1$	q_2	q_2
$*q_2$	q_3	q_3
$*q_3$	q_4	q_4
$*q_4$	q_0	q_0

Q8. Obtain a DFA to accept strings of a's and b's having even number of a's and even number of b's.

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

q_0 is the start state

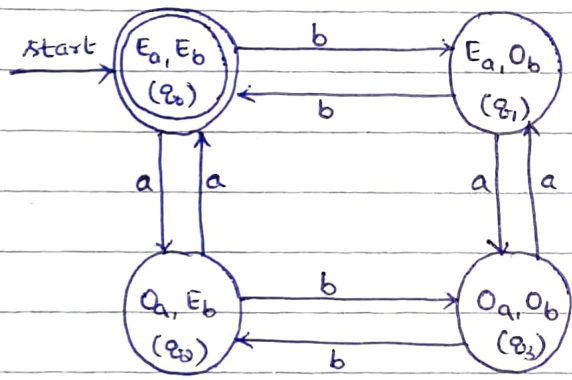
$F = \{q_0\}$

ε, 2, 4, 6

ε, aabb, bbaa,

aaaabbbb, bbbbaaaa

Transition diagram



Transition table

δ	a	b
$\rightarrow *q_0$	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

NOTE : The language accepted by above DFA can also be written as :

$L = \{\omega : \omega$ has even number of a's and even number of b's$\}$

OR

$L = \{\omega :$ Both $N_a(\omega)$ and $N_b(\omega)$ are divisible by 2$\}$

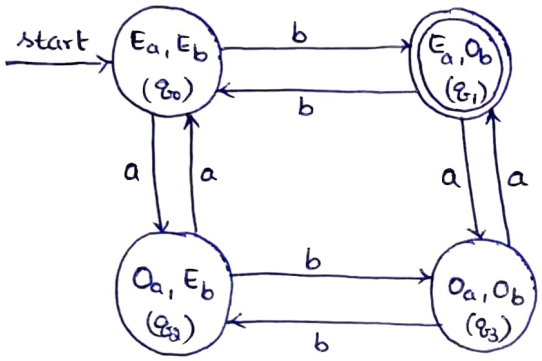
OR

$L = \{\omega :$ Both $N_a(\omega)$ and $N_b(\omega)$ are multiples of 2$\}$

OR

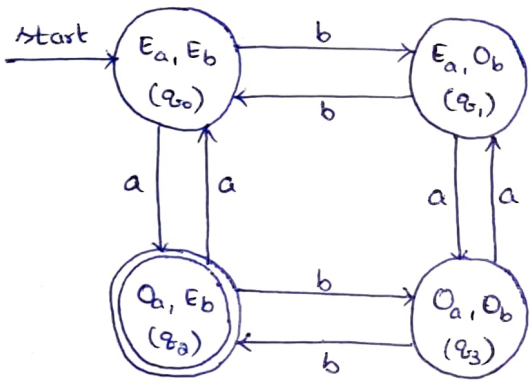
$L = \{\omega \mid N_a(\omega) \bmod 2 = 0$ and $N_b(\omega) \bmod 2 = 0\}$

The DFA to accept even number of a's and odd number of b's



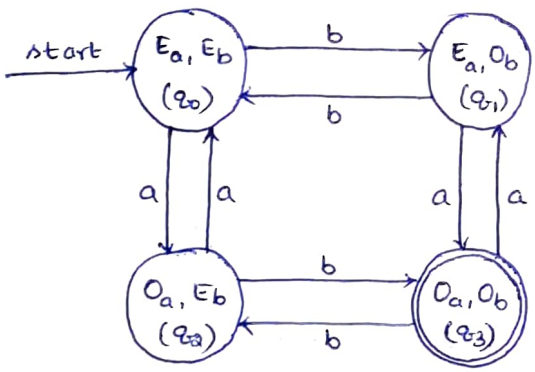
δ	a	b
$\rightarrow q_0$	q_2	q_1
$*q_1$	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

Dec 12 The DFA to accept odd number of a's and even number of b's



δ	a	b
$\rightarrow q_0$	q_2	q_1
q_1	q_3	q_0
$*q_2$	q_0	q_3
q_3	q_1	q_2

The DFA to accept odd numbers of a's and odd number of b's



δ	a	b
$\rightarrow q_0$	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
$*q_3$	q_1	q_2

Q9. Obtain a DFA to accept strings of a's and b's such that

$L = \{ w \mid w \in (a+b)^* \text{ such that } N_a(w) \bmod 3 = 0 \text{ and } N_b(w) \bmod 2 = 0 \}$

The possible remainders for $N_a(w) \bmod 3 = 0$ are

$\{0, 1, 2\}$

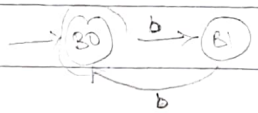
$\downarrow \quad \downarrow \quad \downarrow$

$\{A_0, A_1, A_2\} = Q1$

The possible remainders for $N_b(\omega) \bmod 2 = 0$ are

$\{0, 1\}$

$\{B0, B1\} = Q2$



$Q1 \times Q2 = \{(A0, B0), (A0, B1), (A1, B0), (A1, B1), (A2, B0), (A2, B1)\}$

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

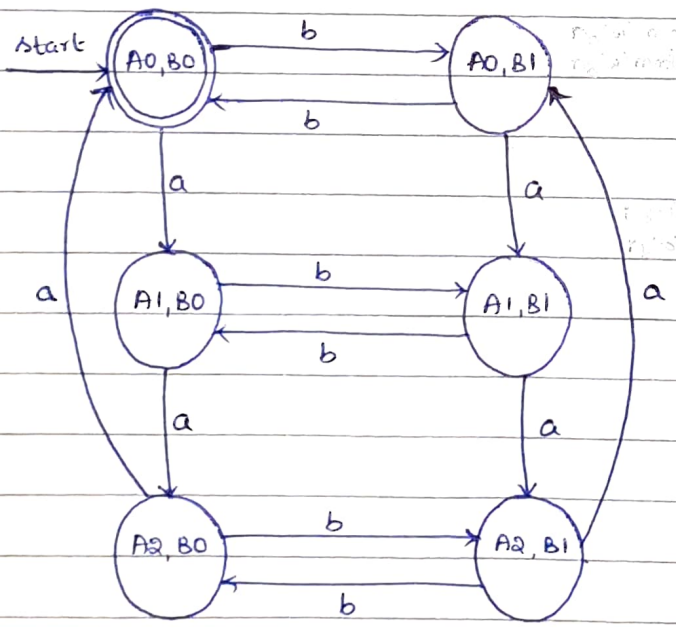
$Q = \{(A0,B0), (A0,B1), (A1,B0), (A1,B1), (A2,B0), (A2,B1)\}$

$\Sigma = \{a, b\}$

$q_0 = (A0, B0)$ is the start state

$F = \{(A0, B0)\}$

Transition diagram



Transition table

δ	a	b
→*(A0,B0)	(A1,B0)	(A0,B1)
(A0,B1)	(A1,B1)	(A0,B0)
(A1,B0)	(A2,B0)	(A1,B1)
(A1,B1)	(A2,B1)	(A1,B0)
(A2,B0)	(A0,B0)	(A2,B1)
(A2,B1)	(A0,B1)	(A2,B0)

NOTE : 1) To accept the language

$L = \{\omega \mid \omega \in (a+b)^*\ N_a(\omega) \bmod 3 = 1 \text{ and } N_b(\omega) \bmod 2 = 0\}$

make (A1,B0) as the final state.

2) To accept the language

$L = \{\omega \mid \omega \in (a+b)^*\ N_a(\omega) \bmod 3 = 2 \text{ and } N_b(\omega) \bmod 2 = 0\}$

make (A2, B0) as the final state.

3) To accept the language $L = \{ w \mid w \in (a+b)^* \; N_a(w) \bmod 3 = 2 \text{ and } N_b(w) \bmod 2 = 1 \}$

make (A2, B1) as the final state and so on.

Q10. Draw a DFA to accept the language

$L = \{ w : w$ has odd number of 1's and followed by even number of 0's$\}$

Completely define DFA and transition function.

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

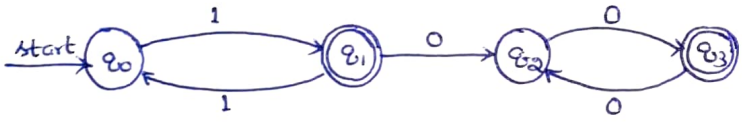
$Q = \{ q_0, q_1, q_2, q_3 \}$

$\Sigma = \{0, 1\}$

q_0 is the start state

$F = \{ q_1, q_3 \}$

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	-	q_1
$*q_1$	q_2	q_0
q_2	q_3	-
$*q_3$	q_2	-

Q11. Obtain a DFA to accept strings of a's and b's such that the number of a's is divisible by 5 and number of b's is divisible by 3.

The possible remainders for $N_a(w) \bmod 5 = 0$ are

$\{ 0, 1, 2, 3, 4 \}$

$\{ A0 \; A1 \; A2 \; A3 \; A4 \} = Q1$

The possible remainders for $N_b(w) \bmod 3 = 0$ are

$\{ 0, 1, 2 \}$

$\{ B0, B1, B2 \} = Q2$

$Q1 \times Q2 = \{ (A0, B0), (A0, B1), (A0, B2),$
$(A1, B0), (A1, B1), (A1, B2),$
$(A2, B0), (A2, B1), (A2, B2),$
$(A3, B0), (A3, B1), (A3, B2),$
$(A4, B0), (A4, B1), (A4, B2) \}$

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

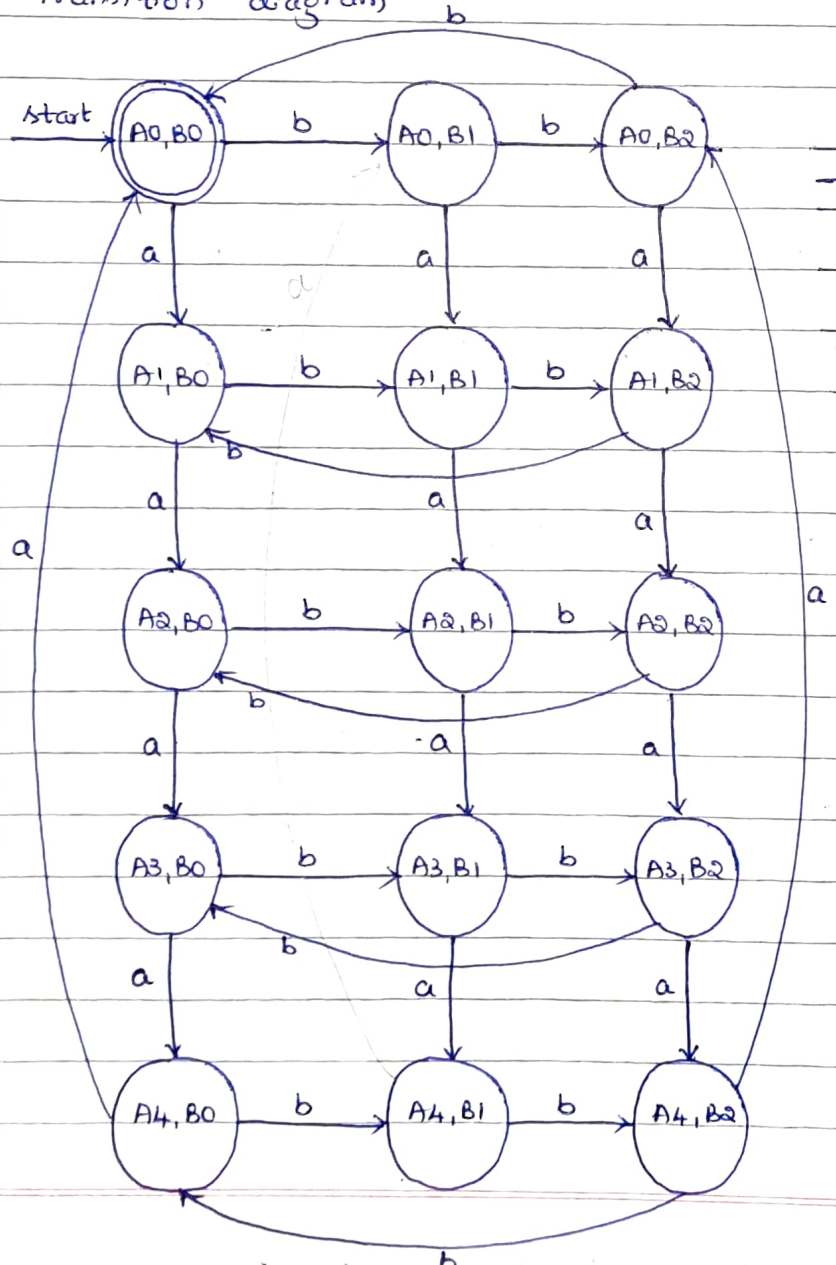
$Q = \{ (A0,B0), (A0,B1), (A0,B2),$
$(A1,B0), (A1,B1), (A1,B2),$
$(A2,B0), (A2,B1), (A2,B3),$
$(A3,B0), (A3,,B1), (A3,B2),$
$(A4,B0), (A4,B1), (A4,B2) \}$

$\Sigma = \{a, b\}$

$q_0 = (A0,B0)$ is the start state

$F = \{(A0,B0)\}$

Transition diagram



Transition table

δ	a	b
→*(A0,B0)	(A1,B0)	(A0,B1)
(A0,B1)	(A1,B1)	(A0,B2)
(A0,B2)	(A1,B2)	(A0,B0)
(A1,B0)	(A2,B0)	(A1,B1)
(A1,B1)	(A2,B1)	(A1,B2)
(A1,B2)	(A2,B2)	(A1,B0)
(A2,B0)	(A3,B0)	(A2,B1)
(A2,B1)	(A3,B1)	(A2,B2)
(A2,B2)	(A3,B2)	(A2,B0)
(A3,B0)	(A4,B0)	(A3,B1)
(A3,B1)	(A4,B1)	(A3,B2)
(A3,B2)	(A4,B2)	(A3,B0)
(A4,B0)	(A0,B0)	(A4,B1)
(A4,B1)	–	(A4,B2)
(A4,B2)	(A0,B2)	(A4,B0)

NOTE : i) To accept the language

$L = \{\omega : n_a(\omega) \bmod 5 \neq n_b(\omega) \bmod 3\}$

except (A0, B0), (A1, B1) and (A2, B2) make all other states as final states.

ii) To accept the language

$L = \{\omega : n_a(\omega) \bmod 5 > n_b(\omega) \bmod 3\}$

except (A0, B0), (A0, B1), (A0, B2), (A1, B1), (A1, B2) and (A2, B2) make all other states as final states.

June 11
June 13

Q12. Obtain a DFA to accept strings of a's and b's such that each blocks of 5 consecutive symbols have at least two a's.

The DFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

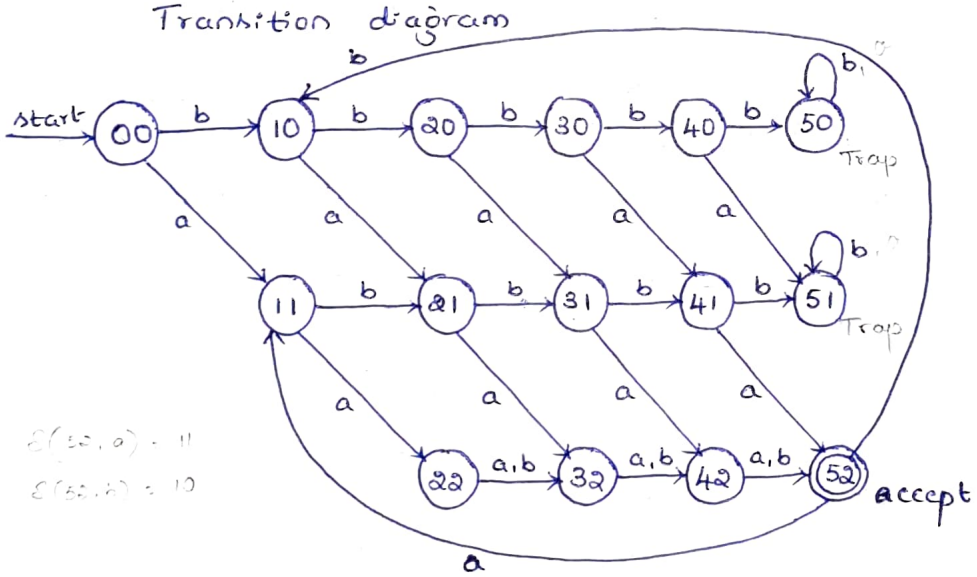
$Q = \{00, 10, 20, 30, 40, 50, 11, 21, 31, 41, 51, 22, 32, 42, 52\}$ (15 states)

$\Sigma = \{a, b\}$

$q_0 = \{(00)\}$ is the start state

$F = \{(52)\}$

Transition diagram



Transition table

δ	a	b
→00	11	10
10	21	20
20	31	30
30	41	40
40	51	50
50	–	50
11	22	21
21	32	31
31	42	41
41	52	51
51	–	51
22	32	32
32	42	42
42	52	52
*52	11	10

i → No of symbols scanned so far

j → No of a's scanned so far

(i, j) => (21) → No of symbols, No of a's

June 11) Use extended transition of function to trace a string $|x| = "aabba"$

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Q13. Prove that $\delta^*(q, xa) = \delta(\delta^*(q, x), a)$ where x is a string and a is the current input symbol.

- It is required to prove that

$$\delta^*(q, xa) = \delta(\delta^*(q, x), a)$$

- From the above statement $w = xa$

where a is the last symbol of w

x is the remaining string of w

q is the current state of the machine.

- From state q on input string w let as assume that the machine enters into state p.

$$\delta^*(q, w) = p \quad \rightarrow (1)$$

Since $w = xa$, $\delta^*(q, xa) = p \quad \rightarrow (2)$

Since $w = xa$, we should find the transition on string x and then on symbol a.

- From state q on input string x, let the machine enters into state r. $\delta^*(q, x) = r \quad \rightarrow (3)$
- From state r on input symbol a, the machine should enter into state p. $\delta(r, a) = p \quad \rightarrow (4)$
- Replacing r, $\delta(\delta^*(q, x), a) = p \rightarrow (5)$
- Comparing equations (2) and (5), R.H.S 's are equal.

Therefore L.H.S's are also equal.

So, $\delta^*(q, w) = \delta^*(q, xa) = \delta(\delta^*(q, x), a)$

Q14. Show that $\delta^*(q, xy) = \delta^*(\delta^*(q, x), y)$ for any state w and strings x and y.

The statement to be proved is

$$\delta^*(q, xy) = \delta^*(\delta^*(q, x), y) \quad \rightarrow (1)$$

By induction on the length of y.

Basis : Let $\delta(q, x) = p \rightarrow (2)$

If $y = \epsilon$ L.H.S. on eqn (1) $= \delta^*(q, xy)$

$= \delta^*(q, x) = \delta(\delta^*(q, \epsilon), x)$

$= p = \delta(q, x) = p$

$\delta(q, x) = p$

$\delta^*(q, x) = p$

R.H.S. on eqn (1) $= \delta^*(\delta^*(q, x), y)$

$= \delta^*(p, y)$

$= \delta^*(p, \epsilon)$ $[\because \delta(q, \epsilon) = q]$

$= p$

Since, L.H.S. is same as R.H.S. the relation (1) holds good.

Induction : Let split the string y into za

$y = za \rightarrow (3)$

where a is the last symbol of y

R.H.S. of eqn (1) $= \delta^*(\delta^*(q, x), y)$

$= \delta^*(\delta^*(q, x), za)$

$= \delta(\delta^*(\delta^*(q, x), z), a)$

$= \delta(\delta^*(q, xz), a)$

$= \delta^*(q, xza)$

$= \delta^*(q, xy)$

$=$ L.H.S. of eqn (1)

Hence, the proof.

Q15. Consider the DFA with the following transition table

δ	0	1
$\rightarrow A$	A	B
$*B$	B	A

Informally describe the language accepted by this DFA and prove by induction on the length of an input string, that your description is correct.

Name of Experiment..... Date.....

Experiment No..... Experiment Result.....

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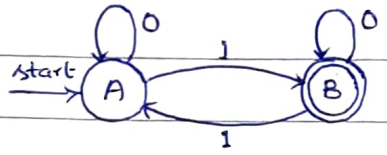
Transition diagram



Informal description of language

The language accepted by the machine is "Strings of 0's and 1's with odd number of 1's".

Prog : It is required to show by induction that $\delta^*(A,\omega) = A$ if and only if ω has even number of 1's. $\delta^*(A,\omega) = B$ iff ω has odd no of 1's

Basis : $|\omega| = 0$

Since ω is an empty string, it has an even number of 1's.

So, $\delta^*(A,\omega) = A$

Induction : let $\omega = za$ where a is either 0 or 1.

case 1 : let $a = 0$

If ω has an odd number of 1's, z also has odd number of 1's.

By inductive hypothesis, $\delta^*(A,z) = B$

Transition of DFA is $\delta^*(A,\omega) = B$

If ω has an even number of 1's, then z also has even number of 1's.

By inductive hypothesis, $\delta^*(A,z) = A$

Transition of DFA is $\delta^*(A,\omega) = A$

Thus,

$\delta^*(A,\omega) = A$ iff ω has an even number of 1's.

$\delta^*(A,\omega) = B$ iff ω has an odd number of 1's.

case 2 : let $a = 1$

if w has an even number of 1's, then 3 has an odd number of 1's.

By inductive hypothesis, $\delta^*(A,3) = B$.

Transition of DFA is $\delta^*(A,w) = A$

if w has an odd number of 1's, then 3 has an even number of 1's.

By inductive hypothesis, $\delta^*(A,3) = A$

Transition of DFA is $\delta^*(A,w) = B$

Thus,

$\delta^*(A,w) = A$ iff w has an even number of 1's.

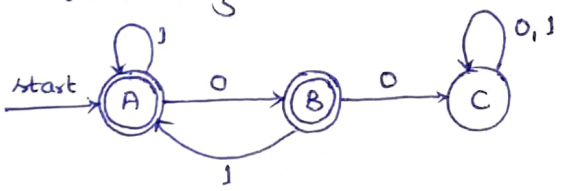
$\delta^*(A,w) = B$ iff w has an odd number of 1's.

Q16. Consider the DFA with the following transition table

δ	0	1
$\rightarrow$*A	B	A
*B	C	A
C	C	C

Informally describe the language accepted by this DFA and prove by induction on the length of an input string, that your description is correct.

Transition diagram



Informal description of the language

The language accepted by DFA is "Strings of 0's and 1's having ε or ending with 1 or ending with 0 but does not have a substring 00".

Proof :

Basis : If $w = \varepsilon$, the machine will be in state A which is the final state. Thus, empty string is accepted by DFA.

Induction : If w has only 1's the machine will be in state A. So, w does not contain the substring 00 and hence the string is accepted by DFA.

N.T.C.S.E. Bangalore

PROBLEMS SOLVED

I	1.	at least one 'a' $\{a\}$
	2.	at least one 'a' $\{a,b\}$
	3.	Exactly one 'a' $\{a,b\}$
	4.	starting with the string ab / 01
	5.	ending with the string abb / 011
	6.	donot end " " " abb
	7.	substring aab / 001
	8.	except substring aab
	9.	three consecutive zero's
	10.	awa $\{a,b\}$
	11.	ending with ab & ba
	12.	four a's
	13.	beginning with a 0 followed by odd numbers of 1's & ending with a 0
	14.	atmost two consecutive b's
	15.	starting with atleast two 0's & ending with at least two 1's $\{a,b\}$
	16.	wbab
	17.	not more than three a's
	18.	begins / ends / substring 01 / ab
	19.	$n_a(w) \geq 1$, $n_b(w) = 2$
	20.	$n_a(w) = 2$, $n_b(w) >= 3$
II	1.	Binary no's - zero modulo five
	2.	beginning with a 1 & multiple of five
	3.	Reverse, divisible by 5
	4.	decimal strings divisible by 3
III	1.	even no of a's & odd no of a's
	2.	$\|w\| \bmod 3 = 0$
	3.	$\|w\| \bmod 3 \geq \|w\| \bmod 2$, $\|w\| \bmod 3 \neq \|w\| \bmod 2$
	4.	$\|w\| \bmod 5 \neq 0$

5. even no of a's & even no of b's

6. $n_a(\omega) \bmod 3 = 0$ and $n_b(\omega) \bmod 2 = 0$

7. odd no of 1's followed by even no of 0's

8. $n_a(\omega) \bmod 5 = 0$ & $n_b(\omega) \bmod 3 = 0$

9. each blocks of 5 consecutive symbols have atleast two a's.

10. $\delta^*(q, xa) = \delta(\delta^*(q, x), a)$

11. $\delta^*(q, xy) = \delta^*(\delta^*(q, x), y)$

12.

δ	0	1
$\rightarrow A$	A	B
$*B$	B	A

13.

δ	0	1
$\rightarrow *A$	B	A
$*B$	C	A
C	C	C

Exercises

1. Draw a DFA to accept string of 0's and 1's ending with the string 011.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

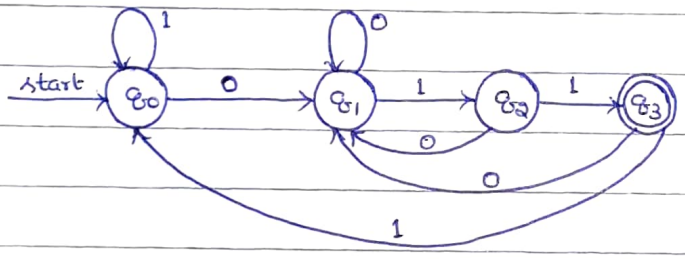
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_3\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_1	q_2
q_2	q_1	q_3
$*q_3$	q_1	q_0

2. Obtain a DFA to accept strings of a's and b's having sub-string aa.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

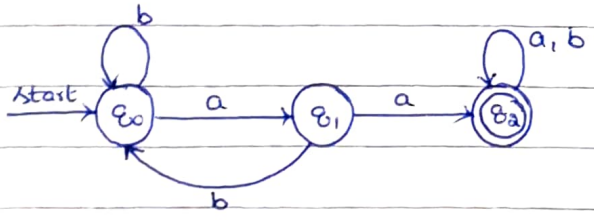
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$q_0 \in Q$ is the start state

$F = \{q_2\}$ is the final state

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
$*q_2$	q_2	q_2

3. Obtain a DFA to accept binary odd numbers.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

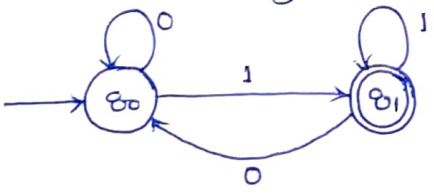
$Q = \{q_0, q_1\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_1\}$ is the final state

Transition diagram

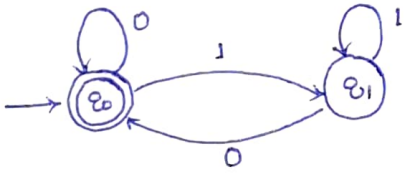


Transition table

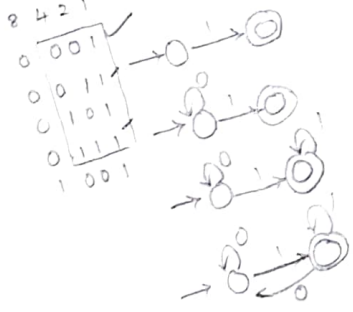
δ	0	1
$\rightarrow q_0$	q_0	q_1
$*q_1$	q_0	q_1

16	8	4	2	1	
0	0	0	0	1	1
0	0	0	1	1	3
0	0	1	0	1	5
0	0	1	1	1	7
0	1	0	0	1	9
0	1	0	1	1	11
0	1	1	0	1	13
0	1	1	1	1	15
1	0	0	0	1	17
		⋮			

NOTE : DFA to accept binary even numbers



δ	0	1
$\rightarrow *q_0$	q_0	q_1
q_1	q_0	q_1



4. Obtain a DFA to accept strings of a's and b's starting with at least two a's and ending with at least two b's.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

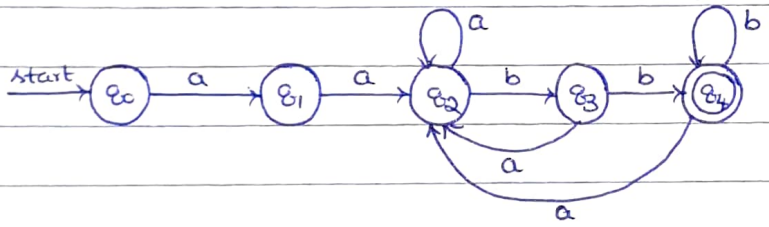
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{a, b\}$

$q_0 \in Q$ is the start state

$F = \{q_4\}$ is the final state

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	-
q_1	q_2	-
q_2	q_2	q_3
q_3	q_2	q_4
$*q_4$	q_2	q_4

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5. Design a DFA to accept the following language over the alphabet $\{0,1\}$ (i) $L = \{(01)^i 1^{2j} \mid i \geq 1, j \geq 1\}$

(ii) The set of strings either start with 01 or end with 01.

(i) The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

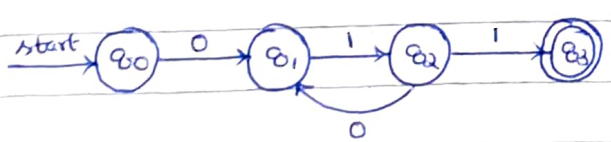
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_3\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	-
q_1	-	q_2
q_2	q_1	q_3
$*q_3$	-	-

(ii) The set of strings either start with 01 or end with 01

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

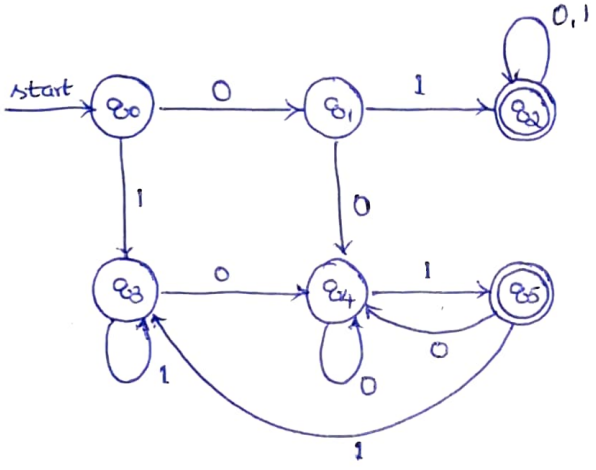
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_2, q_5\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	q_3
q_1	q_4	q_2
$*q_2$	q_2	q_2
q_3	q_4	q_3
q_4	q_4	q_5
$*q_5$	q_4	q_3

6. Give DFA's accepting the following strings over the alphabet {0,1}

(a) The set of all strings beginning with 101

(b) The set of all strings containing 1101 as a substring

(c) The set of all strings not containing 110

(d) The set of all strings that begin with 01 and end with 11.

(a) The set of all strings beginning with 101

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

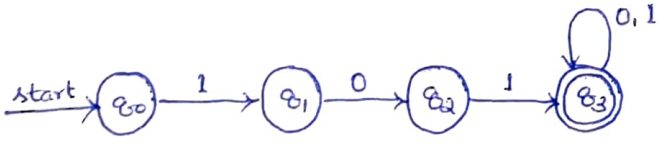
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_3\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	–	q_1
q_1	q_2	–
q_2	–	q_3
$*q_3$	q_3	q_3

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(b) The set of all strings containing 1101 as a substring

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

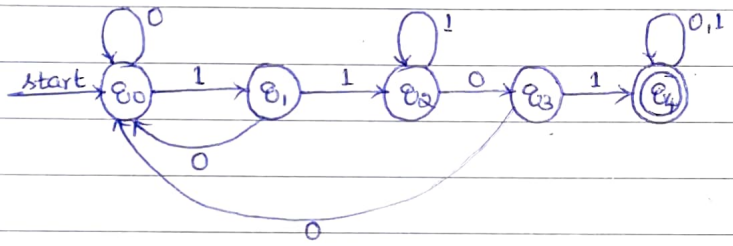
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_4\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_0	q_1
q_1	q_0	q_2
q_2	q_3	q_2
q_3	q_0	q_4
$*q_4$	q_4	q_4

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(c) The set of all strings not containing 110

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

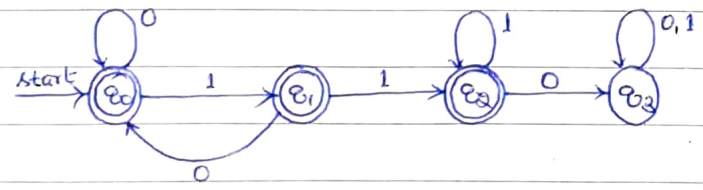
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_0, q_1, q_2\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow *q_0$	q_0	q_1
$*q_1$	q_0	q_2
$*q_2$	q_3	q_2
q_3	q_3	q_3

(d) The set of all strings that begin with 01 and end with 11.

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

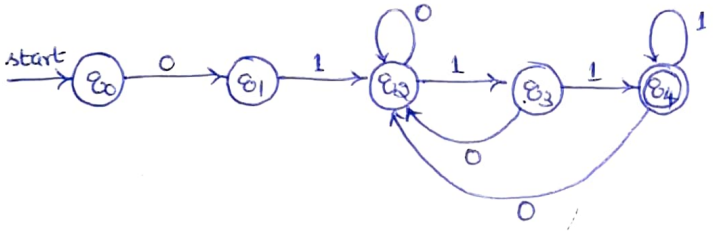
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_4\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	–
q_1	–	q_2
q_2	q_2	q_3
q_3	q_2	q_4
$*q_4$	q_2	q_4

7. Consider the DFA's with the following transition tables,

(a)

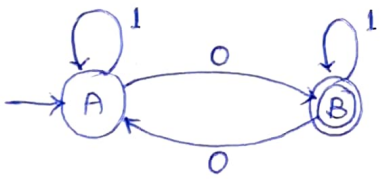
δ	0	1
$\rightarrow A$	B	A
$*B$	A	B

(b)

δ	0	1
$\rightarrow A$	B	A
B	C	A
$*C$	C	C

Informally describe the language accepted by these DFA's and prove by induction on the length of an input string that your description is correct.

(a)



Informal description of the language

The language accepted by DFA is "Strings of 0's and 1's having odd number of 0's".

Proof:

Basis: $|w| = 0$ Since w is an empty string, surely has an even number of 0's, namely zero 0's and so $\delta^*(A, w) = A$.

Induction : Let $w = za$ where a is either 0 or 1

case 1 : let $a = 1$

If w has an odd number of 0's, z also has odd number of 0's.

By inductive hypothesis, $\delta^*(A, z) = B$

Transition of DFA is, $\delta^*(A, w) = B$

If w has an even number of 0's, then z also has even number of 0's.

By inductive hypothesis $\delta^*(A, z) = A$

Transition of DFA is $\delta^*(A, w) = A$

Thus,

$\delta^*(A, w) = A$ iff w has an even number of 0's

$\delta^*(A, w) = B$ iff w has an odd number of 0's

case 2 : let $a = 0$

If w has an even number of 0's, then z has an odd number of 0's.

By inductive hypothesis, $\delta^*(A, z) = B$

Transition of DFA is $\delta^*(A, w) = A$

If w has an odd number of 0's, then z has an even number of 0's.

By inductive hypothesis, $\delta^*(A, z) = A$

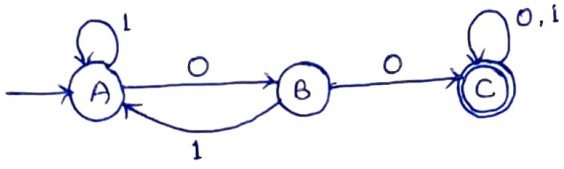
Transition of DFA is $\delta^*(A, w) = B$

Thus,

$\delta^*(A, w) = A$ iff w has an even number of 0's.

$\delta^*(A, w) = B$ iff w has an odd number of 0's

(b)



Informal description of the Language

The language accepted by DFA is "Strings of 0's and 1's having a substring 00".

Proof :

Basis : If $w = \varepsilon$, the machine will be in state A which is not the final state. Thus, empty string is not accepted by DFA. Since it doesnot have substring 00.

Induction : If w has only 1's the machine will be in state A. So, w does not contain the substring 00 and hence the string is not accepted by DFA.

$w = 10$ If w ends with $\overset{\text{one}}{\wedge}0$, the machine goes to state B and hence string ends with 0. But on 1 the machine enters into state A and thus the string is not accepted since it doesnot have substring 00.

If w is 100, then the machine goes to state C which is the final state. Thus the string 100 is accepted by the machine|DFA. Since string has substring 00.

Dec08 8. Design the DFA for the following languages on $\Sigma = \{a, b\}$

$L = \{ w : |w| \bmod 5 <> 0 \}$

The DFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

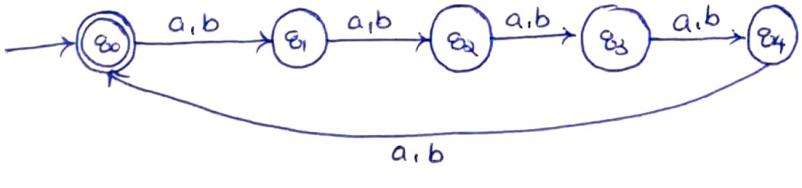
$Q = \{ q_0, q_1, q_2, q_3, q_4 \}$

$\Sigma = \{a, b\}$

$q_0 \in Q$ is the start state

$F = \{q_0\}$ is the final state

Transition diagram



Transition table

δ	a	b
$\rightarrow *q_0$	q_1	q_1
q_1	q_2	q_2
q_2	q_3	q_3
q_3	q_4	q_4
q_4	q_0	q_0

June 14 9. Write DFA for the following

(i) Set of all strings ending with a & b

The DFA is defined as $M = (Q, \Sigma, S, q_0, F)$

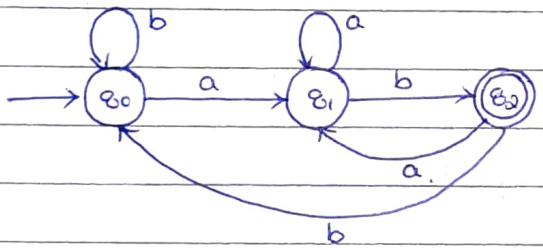
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$q_0 \in Q$ is the start state

$F = \{q_2\}$ is the final state

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_1	q_2
$*q_2$	q_1	q_0

June 12
June 14 (ii) Set of all strings with exactly three consecutive 0's.

The DFA is defined as $M = (Q, \Sigma, S, q_0, F)$

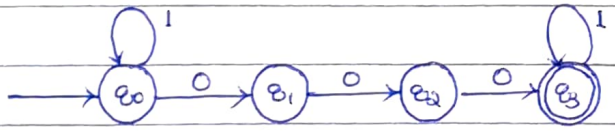
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_3\}$ is the final state

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	-
q_2	q_3	-
$*q_3$	-	q_3

10. The set of all strings such that the number of 1's is even and the number of 0's is multiple of 3.

The possible remainders for $n_1(\omega) \bmod 2 = 0$ are $\{0,1\}$ $\quad Q1 = \{A_0, A_1\}$

The possible remainders for $n_0(\omega) \bmod 3 = 0$ are $\{0,1,2\}$ $\quad Q2 = \{B_0, B_1, B_2\}$

$Q1 \times Q2 = \{(A_0,B_0), (A_0,B_1), (A_0,B_2), (A_1,B_0), (A_1,B_1), (A_1,B_2)\}$

The DFA can be defined as $M = (Q, \Sigma, S, q_0, F)$

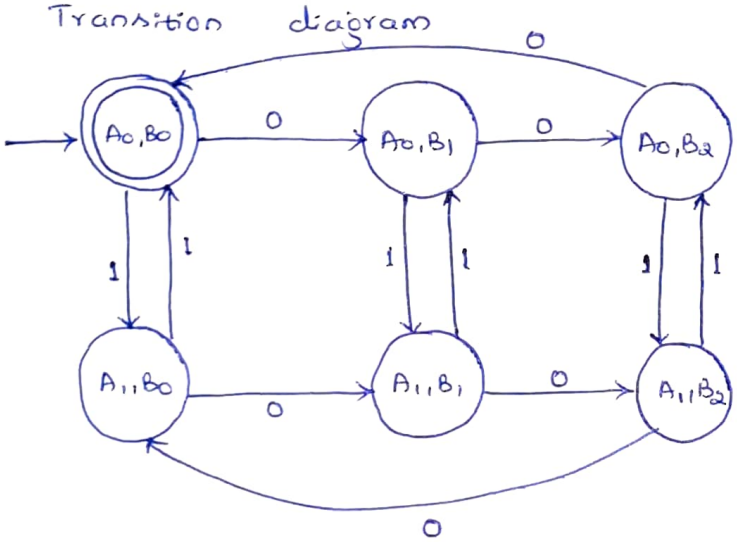
$Q = \{(A_0,B_0), (A_0,B_1), (A_0,B_2), (A_1,B_0), (A_1,B_1), (A_1,B_2)\}$

$\Sigma = \{0,1\}$

$q_0 = \{(A_0,B_0)\} \in Q$ is the start state

$F = \{(A_0,B_0)\}$ is the final state

Transition diagram



Transition table

δ	0	1
→*(A0,B0)	(A0,B1)	(A1,B0)
(A0,B1)	(A0,B2)	(A1,B1)
(A0,B2)	(A0,B0)	(A1,B2)
(A1,B0)	(A1,B1)	(A0,B0)
(A1,B1)	(A1,B2)	(A0,B1)
(A1,B2)	(A1,B0)	(A0,B2)

11. The set of all strings which when interpreted as a binary integer is a multiple of 3.

STEP 1 : $r = 2$ Radix binary 2

$\Sigma = \{0,1\} = d$ digit — 0,1

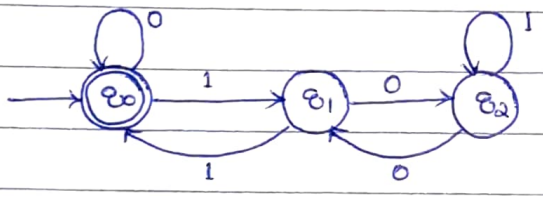
$k = 3$ k divisor

STEP 2 : $i = 0,1,2$ Remainder

STEP 3 : $\delta(q_i, a) = q_j$ where $j = (r * i + d) \bmod k$

remainder	d	$(2 * i + d) \bmod 3 = j$	$\delta(q_i, d) = q_j$
$i = 0$	0	$(2*0+0) \bmod 3 = 0$	$\delta(q_0, 0) = q_0$
	1	$(2*0+1) \bmod 3 = 1$	$\delta(q_0, 1) = q_1$
$i = 1$	0	$(2*1+0) \bmod 3 = 2$	$\delta(q_1, 0) = q_2$
	1	$(2*1+1) \bmod 3 = 0$	$\delta(q_1, 1) = q_0$
$i = 2$	0	$(2*2+0) \bmod 3 = 1$	$\delta(q_2, 0) = q_1$
	1	$(2*2+1) \bmod 3 = 2$	$\delta(q_2, 1) = q_2$

Transition diagram



Transition table

δ	0	1
→* q_0	q_0	q_1
q_1	q_2	q_0
q_2	q_1	q_2

8	4	2	1
0	0	0	0
0	0	1	1
0	1	1	0
1	0	0	1
1	1	0	0
1	1	1	1

If w ends with 0, the machine goes to state B and hence string ends with 0. But on 1 the machine enters into state A and thus the substring 00 is not accepted.

Dec 11
Dec 12

Applications of finite Automata

* Design of digital circuits

The FA is used during designing and checking the behavior of the digital circuits using software. It will be useful in hardware design such as circuit verification, in design of automatic traffic signals and so on.

* Compiler construction

Used in the design of lexical analyzer, which breaks the input text into various units such as identifiers, keywords, punctuation and so on.

* String matching

In designing a software for identifying the words, phrases and other patterns in large bodies of text.

Example : collection of web pages

* String processing

To write software for processing the natural language. Large natural vocabularies can be described which includes the applications such as spelling checkers and advisers, multi-language dictionaries, indenting the documents etc.

* Software design

In building the software to verify the systems having finite number of states.

Example : Communication protocols in computer networks

* Other applications
  - In Artificial Intelligence and knowledge engineering
  - In game theory and games
  - Computer graphics
  - linguistics etc.,

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Disadvantages of DFA

1. * Constructing a DFA is difficult
2. * DFA can not guess about its input
3. * DFA is not very powerful
4. * At any point of time, the DFA is in only one state. So, a DFA does not have the power to be in several states at once.

NON-DETERMINISTIC FINITE AUTOMATA (NFA / NDFA)

Advantages of NFA

1. * Very easy to construct
2. * A "non-deterministic" finite automaton has the ability to guess something about its input.
3. * A "non-deterministic" finite automaton is more powerful than DFA.
4. * It has the power to be in several states at once.
   * An NFA is an efficient mechanism to describe some complicated languages concisely.

June 09
June 13
June 14

Definition

The Non-deterministic finite Automaton is 5-tuple or quintuple indicating five components

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

M is the name of the machine.

Q is non-empty finite set of states

Σ is non-empty finite set of input alphabets

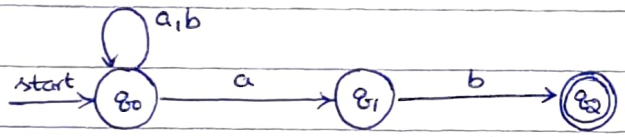
$\delta : Q \times \Sigma$ to 2^Q i.e., δ is transition function which is a mapping from $Q \times \Sigma$ to 2^Q. Based on the current state and input symbol, the machine enters into one or more states.

$q_0 \in Q$ is the start state

$F \subseteq Q$ is set of accepting or final states.

Example : NFA to accept strings of a's and b's ending with ab

is



The NFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

q_0 is the start state

$F = \{q_2\}$ is the final state

Transition table

δ	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	ϕ	$\{q_2\}$
$*q_2$	ϕ	ϕ

2^Q Power set

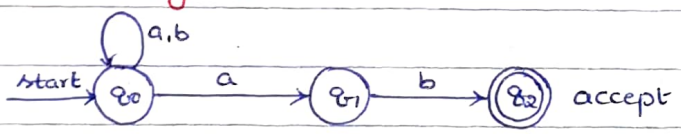
Set of subsets of set Q.

$Q = \{q_0, q_1, q_2\}$

$2^Q = \{\phi, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\} = 8$

$\delta : Q \times \Sigma \to 2^Q$

Moves made by NFA



Processing of input sequence abaab

$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2$ (accept)

- q_0 —a→ q_1 —b→ q_2 (stuck)
- q_0 —a→ q_1 (stuck)
- q_0 —a→ q_0 —b→ q_0 (Not accept)

The input string abaab is accepted by the machine.

Processing of input sequence abb

$q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2$ (stucks)

a ↓ q_0

b ↓ q_0

b ↓ q_0 (Not accept)

The input string abb is rejected by the machine.

Extended transition function of NFA

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a NFA. The extended transition function $\delta^* : Q \times \Sigma^* \to 2^Q$ is defined recursively as

Basis : $\delta^*(q_i, \varepsilon) = \{q_i\}$

This indicates that if the machine is in state q and read no input, then the machine is still in state q.

Induction : Let $\omega = xa$

where a is the last symbol of ω and x is the remaining string of ω.

Let q is the current state and x is the string to be processed and after consuming the string x, let the states of the machine is $\{p_1, p_2, p_3, \cdots, p_m\}$

i.e., $\delta^*(q_i, x) = \{p_1, p_2, p_3, \cdots, p_m\}$

Let the transition from $\{p_1, p_2, p_3, \cdots, p_m\}$ on input symbol a is

$$\delta(\{p_1, p_2, p_3, \cdots, p_m\}, a) = \{r_1, r_2, r_3, \cdots, r_k\}$$

i.e., $$\bigcup_{i=1}^{m} \delta(p_i, a) = \{r_1, r_2, r_3, \cdots, r_k\}$$

Then,

$$\delta^*(q_i, \omega) = \delta^*(q_i, xa) = \{r_1, r_2, r_3, \cdots, r_k\}$$

where

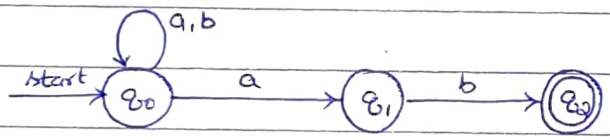
δ^* is a extended transition function

q is the first parameter representing the current state of the machine.

w is the second parameter representing the current input string being read.

$\{r_1, r_2, r_3, \ldots, r_k\}$ represent the possible states of the machine which is returned by the transition function.

Example: Processing the string abaab using the extended transition function



Taking the prefixes of abaab,

for prefix ε : $\delta^*(q_0, \varepsilon) = \{q_0\}$

for prefix a : $\delta^*(q_0, a) = \delta(\delta^*(q_0, \varepsilon), a)$

$= \delta(q_0, a)$

$= \{q_0, q_1\}$

for prefix ab : $\delta^*(q_0, ab) = \delta(\delta^*(q_0, a), b)$

$= \delta(\{q_0, q_1\}, b)$

$= \delta(q_0, b) \cup \delta(q_1, b)$

$= \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

for prefix aba : $\delta^*(q_0, aba) = \delta(\delta^*(q_0, ab), a)$

$= \delta(\{q_0, q_2\}, a)$

$= \delta(q_0, a) \cup \delta(q_2, a)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

for prefix abaa : $\delta^*(q_0, abaa) = \delta(\delta^*(q_0, aba), a)$

$= \delta(\{q_0, q_1\}, a)$

$= \delta(q_0, a) \cup \delta(q_1, a)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

for prefix abaab : $\delta^*(q_0, abaab) = \delta(\delta^*(q_0, abaa), b)$

$= \delta(\{q_0, q_1\}, b)$

$= \delta(q_0, b) \cup \delta(q_1, b)$

$= \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

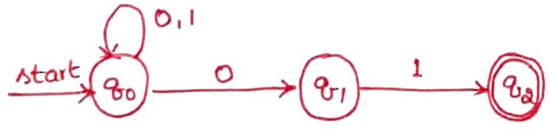
After consuming the string abaab, the states of NFA = $\{q_0, q_2\}$
Since the possible states of NFA has one final state, the string abaab is accepted by the machine.

Language accepted by a NFA

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a NFA. A string ω is accepted by the machine M, if it takes the initial state q_0 to final state i.e., $\delta^*(q_0, \omega)$ is $\in F$.
Thus, the languages accepted by NFA represented as L(M) can be formally written as

$$L(M) = \{\omega | \omega \in \Sigma^* \text{ and } \delta^*(q_0, \omega) \text{ is in } F\}$$

Q1. Prove formally that the following NFA accepts the language
$L = \{\omega : \omega \text{ ends in } 01\}$



The above NFA accepts the language $L = \{\omega : \omega \text{ ends in } 01\}$
The statement can be proved by mutual induction on the length of string ω accepted by the three states in three cases as :

1) $\delta^*(q_0, \omega) = q_0$ for every ω
2) $\delta^*(q_0, \omega) = q_1$ for every ω ending in 0
3) $\delta^*(q_0, \omega) = q_2$ for every ω ending in 01

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Basis : Let $\omega = \varepsilon$ where $|\omega| = 0$

1) $\delta(q_{00}, \varepsilon) = q_{00}$ by definition

2) $\delta(q_{00}, \varepsilon) = q_{00}$ by definition.

Here, ε do not end with 0 and hence $\delta(q_{00}, \varepsilon)$ does not contain q_1.

3) $\delta(q_{00}, \varepsilon) = q_{00}$ by definition.

Here, ε do not end with 01 and hence $\delta(q_{00}, \varepsilon)$ does not contain q_2.

Induction hypotheses : Let $\omega = xa$ where a is either 0 or 1.

let $|x| = n$ so, $|\omega| = n+1$

We need to prove that the above mentioned three cases are true for $n+1$

1) $\delta^*(q_{00}, \omega) = q_{00}$

This is true because on any of the input symbols 0 or 1, the NFA may stay in q_{00}.

2) Assume ω ends in 0. i.e., $a = 0$

$\delta^*(q_{00}, \omega) = q_{00}$ is true.

So, $\delta^*(q_{00}, x) = q_{00}$

Since there is a transition from q_{00} to q_1 on 0.

We conclude that, $\delta^*(q_{00}, \omega) = q_1$

3) Assume ω ends in 01 where $a = 1$

and x is a string such that $\delta^*(q_{00}, x) = q_1$ since x ends in 0.

Since there is a transition from q_1 to q_2 on 1.

We conclude that, $\delta^*(q_{00}, \omega) = q_{02}$

NOTE : $\delta(q_i, \varepsilon) = \delta^*(q_i, \varepsilon) = q_i$

$\delta(q_i, \omega a) = \delta(\delta^*(q_i, \omega), a) = P_j$

$\delta(q_i, a\omega) = \delta^*(\delta(q_i, a), \omega) = P_p$

where q_i is in Q, a is in Σ, ω is in Σ^*, P_j and P_p are the set of states which are reachable from q_i

Q2. Obtain a NFA to accept the following Language

$L = \{ w \mid w \in abab^n \text{ or } aba^n \text{ where } n \geq 0 \}$

The NFA to accept $abab^n$ or aba^n where $n \geq 0$ is $M = (Q, \Sigma, \delta, q_0, F)$

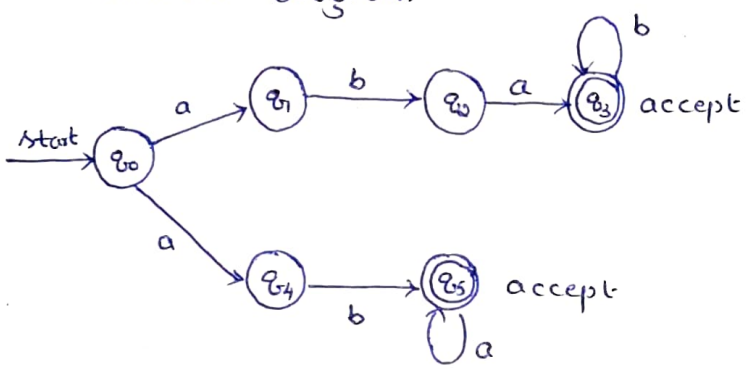
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{a, b\}$

$\{q_0\}$ is the start state

$F = \{\{q_3\}, \{q_5\}\}$

Transition diagram



Transition table

δ	a	b
$\rightarrow q_0$	$\{q_1, q_4\}$	ϕ
q_1	ϕ	$\{q_2\}$
q_2	$\{q_3\}$	ϕ
$*q_3$	ϕ	$\{q_3\}$
q_4	ϕ	$\{q_5\}$
$*q_5$	$\{q_5\}$	ϕ

Dec08 Q3. Design a NFA to recognize the following set of strings abc, abd and aacd.

The NFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

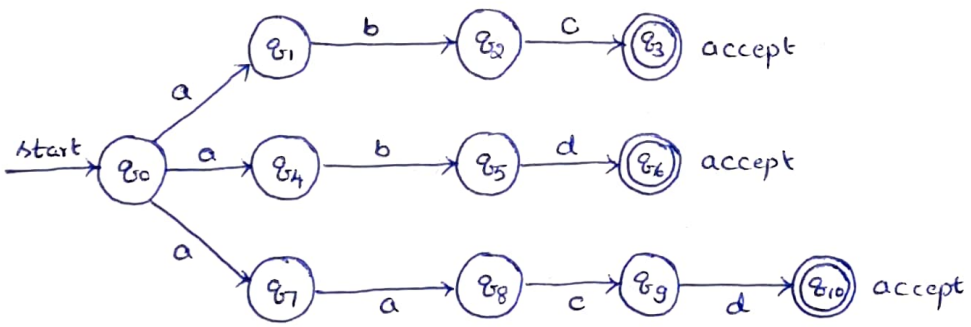
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}\}$

$\Sigma = \{a, b, c, d\}$

$\{q_0\}$ is the start state

$F = \{\{q_3\}, \{q_6\}, \{q_{10}\}\}$

Transition diagram



Transition table

δ	a	b	c	d
$\rightarrow q_0$	$\{q_1, q_4, q_7\}$	ϕ	ϕ	ϕ
q_1	ϕ	$\{q_2\}$	ϕ	ϕ
q_2	ϕ	ϕ	$\{q_3\}$	ϕ
$*q_3$	ϕ	ϕ	ϕ	ϕ
q_4	ϕ	$\{q_5\}$	ϕ	ϕ
q_5	ϕ	ϕ	ϕ	$\{q_6\}$
$*q_6$	ϕ	ϕ	ϕ	ϕ
q_7	$\{q_8\}$	ϕ	ϕ	ϕ
q_8	ϕ	ϕ	$\{q_9\}$	ϕ
q_9	ϕ	ϕ	ϕ	$\{q_{10}\}$
$*q_{10}$	ϕ	ϕ	ϕ	ϕ

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Q4. Obtain a NFA to recognize the following set of strings 0101, 101 and 011.

The NFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

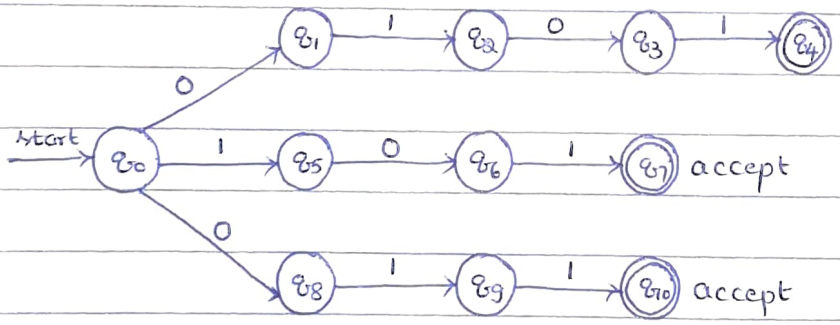
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}\}$

$\Sigma = \{0, 1\}$

$\{q_0\}$ is the start state

$F = \{\{q_4\}, \{q_7\}, \{q_{10}\}\}$

Transition diagram



Transition table

δ	0	1
$\rightarrow q_0$	$\{q_1, q_8\}$	$\{q_5\}$
q_1	ϕ	$\{q_2\}$
q_2	$\{q_3\}$	ϕ
q_3	ϕ	$\{q_4\}$
$*q_4$	ϕ	ϕ
q_5	$\{q_6\}$	ϕ
q_6	ϕ	$\{q_7\}$
$*q_7$	ϕ	ϕ
q_8	ϕ	$\{q_9\}$
q_9	ϕ	$\{q_{10}\}$
$*q_{10}$	ϕ	ϕ

Conversion from NFA to DFA

Practically, non-deterministic machines will not exist. So, we convert a NFA in to a DFA, by using two methods

1) Subset construction
2) Lazy evaluation

Prove that if $D = (Q_D, \Sigma, \delta_D, q_0, F_D)$ is the DFA constructed from NFA $N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ by subset construction then $L(D) = L(N)$.

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Given a NFA $M_N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ which accepts the language $L(M_N)$, we can find an equivalent DFA $M_D = (Q_D, \Sigma, \delta_D, q_0, F_D)$ such that $L(M_D) = L(M_N)$.

STEP 1 : Identify the start state of DFA.

Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : Identify the alphabets of DFA.

The input alphabets of DFA are the input alphabets of NFA.

STEP 3 : Identify Q_D which are the states of DFA.

The set of subsets of Q_N will be the states of DFA Q_D. So, if Q_N has n states then Q_D will have 2^n states.

STEP 4 : Identify the final states of DFA.

If $\{q_i, q_j, \ldots, q_k\}$ is a state in Q_D then $\{q_i, q_j, \ldots, q_k\}$ will be final state of DFA provided one of $q_i, q_j, \ldots, q_k$ is the final state of NFA.

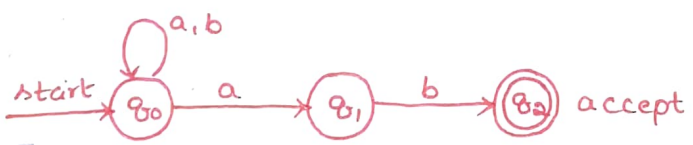
STEP 5 : Identify the transitions i.e., δ_D of DFA.

For each state $\{q_i, q_j, \ldots, q_k\}$ in Q_D and for each input symbol a in Σ, the transition can be obtained as shown below

$$\delta_D(\{q_i, q_j, \ldots, q_k\}, a) = \delta_N(q_i, a) \cup \delta_N(q_j, a) \cup \ldots \delta_N(q_k, a)$$

Thus, DFA can be obtained using subset construction method.

Q1. Obtain the DFA for the following NFA using subset construction method.



Transition table

δ	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	ϕ	$\{q_2\}$
$*q_2$	ϕ	ϕ

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{a, b\}$

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STEP 3 : $Q_N = \{q_0, q_1, q_2\}$

$Q_D = \{\phi, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$

STEP 4 : Since q_2 is the final state of NFA.

So, $F_D = \{\{q_2\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$

STEP 5 : For state ϕ :

Input symbol = a

$\delta_D(\phi, a) = \phi$

For state $\{q_0\}$:

$\delta_D(\{q_0\}, a) = \{q_0, q_1\}$

For state $\{q_1\}$:

$\delta_D(\{q_1\}, a) = \phi$

For state $\{q_2\}$:

$\delta_D(\{q_2\}, a) = \phi$

For state $\{q_0, q_1\}$:

$\delta_D(\{q_0, q_1\}, a) = \delta_N(\{q_0, q_1\}, a)$

$= \delta_N(q_0, a) \cup \delta_N(q_1, a)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

for state $\{q_0, q_2\}$:

$\delta_D(\{q_0, q_2\}, a)$

$= \delta_N(\{q_0, q_2\}, a)$

$= \delta_N(q_0, a) \cup \delta_N(q_2, a)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

Input symbol = b

$\delta_D(\phi, b) = \phi$

For state $\{q_0\}$:

$\delta_D(\{q_0\}, b) = \{q_0\}$

For state $\{q_1\}$:

$\delta_D(\{q_1\}, b) = \{q_2\}$

For state $\{q_2\}$:

$\delta_D(\{q_2\}, b) = \phi$

For state $\{q_0, q_1\}$:

$\delta_D(\{q_0, q_1\}, b) = \delta_N(\{q_0, q_1\}, b)$

$= \delta_N(q_0, b) \cup \delta_N(q_1, b)$

$= \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

for state $\{q_0, q_2\}$:

$\delta_D(\{q_0, q_2\}, b)$

$= \delta_N(\{q_0, q_2\}, b)$

$= \delta_N(q_0, b) \cup \delta_N(q_2, b)$

$= \{q_0\} \cup \phi$

$= \{q_0\}$

for state $\{q_1, q_2\}$:

$\delta_D(\{q_1,q_2\}, a) = \delta_N(\{q_1,q_2\}, a)$

$= \delta_N(q_1,a) \cup \delta_N(q_2,a)$

$= \phi \cup \phi$

$= \phi$

$\delta_D(\{q_1,q_2\}, b) = \delta_N(\{q_1,q_2\}, b)$

$= \delta_N(q_1,b) \cup \delta_N(q_2,b)$

$= \{q_2\} \cup \phi$

$= \{q_2\}$

for state $\{q_0, q_1, q_2\}$:

$\delta_D(\{q_0,q_1,q_2\}, a)$

$= \delta_N(\{q_0,q_1,q_2\}, a)$

$= \delta_N(q_0,a) \cup \delta_N(q_1,a) \cup \delta_N(q_2,a)$

$= \{q_0,q_1\} \cup \phi \cup \phi$

$= \{q_0,q_1\}$

$\delta_D(\{q_0,q_1,q_2\}, b)$

$= \delta_N(\{q_0,q_1,q_2\}, b)$

$= \delta_N(q_0,b) \cup \delta_N(q_1,b) \cup \delta_N(q_2,b)$

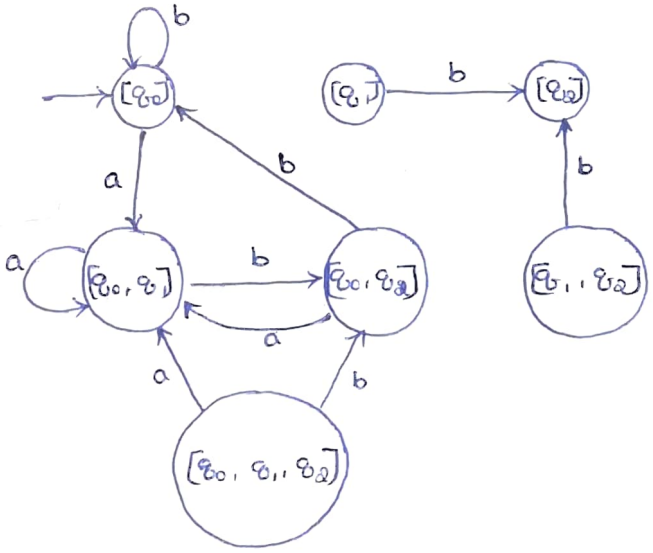
$= \{q_0\} \cup \{q_2\} \cup \phi$

$= \{q_0,q_2\}$

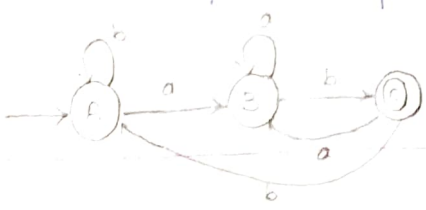
Transition table :

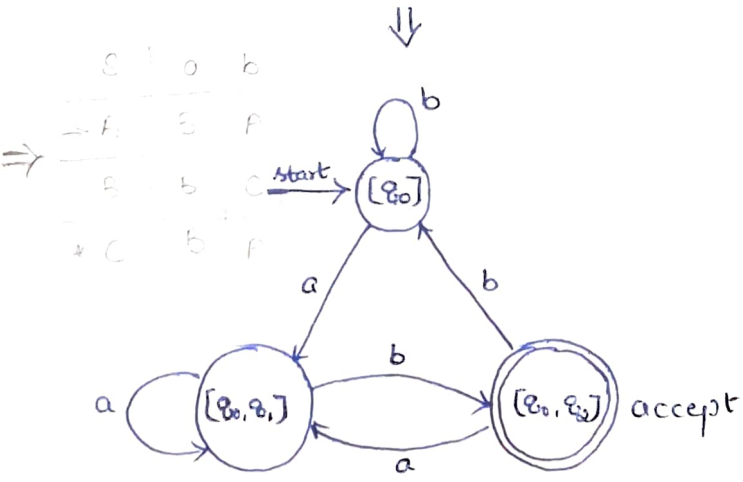
δ	a	b
ϕ	ϕ	ϕ
$\rightarrow \{q_0\}$	$\{q_0,q_1\}$	$\{q_0\}$
$\{q_1\}$	ϕ	$\{q_2\}$
$*\{q_2\}$	ϕ	ϕ
$\{q_0,q_1\}$	$\{q_0,q_1\}$	$\{q_0,q_2\}$
$*\{q_0,q_2\}$	$\{q_0,q_1\}$	$\{q_0\}$
$*\{q_1,q_2\}$	ϕ	$\{q_2\}$
$*\{q_0,q_1,q_2\}$	$\{q_0,q_1\}$	$\{q_0,q_2\}$

Transition diagram



δ	a	b
$\rightarrow \{q_0\}$	$\{q_0,q_1\}$	$\{q_0\}$
$\{q_0,q_1\}$	$\{q_0,q_1\}$	$\{q_0,q_2\}$
$*\{q_0,q_2\}$	$\{q_0,q_1\}$	$\{q_0\}$





Disadvantage of Subset Construction Method

- The time complexity to convert a NFA to DFA is $\Sigma * 2^n$. Because there are 2^n states and from each state we have the transition from input symbol in Σ.
- Since the time complexity is exponential, the procedure takes very long time to construct the table. This exponential time complexity can be avoided using another technique called "Lazy Evaluation" on the subsets.

Lazy Evaluation Method

Given a NFA $M_N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ which accepts the language $L(M_N)$, find an equivalent DFA $M_D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ such that $L(M_D) = L(M_N)$.

STEP 1 : Identify the start state of DFA.

Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : Identify the alphabets of DFA.

The input alphabets of DFA are the input alphabets of NFA.

STEP 3 : Identify the transitions (i.e., δ_D) of DFA.

For each state $\{q_i, q_j, \ldots, q_k\}$ in Q_D and for each input symbol a in Σ, the transition can be obtained as

$$\delta_D(\{q_i, q_j, \ldots, q_k\}, a) = \delta_N(q_i, a) \cup \delta_N(q_j, a) \cup \ldots \delta_N(q_k, a)$$
$$= \{q_l, q_m, \ldots q_n\}$$

Add the state $\{q_l, q_m, \ldots q_n\}$ to Q_D, if it is not already in Q_D.

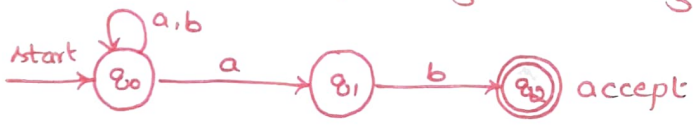
Add the transitions from $\{q_i, q_j, \ldots q_k\}$ to $\{q_l, q_m, \ldots q_n\}$ on the input symbol a.

The step3 has to be repeated for each state that is added to Q_D.

STEP 4 : Identify the final states of DFA.

If $\{q_i, q_j, \ldots q_k\}$ is a state in Q_D and if one of $q_i, q_j, \ldots, q_k$ is the final state of NFA, then $\{q_i, q_j, \ldots, q_k\}$ will be the final state of DFA.

Q1. Obtain the DFA for the following NFA using lazy evaluation method.



Transition table

δ	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	ϕ	$\{q_2\}$
$*q_2$	ϕ	ϕ

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{a, b\}$

STEP 3 : for state $\{q_0\}$

$\delta_D(\{q_0\}, a) = \{q_0, q_1\}$

$\delta_D(\{q_0\}, b) = \{q_0\}$

for state $\{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, a) = \delta_N(\{q_0, q_1\}, a)$

$= \delta_N(q_0, a) \cup \delta_N(q_1, a)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, b) = \delta_N(\{q_0, q_1\}, b)$

$= \delta_N(q_0, b) \cup \delta_N(q_1, b)$

$= \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

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For state $\{q_0, q_2\}$

$$\delta_D(\{q_0, q_2\}, a) = \delta_N(\{q_0, q_2\}, a)$$
$$= \delta_N(q_0, a) \cup \delta_N(q_2, a)$$
$$= \{q_0, q_1\} \cup \phi$$
$$= \{q_0, q_1\}$$

$$\delta_D(\{q_0, q_2\}, b) = \delta_N(\{q_0, q_2\}, b)$$
$$= \delta_N(q_0, b) \cup \delta_N(q_2, b)$$
$$= \{q_0\} \cup \phi$$
$$= \{q_0\}$$

STEP 4 : The final state is $\{q_0, q_2\}$

The DFA is given by $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{A, B, C\}$

$\Sigma = \{a, b\}$

$q_0 = A$

$F = \{C\}$

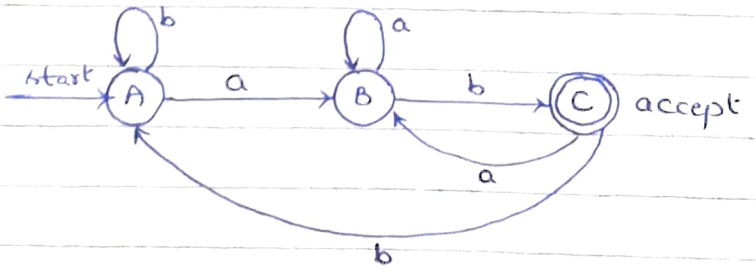
Transition table

δ	a	b
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$* \{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$

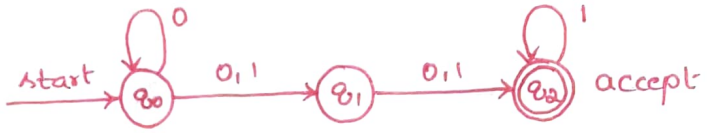
$\Rightarrow$

δ	a	b
$\rightarrow A$	B	A
B	B	C
$* C$	B	A

Transition diagram



Q2. Convert the following NFA to its equivalent DFA.



Transition table

δ	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
$*q_2$	ϕ	$\{q_2\}$

STEP 1: Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2: The input alphabets of NFA are the input alphabets of DFA.

$\Sigma = \{0, 1\}$

STEP 3: for state $\{q_0\}$

$\delta_D(\{q_0\}, 0) = \delta_N(\{q_0\}, 0) = \{q_0, q_1\}$

$\delta_D(\{q_0\}, 1) = \delta_N(\{q_0\}, 1) = \{q_1\}$

for state $\{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, 0) = \delta_N(\{q_0, q_1\}, 0)$

$= \delta_N(q_0, 0) \cup \delta_N(q_1, 0)$

$= \{q_0, q_1\} \cup \{q_2\}$

$= \{q_0, q_1, q_2\}$

$\delta_D(\{q_0, q_1\}, 1) = \delta_N(\{q_0, q_1\}, 1)$

$= \delta_N(q_0, 1) \cup \delta_N(q_1, 1)$

$= \{q_1\} \cup \{q_2\}$

$= \{q_1, q_2\}$

for state $\{q_1\}$

$\delta_D(\{q_1\}, 0) = \delta_N(\{q_1\}, 0) = \{q_2\}$

$\delta_D(\{q_1\}, 1) = \delta_N(\{q_1\}, 1) = \{q_2\}$

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$\delta_D(\{q_0,q_1,q_2\},0) = \delta_N(\{q_0,q_1,q_2\},0)$

$= \delta_N(q_0,0) \cup \delta_N(q_1,0) \cup \delta_N(q_2,0)$

$= \{q_0,q_1\} \cup \{q_2\} \cup \phi$

$= \{q_0,q_1,q_2\}$

$\delta_D(\{q_0,q_1,q_2\},1) = \delta_N(\{q_0,q_1,q_2\},1)$

$= \delta_N(q_0,1) \cup \delta_N(q_1,1) \cup \delta_N(q_2,1)$

$= \{q_1\} \cup \{q_2\} \cup \{q_2\}$

$= \{q_1,q_2\}$

for state $\{q_1,q_2\}$

$\delta_D(\{q_1,q_2\},0) = \delta_N(\{q_1,q_2\},0)$

$= \delta_N(q_1,0) \cup \delta_N(q_2,0)$

$= \{q_2\} \cup \phi$

$= \{q_2\}$

$\delta_D(\{q_1,q_2\},1) = \delta_N(\{q_1,q_2\},1)$

$= \delta_N(q_1,1) \cup \delta_N(q_2,1)$

$= \{q_2\} \cup \{q_2\}$

$= \{q_2\}$

for state $\{q_2\}$

$\delta_D(\{q_2\},0) = \delta_N(\{q_2\},0) = \phi$

$\delta_D(\{q_2\},1) = \delta_N(\{q_2\},1) = \{q_2\}$

The DFA is given by $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{A,B,C,D,E,F\}$

$\Sigma = \{0,1\}$

$q_0 = A$

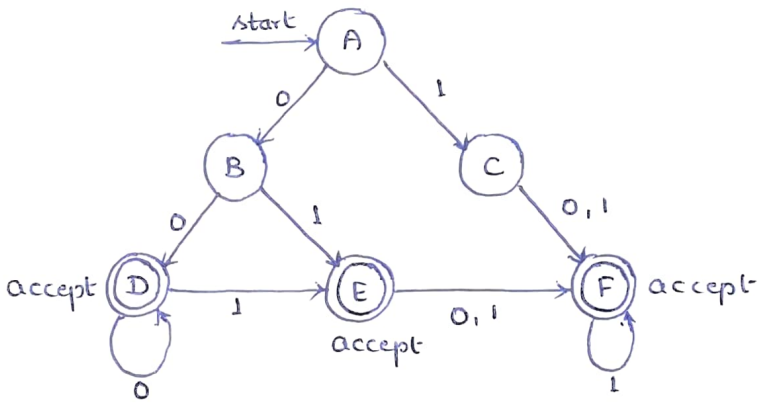
$F = \{\{D\},\{E\},\{F\}\}$

Transition table

δ	0	1
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_1\}$
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$\{q_1\}$	$\{q_2\}$	$\{q_2\}$
$*\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$*\{q_1, q_2\}$	$\{q_2\}$	$\{q_2\}$
$*\{q_2\}$	ϕ	$\{q_2\}$

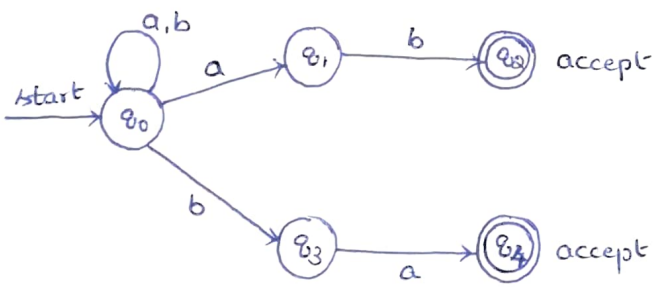
$\Rightarrow$

δ	0	1
$\rightarrow A$	B	C
B	D	E
C	F	F
$*$ D	D	E
$*$ E	F	F
$*$ F	ϕ	F



Q3. Obtain a NFA to accept strings of a's and b's ending with ab or ba. From this obtain an equivalent DFA.

The NFA to accept strings of a's and b's ending with ab or ba is



Transition Diagram

δ	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	ϕ	$\{q_2\}$
$*q_2$	ϕ	ϕ
q_3	$\{q_4\}$	ϕ
$*q_4$	ϕ	ϕ

Transition table

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA. So, $\Sigma = \{a, b\}$

STEP 3 : For state $\{q_0\}$

$\delta_D(\{q_0\}, a) = \delta_N(\{q_0\}, a) = \{q_0, q_1\}$

$\delta_D(\{q_0\}, b) = \delta_N(\{q_0\}, b) = \{q_0, q_3\}$

For state $\{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, a) = \delta_N(\{q_0, q_1\}, a)$

$= \delta_N(q_0, a) \cup \delta_N(q_1, a)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, b) = \delta_N(\{q_0, q_1\}, b)$

$= \delta_N(q_0, b) \cup \delta_N(q_1, b)$

$= \{q_0, q_3\} \cup \{q_2\}$

$= \{q_0, q_2, q_3\}$

For state $\{q_0, q_3\}$

$\delta_D(\{q_0, q_3\}, a) = \delta_N(\{q_0, q_3\}, a)$

$= \delta_N(q_0, a) \cup \delta_N(q_3, a)$

$= \{q_0, q_1\} \cup \{q_4\}$

$= \{q_0, q_1, q_4\}$

$\delta_D(\{q_0, q_3\}, b) = \delta_N(\{q_0, q_3\}, b)$

$= \delta_N(q_0, b) \cup \delta_N(q_3, b)$

$= \{q_0, q_3\} \cup \phi$

$= \{q_0, q_3\}$

For state $\{q_0, q_2, q_3\}$

$\delta_D(\{q_0, q_2, q_3\}, a) = \delta_N(\{q_0, q_2, q_3\}, a)$

$= \delta_N(q_0, a) \cup \delta_N(q_2, a) \cup \delta_N(q_3, a)$

$= \{q_0, q_1\} \cup \phi \cup \{q_4\}$

$= \{q_0, q_1, q_4\}$

$\delta_D(\{q_0, q_2, q_3\}, b) = \delta_N(\{q_0, q_2, q_3\}, b)$

$= \delta_N(q_0, b) \cup \delta_N(q_2, b) \cup \delta_N(q_3, b)$

$= \{q_0, q_3\} \cup \phi \cup \phi = \{q_0, q_3\}$

For state $\{q_0, q_1, q_4\}$

$$\delta_D(\{q_0, q_1, q_4\}, a) = \delta_N(\{q_0, q_1, q_4\}, a)$$
$$= \delta_N(q_0, a) \cup \delta_N(q_1, a) \cup \delta_N(q_4, a)$$
$$= \{q_0, q_1\} \cup \phi \cup \phi$$
$$= \{q_0, q_1\}$$

$$\delta_D(\{q_0, q_1, q_4\}, b) = \delta_N(\{q_0, q_1, q_4\}, b)$$
$$= \delta_N(q_0, b) \cup \delta_N(q_1, b) \cup \delta_N(q_4, b)$$
$$= \{q_0, q_3\} \cup \{q_2\} \cup \phi$$
$$= \{q_0, q_2, q_3\}$$

The DFA is given by $M = (Q, \Sigma, \delta, q_0, F)$

Q: $\{A, B, C, D, E\}$

Σ: $\{a, b\}$

q_0: $\{A\}$

F: $\{\{D\}, \{E\}\}$

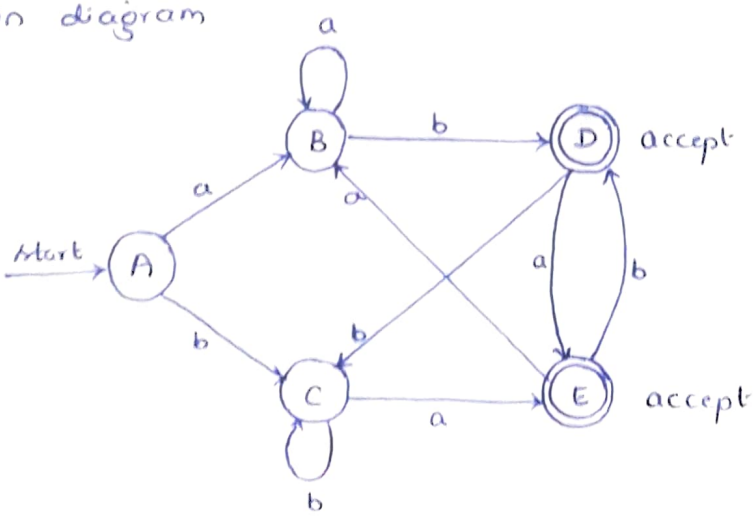
Transition table

δ	a	b
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0, q_3\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2, q_3\}$
$\{q_0, q_3\}$	$\{q_0, q_1, q_4\}$	$\{q_0, q_3\}$
$*\{q_0, q_2, q_3\}$	$\{q_0, q_1, q_4\}$	$\{q_0, q_3\}$
$*\{q_0, q_1, q_4\}$	$\{q_0, q_1\}$	$\{q_0, q_2, q_3\}$

$\Rightarrow$

δ	a	b
$\rightarrow A$	B	C
B	B	D
C	E	C
$*D$	E	C
$*E$	B	D

Transition diagram



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N.I.T.C.S.E. Jamshedpur

Theorem :

If there exists NFA $M_N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ which accepts the language $L(M_N)$, there exists an equivalent DFA $M_D = (Q_D, \Sigma, \delta_D, q_0, F_D)$ such that $L(M_D) = L(M_N)$.

It is required to prove that $\delta^*_D(q_0, \omega) = \delta^*_N(q_0, \omega)$

Basis : Consider a string $\omega = \varepsilon$ where $|\omega| = 0$

$\delta^*_D(\{q_0\}, \varepsilon) = \{q_0\}$ by definition of extended transition function of DFA.

$\delta^*_N(\{q_0\}, \varepsilon) = \{q_0\}$ by definition of extended transition function of NFA.

Hence $\delta^*_D(q_0, \varepsilon) = \delta^*_N(q_0, \omega)$ is proved when $\omega = \varepsilon$

Induction hypothesis : Let us assume that

$\delta^*_D(q_0, \omega) = \delta^*_N(q_0, \omega)$ for some ω where $|\omega| = n$

it is required to prove that

$\delta^*_D(q_0, \omega) = \delta^*_N(q_0, \omega)$ is true for some ω where $|\omega| = n+1$

Inductive proof : Let $\omega = xa$ where a is the last symbol of ω and x is the remaining string of ω.

So, $|x| = n$ and $|xa| = |\omega| = n+1$

By Extended definition δ^* of NFA, $\delta^*_N(q_0, \omega)$

$= \delta^*_N(q_0, xa)$

$= \delta_N(\delta_N^*(q_0, x), a)$

x is the string to be processed and after consuming the string x, let the states of the machine be $\{p_i, p_j, \ldots, p_k\}$

i.e., $\delta_N^*(q_0, x) = \{p_i, p_j, \ldots p_k\}$

Substituting this in previous equation,

$\delta_N^*(q_0, \omega) = \delta_N(\{p_i, p_j, \ldots, p_k\}, a)$

$= \delta_N(p_i, a) \cup \delta_N(p_j, a) \ldots \cup \delta_N(p_k, a)$

By extended definition δ^* of DFA,

$$\delta_D^*(q_0, w) = \delta_D^*(q_0, xa) = \delta_D(\delta_D^*(q_0, x), a)$$

x is the string to be processed and after consuming the string x, let the states of the machine be $\{p_i, p_j, \ldots, p_k\}$.

i.e., $\delta_D^*(q_0, x) = \{p_i, p_j, \ldots, p_k\}$

$$\delta_D^*(q_0, w) = \delta_D^*(q_0, xa) = \delta_D(\delta_D^*(q_0, x), a)$$

$$= \delta_D(\{p_i, p_j, \ldots, p_k\}, a)$$

$$= \delta_N(\{p_i, p_j, \ldots p_k\}, a)$$

$$= \delta_N(p_i, a) \cup \delta_N(p_j, a) \cup \ldots \delta_N(p_k, a)$$

By comparing both equations of NFA & DFA

$$\delta_D^*(\{q_0\}, w) = \delta_N^*(\{q_0\}, w)$$

So, if $\delta_D^*(\{q_0\}, w)$ is in F_D and $\delta_N^*(\{q_0\}, w)$ is in F_N, then both enters into final state accepting the same language.

Thus, $L(M_N) = L(M_D)$ Hence, the proof.

Exercises

Dec 10

Q1. Convert the following NFA to its equivalent DFA

June 14

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{0, 1\}$

STEP 3 : for state $\{q_0\}$

$$\delta_D(\{q_0\}, 0) = \delta_N(\{q_0\}, 0) = \{q_0, q_1\}$$

$$\delta_D(\{q_0\}, 1) = \delta_N(\{q_0\}, 1) = \{q_1\}$$

for state $\{q_0, q_1\}$

$$\delta_D(\{q_0, q_1\}, 0) = \delta_N(\{q_0, q_1\}, 0) = \delta_N(q_0, 0) \cup \delta_N(q_1, 0)$$

$$= \{q_0, q_1\} \cup \{q_2\}$$

$$= \{q_0, q_1, q_2\}$$

$\delta_D(\{q_0, q_1\}, 1) = \delta_N(\{q_0, q_1\}, 1) = \delta_N(q_0, 1) \cup \delta_N(q_1, 1)$

$= \{q_1\} \cup \{q_2\}$

$= \{q_1, q_2\}$

for state $\{q_1\}$

$\delta_D(\{q_1\}, 0) = \delta_N(\{q_1\}, 0) = \{q_2\}$

$\delta_D(\{q_1\}, 1) = \delta_N(\{q_1\}, 1) = \{q_2\}$

for state $\{q_0, q_1, q_2\}$

$\delta_D(\{q_0, q_1, q_2\}, 0) = \delta_N(\{q_0, q_1, q_2\}, 0) = \delta_N(q_0, 0) \cup \delta_N(q_1, 0) \cup \delta_N(q_2, 0)$

$= \{q_0, q_1\} \cup \{q_2\} \cup \phi$

$= \{q_0, q_1, q_2\}$

$\delta_D(\{q_0, q_1, q_2\}, 1) = \delta_N(\{q_0, q_1, q_2\}, 1) = \delta_N(q_0, 1) \cup \delta_N(q_1, 1) \cup \delta_N(q_2, 1)$

$= \{q_1\} \cup \{q_2\} \cup \{q_2\}$

$= \{q_1, q_2\}$

for state $\{q_1, q_2\}$

$\delta_D(\{q_1, q_2\}, 0) = \delta_N(\{q_1, q_2\}, 0) = \delta_N(q_1, 0) \cup \delta_N(q_2, 0)$

$= \{q_2\} \cup \phi$

$= \{q_2\}$

$\delta_D(\{q_1, q_2\}, 1) = \delta_N(\{q_1, q_2\}, 1) = \delta_N(q_1, 1) \cup \delta_N(q_2, 1)$

$= \{q_2\} \cup \{q_2\}$

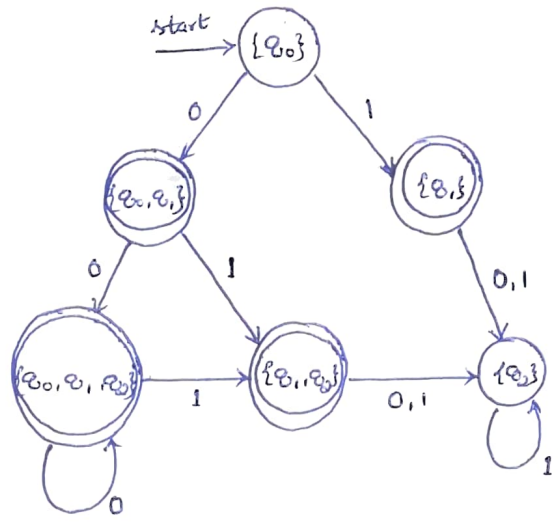
$= \{q_2\}$

for state $\{q_2\}$

$\delta_D(\{q_2\}, 0) = \delta_N(\{q_2\}, 0) = \phi$

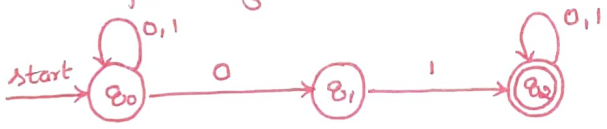
$\delta_D(\{q_2\}, 1) = \delta_N(\{q_2\}, 1) = \{q_2\}$

δ	0	1
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_1\}$
$* \{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$* \{q_1\}$	$\{q_2\}$	$\{q_2\}$
$* \{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
$* \{q_1, q_2\}$	$\{q_2\}$	$\{q_2\}$
$\{q_2\}$	ϕ	$\{q_2\}$



Dec 12

Q2. Convert the following NFA to DFA



STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

q_0, $\Sigma = \{0, 1\}$

STEP 3 : for state $\{q_0\}$

$\delta_D(\{q_0\}, 0) = \delta_N(\{q_0\}, 0) = \{q_0, q_1\}$

$\delta_D(\{q_0\}, 1) = \delta_N(\{q_0\}, 1) = \{q_0\}$

For state $\{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, 0) = \delta_N(\{q_0, q_1\}, 0) = \delta_N(q_0, 0) \cup \delta_N(q_1, 0)$

$= \{q_0, q_1\} \cup \phi$

$= \{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, 1) = \delta_N(\{q_0, q_1\}, 1) = \delta_N(q_0, 1) \cup \delta_N(q_1, 1)$

$= \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

for state $\{q_0, q_2\}$

$\delta_D(\{q_0, q_2\}, 0) = \delta_N(\{q_0, q_2\}, 0) = \delta_N(q_0, 0) \cup \delta_N(q_2, 0)$

$= \{q_0, q_1\} \cup \{q_2\}$

$= \{q_0, q_1, q_2\}$

$\delta_D(\{q_0, q_2\}, 1) = \delta_N(\{q_0, q_2\}, 1) = \delta_N(q_0, 1) \cup \delta_N(q_2, 1)$

$= \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

for state $\{q_0, q_1, q_2\}$

$\delta_D(\{q_0, q_1, q_2\}, 0) = \delta_N(\{q_0, q_1, q_2\}, 0) = \delta_N(q_0, 0) \cup \delta_N(q_1, 0) \cup \delta(q_2, 0)$

$= \{q_0, q_1\} \cup \phi \cup \{q_2\}$

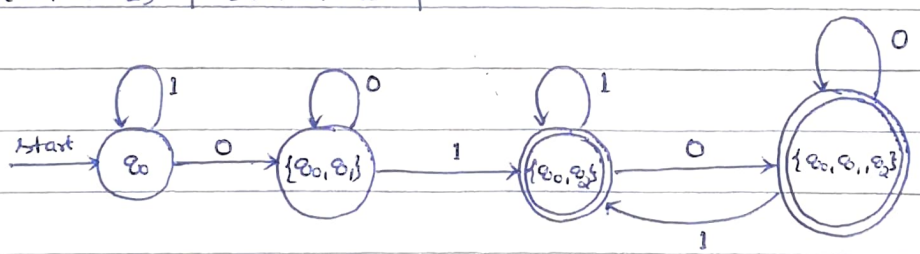
$= \{q_0, q_1, q_2\}$

$\delta_D(\{q_0, q_1, q_2\}, 1) = \delta_N(\{q_0, q_1, q_2\}, 1) = \delta_N(q_0, 1) \cup \delta_N(q_1, 1) \cup \delta_N(q_2, 1)$

$= \{q_0\} \cup \{q_2\} \cup \{q_2\}$

$= \{q_0, q_2\}$

δ	0	1
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$*\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$
$*\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$



Dec 09 Q3. Convert the following NFA to its equivalent DFA using subset construction

δ	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r\}$	$\{r\}$
r	$\{s\}$	ϕ
$*s$	$\{s\}$	$\{s\}$

STEP 1 : Since p is the start state of NFA, {p} is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{0, 1\}$

STEP 3 : $Q_N = \{p, q, r, s\}$

$Q_D = \{\phi, \{p\}, \{q\}, \{r\}, \{s\}, \{p,q\}, \{p,r\}, \{p,s\}, \{q,r\}, \{q,s\}, \{r,s\}, \{p,q,r\}, \{p,r,s\}, \{q,r,s\}, \{p,q,s\}, \{p,q,r,s\}\}$

STEP 4 : Since s is the final state of NFA.

$F_D = \{\{s\}, \{p,s\}, \{q,s\}, \{r,s\}, \{p,r,s\}, \{q,r,s\}, \{p,q,s\}, \{p,q,r,s\}\}$

STEP 5 : For state ϕ

$\delta_D(\phi, 0) = \phi$

$\delta_D(\phi, 1) = \phi$

For state $\{p\}$

$\delta_D(\{p\}, 0) = \delta_N(\{p\}, 0) = \{p, q\}$

$\delta_D(\{p\}, 1) = \delta_N(\{p\}, 1) = \{p\}$

For state $\{q\}$

$\delta_D(\{q\}, 0) = \delta_N(\{q\}, 0) = \{r\}$

$\delta_D(\{q\}, 1) = \delta_N(\{q\}, 1) = \{r\}$

For state $\{r\}$

$\delta_D(\{r\}, 0) = \delta_N(\{r\}, 0) = \{s\}$

$\delta_D(\{r\}, 1) = \delta_N(\{r\}, 1) = \phi$

For state $\{s\}$

$\delta_D(\{s\}, 0) = \delta_N(\{s\}, 0) = \{s\}$

$\delta_D(\{s\}, 1) = \delta_N(\{s\}, 1) = \{s\}$

For state $\{p,q\}$

$\delta_D(\{p,q\}, 0) = \delta_N(\{p,q\}, 0) = \delta_N(p, 0) \cup \delta_N(q, 0)$

$= \{p, q\} \cup \{r\}$

$= \{p, q, r\}$

$\delta_D(\{p,q\}, 1) = \delta_N(\{p,q\}, 1) = \delta_N(p, 1) \cup \delta_N(q, 1)$

$= \{p\} \cup \{r\} = \{p, r\}$

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For state $\{p,r\}$

$\delta_D(\{p,r\},a) = \delta_N(p,a) \cup \delta_N(r,a)$

$= \{p,q\} \cup \{s\}$

$= \{p,q,s\}$

$\delta_D(\{p,r\},1) = \delta_N(p,1) \cup \delta_N(r,1)$

$= \{p\} \cup \phi$

$= \{p\}$

For state $\{p,s\}$

$\delta_D(\{p,s\},0) = \delta_N(\{p,s\},0) = \delta_N(p,0) \cup \delta_N(s,0) = \{p,q\} \cup \{s\} = \{p,q,s\}$

$\delta_D(\{p,s\},1) = \delta_N(\{p,s\},1) = \delta_N(p,1) \cup \delta_N(s,1) = \{p\} \cup \{s\} = \{p,s\}$

For state $\{q,r\}$

$\delta_D(\{q,r\},0) = \delta_N(\{q,r\},0) = \delta_N(q,0) \cup \delta_N(r,0) = \{q\} \cup \{s\} = \{r,s\}$

$\delta_D(\{q,r\},1) = \delta_N(\{q,r\},1) = \delta_N(q,1) \cup \delta_N(r,1) = \{r\} \cup \{\phi\} = \{r\}$

For state $\{q,s\}$

$\delta_D(\{q,s\},0) = \delta_N(\{q,s\},0) = \delta_N(q,0) \cup \delta_N(s,0) = \{r\} \cup \{s\} = \{r,s\}$

$\delta_D(\{q,s\},1) = \delta_N(\{q,s\},1) = \delta_N(q,1) \cup \delta_N(s,1) = \{r\} \cup \{s\} = \{r,s\}$

For state $\{r,s\}$

$\delta_D(\{r,s\},0) = \delta_N(\{r,s\},0) = \delta_N(r,0) \cup \delta_N(s,0) = \{s\} \cup \{s\} = \{s\}$

$\delta_D(\{r,s\},1) = \delta_N(\{r,s\},1) = \delta_N(r,1) \cup \delta_N(s,1) = \{\phi\} \cup \{s\} = \{s\}$

for state $\{p,q,r\}$

$\delta_D(\{p,q,r\},0) = \delta_N(\{p,q,r\},0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(r,0)$

$= \{p,q\} \cup \{r\} \cup \{s\} = \{p,q,r,s\}$

$\delta_D(\{p,q,r\},1) = \delta_N(\{p,q,r\},1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(r,1)$

$= \{p\} \cup \{r\} \cup \{\phi\} = \{p,r\}$

For state $\{p,r,s\}$

$\delta_D(\{p,r,s\},0) = \delta_N(\{p,r,s\},0) = \delta_N(p,0) \cup \delta_N(r,0) \cup \delta_N(s,0)$

$= \{p,q\} \cup \{s\} \cup \{s\} = \{p,q,s\}$

$\delta_D(\{p,r,s\},1) = \delta_N(\{p,r,s\},1) = \delta_N(p,1) \cup \delta_N(r,1) \cup \delta_N(s,1)$

$= \{p\} \cup \{\phi\} \cup \{s\} = \{p,s\}$

For state $\{q, r, s\}$

$$\delta_D(\{q,r,s\},0) = \delta_N(\{q,r,s\},0) = \delta_N(q,0) \cup \delta_N(r,0) \cup \delta_N(s,0)$$
$$= \{r\} \cup \{s\} \cup \{s\} = \{r,s\}$$
$$\delta_D(\{q,r,s\},1) = \delta_N(\{q,r,s\},1) = \delta_N(q,1) \cup \delta_N(r,1) \cup \delta_N(s,1)$$
$$= \{r\} \cup \{\phi\} \cup \{s\} = \{r,s\}$$

For state $\{p, q, s\}$

$$\delta_D(\{p,q,s\},0) = \delta_N(\{p,q,s\},0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(s,0)$$
$$= \{p,q\} \cup \{r\} \cup \{s\} = \{p,q,r,s\}$$
$$\delta_D(\{p,q,s\},1) = \delta_N(\{p,q,s\},1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(s,1)$$
$$= \{p\} \cup \{r\} \cup \{s\} = \{p,r,s\}$$

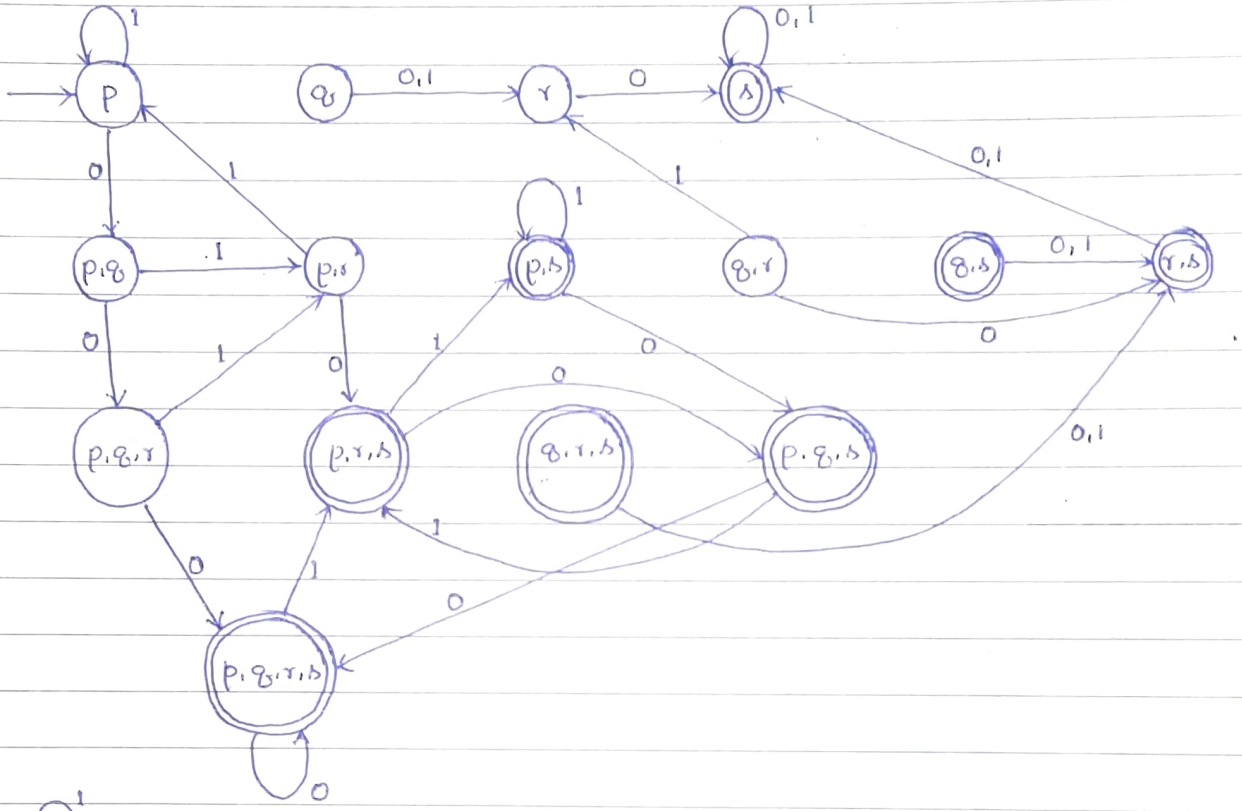
For state $\{p, q, r, s\}$

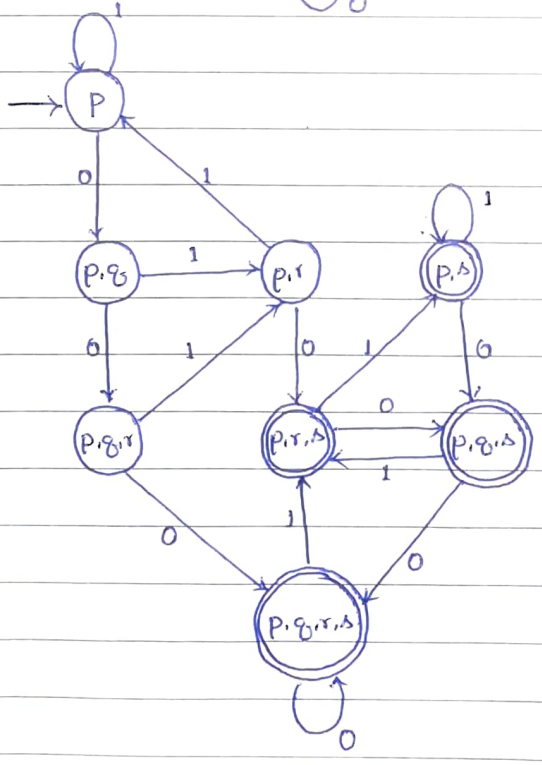
$$\delta_D(\{p,q,r,s\},0) = \delta_N(\{p,q,r,s\},0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(r,0) \cup \delta_N(s,0)$$
$$= \{p,q\} \cup \{r\} \cup \{s\} \cup \{s\} = \{p,q,r,s\}$$
$$\delta_D(\{p,q,r,s\},1) = \delta_N(\{p,q,r,s\},1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(r,1) \cup \delta_N(s,1)$$
$$= \{p\} \cup \{r\} \cup \{\phi\} \cup \{s\} = \{p,r,s\}$$

δ	0	1
ϕ	ϕ	ϕ
→ $\{p\}$	$\{p,q\}$	$\{p\}$
$\{q\}$	$\{r\}$	$\{r\}$
$\{r\}$	$\{s\}$	ϕ
* $\{s\}$	$\{s\}$	$\{s\}$
$\{p,q\}$	$\{p,q,r\}$	$\{p,r\}$
$\{p,r\}$	$\{p,q,s\}$	$\{p\}$
* $\{p,s\}$	$\{p,q,s\}$	$\{p,s\}$
$\{q,r\}$	$\{r,s\}$	$\{r\}$
* $\{q,s\}$	$\{r,s\}$	$\{r,s\}$
* $\{r,s\}$	$\{s\}$	$\{s\}$
$\{p,q,r\}$	$\{p,q,r,s\}$	$\{p,r\}$
* $\{p,r,s\}$	$\{p,q,s\}$	$\{p,s\}$
* $\{q,r,s\}$	$\{r,s\}$	$\{r,s\}$
* $\{p,q,s\}$	$\{p,q,r,s\}$	$\{p,r,s\}$
* $\{p,q,r,s\}$	$\{p,q,r,s\}$	$\{p,r,s\}$

Transition table

Transition diagram





δ	0	1
→{P}	{P,Q}	{P}
{P,Q}	{P,Q,R}	{P,R}
{P,R}	{P,R,S}	{P}
*{P,S}	{P,Q,S}	{P,S}
{P,Q,R}	{P,Q,R,S}	{P,R}
*{P,R,S}	{P,Q,S}	{P,S}
*{P,Q,S}	{P,Q,R,S}	{P,R,S}
*{P,Q,R,S}	{P,Q,R,S}	{P,R,S}

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Q4. Convert to a DFA the following NFA

δ	0	1
→p	{p,q}	{p}
q	ϕ	{r}
*r	{p,r}	{q}

STEP 1 : Since p is the start state of NFA, {p} is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{0,1\}$

STEP 3 : for state {p}

$\delta_D(\{p\},0) = \delta_N(\{p\},0) = \{p,q\}$

$\delta_D(\{p\},1) = \delta_N(\{p\},1) = \{p\}$

for state {p,q}

$\delta_D(\{p,q\},0) = \delta_N(\{p,q\},0) = \delta_N(p,0) \cup \delta_N(q,0) = \{p,q\} \cup \phi = \{p,q\}$

$\delta_D(\{p,q\},1) = \delta_N(\{p,q\},1) = \delta_N(p,1) \cup \delta_N(q,1) = \{p\} \cup \{r\} = \{p,r\}$

for state {p,r}

$\delta_D(\{p,r\},0) = \delta_N(\{p,r\},0) = \delta_N(p,0) \cup \delta_N(r,0) = \{p,q\} \cup \{p,r\} = \{p,q,r\}$

$\delta_D(\{p,r\},1) = \delta_N(\{p,r\},1) = \delta_N(p,1) \cup \delta_N(r,1) = \{p\} \cup \{q\} = \{p,q\}$

for state {p,q,r}

$\delta_D(\{p,q,r\},0) = \delta_N(\{p,q,r\},0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(r,0)$

$= \{p,q\} \cup \{\phi\} \cup \{p,r\}$

$= \{p,q,r\}$

$\delta_D(\{p,q,r\},1) = \delta_N(\{p,q,r\},1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(r,1)$

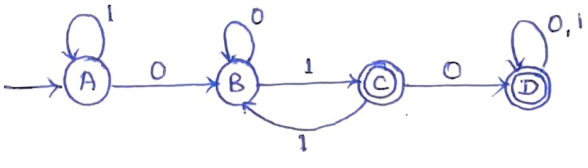
$= \{p\} \cup \{r\} \cup \{q\}$

$= \{p,r,q\}$

δ	0	1
→{p}	{p,q}	{p}
{p,q}	{p,q}	{p,r}
*{p,r}	{p,q,r}	{p,q}
*{p,q,r}	{p,q,r}	{p,q,r}

⇒

δ	0	1
→A	B	A
B	B	C
*C	D	B
*D	D	D



Q5. Convert to a DFA the following NFA

δ	0	1
→p	{p,r}	{q}
q	{r,s}	{p}
*r	{p,s}	{r}
*s	{q,r}	ϕ

STEP 1: Since p is the start state of NFA, {p} is the start state of DFA.

STEP 2: The input alphabets of NFA are the input alphabets of DFA.

δ_0, $\Sigma = \{0,1\}$

STEP 3: for state {p}

$\delta_D(\{p\},0) = \delta_N(\{p\},0) = \delta_N(p,0) = \{p,r\}$

$\delta_D(\{p\},1) = \delta_N(\{p\},1) = \delta_N(p,1) = \{q\}$

for state {q}

$\delta_D(\{q\},0) = \delta_N(\{q\},0) = \{r,s\}$

$\delta_D(\{q\},1) = \delta_N(\{q\},1) = \{p\}$

for state {p,r}

$\delta_D(\{p,r\},0) = \delta_N(\{p,r\},0) = \delta_N(p,0) \cup \delta_N(r,0) = \{p,r\} \cup \{p,s\} = \{p,r,s\}$

$\delta_D(\{p,r\},1) = \delta_N(\{p,r\},1) = \delta_N(p,1) \cup \delta_N(r,1) = \{q\} \cup \{r\} = \{q,r\}$

for state {r,s}

$\delta_D(\{r,s\},0) = \delta_N(\{r,s\},0) = \delta_N(r,0) \cup \delta_N(s,0) = \{p,s\} \cup \{q,r\} = \{p,q,r,s\}$

$\delta_D(\{r,s\},1) = \delta_N(\{r,s\},1) = \delta_N(r,1) \cup \delta_N(s,1) = \{r\} \cup \{\phi\} = \{r\}$

for state {q,r}

$\delta_D(\{q,r\},0) = \delta_N(\{q,r\},0) = \delta_N(q,0) \cup \delta_N(r,0) = \{r,s\} \cup \{p,s\} = \{p,r,s\}$

$\delta_D(\{q,r\},1) = \delta_N(\{q,r\},1) = \delta_N(q,1) \cup \delta_N(r,1) = \{p\} \cup \{r\} = \{p,r\}$

for state {p,r,s}

$\delta_D(\{p,r,s\},0) = \delta_N(\{p,r,s\},0) = \delta_N(p,0) \cup \delta_N(r,0) \cup \delta_N(s,0)$

$= \{p,r\} \cup \{p,s\} \cup \{q,r\}$

$= \{p,q,r,s\}$

$\delta_D(\{p,r,s\},1) = \delta_N(\{p,r,s\},1) = \delta_N(p,1) \cup \delta_N(r,1) \cup \delta_N(s,1)$

$= \{q\} \cup \{r\} \cup \{\phi\}$

$= \{q,r\}$

for state $\{p, q, r, s\}$

$\delta_D(\{p,q,r,s\}, 0) = \delta_N(\{p,q,r,s\}, 0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(r,0) \cup \delta_N(s,0)$

$= \{p,q\} \cup \{r,s\} \cup \{p,s\} \cup \{q,r\}$

$= \{p,q,r,s\}$

$\delta_D(\{p,q,r,s\}, 1) = \delta_N(\{p,q,r,s\}, 1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(r,1) \cup \delta_N(s,1)$

$= \{q\} \cup \{p\} \cup \{r\} \cup \{\phi\}$

$= \{p,q,r\}$

for state $\{r\}$

$\delta_D(\{r\}, 0) = \delta_N(\{r\}, 0) = \{p,s\}$

$\delta_D(\{r\}, 1) = \delta_N(\{r\}, 1) = \{r\}$

for state $\{p,q,r\}$

$\delta_D(\{p,q,r\}, 0) = \delta_N(\{p,q,r\}, 0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(r,0) = \{p,q\} \cup \{r,s\} \cup \{p,s\}$

$= \{p,q,r,s\}$

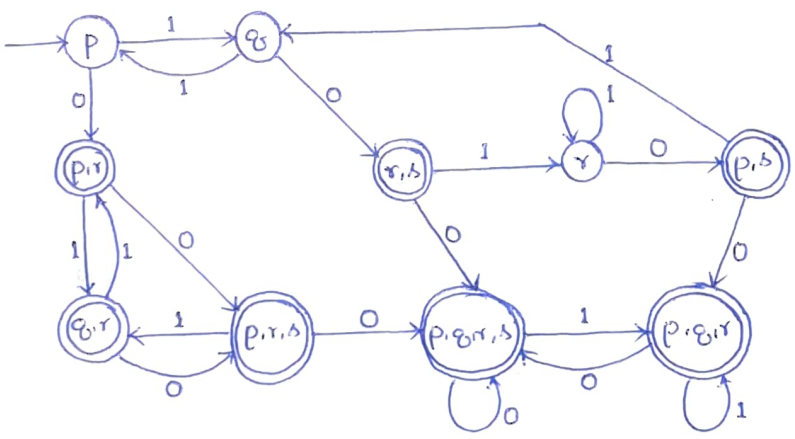
$\delta_D(\{p,q,r\}, 1) = \delta_N(\{p,q,r\}, 1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(r,1) = \{q\} \cup \{p\} \cup \{r\} = \{p,q,r\}$

for state $\{p,s\}$

$\delta_D(\{p,s\}, 0) = \delta_N(\{p,s\}, 0) = \delta_N(p,0) \cup \delta_N(s,0) = \{p,q\} \cup \{r,s\} = \{p,q,r,s\}$

$\delta_D(\{p,s\}, 1) = \delta_N(\{p,s\}, 1) = \delta_N(p,1) \cup \delta_N(s,1) = \{q\} \cup \{\phi\} = \{q\}$

δ	0	1
$\rightarrow\{p\}$	$\{p,r\}$	$\{q\}$
$*\{p,r\}$	$\{p,r,s\}$	$\{q,r\}$
$\{q\}$	$\{r,s\}$	$\{p\}$
$*\{p,r,s\}$	$\{p,q,r,s\}$	$\{q,r\}$
$*\{q,r\}$	$\{p,r,s\}$	$\{p,r\}$
$*\{r,s\}$	$\{p,q,r,s\}$	$\{r\}$
$*\{p,q,r,s\}$	$\{p,q,r,s\}$	$\{p,q,r\}$
$*\{r\}$	$\{p,s\}$	$\{r\}$
$*\{p,q,r\}$	$\{p,q,r,s\}$	$\{p,q,r\}$
$*\{p,s\}$	$\{p,q,r\}$	$\{q\}$



Q6. Convert the following NFA to a DFA and informally describe the language it accepts

δ	0	1
→p	{p,q}	{p}
q	{r,s}	{t}
r	{p,r}	{t}
*s	ϕ	ϕ
*t	ϕ	ϕ

STEP 1 : Since p is the start state of NFA, {p} is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

δ_0, $\Sigma = \{0,1\}$

STEP 3 : for state {p}

$\delta_D(\{p\},0) = \delta_N(\{p\},0) = \delta_N(p,0) = \{p,q\}$

$\delta_D(\{p\},1) = \delta_N(\{p\},1) = \delta_N(p,1) = \{p\}$

for state {p,q}

$\delta_D(\{p,q\},0) = \delta_N(\{p,q\},0) = \{p,q,r,s\}$

$\delta_D(\{p,q\},1) = \delta_N(\{p,q\},1) = \delta_N(p,1) \cup \delta_N(q,1) = \{p\} \cup \{t\} = \{p,t\}$

for state {p,t}

$\delta_D(\{p,t\},0) = \delta_N(\{p,t\},0) = \delta_N(p,0) \cup \delta_N(t,0) = \{p,q\} \cup \phi = \{p,q\}$

$\delta_D(\{p,t\},1) = \delta_N(\{p,t\},1) = \delta_N(p,1) \cup \delta_N(t,1) = \{p\} \cup \phi = \{p\}$

for state {p,q,r,s}

$\delta_D(\{p,q,r,s\},0) = \delta_N(\{p,q,r,s\},0) = \delta_N(p,0) \cup \delta_N(q,0) \cup \delta_N(r,0) \cup \delta_N(s,0)$

$= \{p,q\} \cup \{r,s\} \cup \{p,r\} \cup \phi$

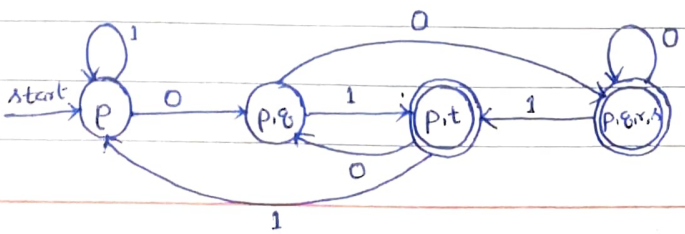
$= \{p,q,r,s\}$

$\delta_D(\{p,q,r,s\},1) = \delta_N(\{p,q,r,s\},1) = \delta_N(p,1) \cup \delta_N(q,1) \cup \delta_N(r,1) \cup \delta_N(s,1)$

$= \{p\} \cup \{t\} \cup \{t\} \cup \phi$

$= \{p,t\}$

δ	0	1
→{p}	{p,q}	{p}
{p,q}	{p,q,r,s}	{p,t}
*{p,t}	{p,q}	{p}
*{p,q,r,s}	{p,q,r,s}	{p,t}



The language accepts strings of a's & b's with substrings 01 or 00.

Q7. Convert the following NFA to DFA.

June 12

δ	0	1
→{q0}	{q0}	{q0, q1}
{q0, q1}	{q0, q2}	{q2}
*{q2}	ϕ	ϕ

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA. So, $\Sigma = \{0,1\}$

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

STEP 3 : for state $\{q_0\}$

$\delta_D(\{q_0\},0) = \delta_N(\{q_0\},0) = \{q_0\}$

$\delta_D(\{q_0\},1) = \delta_N(\{q_0\},1) = \{q_0, q_1\}$

for state $\{q_0, q_1\}$

$\delta_D(\{q_0,q_1\},0) = \delta_N(\{q_0,q_1\},0) = \delta_N(q_0,0) \cup \delta_N(q_1,0) = \{q_0\} \cup \{q_2\} = \{q_0, q_2\}$

$\delta_D(\{q_0,q_1\},1) = \delta_N(\{q_0,q_1\},1) = \delta_N(q_0,1) \cup \delta_N(q_1,1) = \{q_0,q_1\} \cup \{q_2\} = \{q_0,q_1,q_2\}$

for state $\{q_0, q_2\}$

$\delta_D(\{q_0,q_2\},0) = \delta_N(\{q_0,q_2\},0) = \delta_N(q_0,0) \cup \delta_N(q_2,0) = \{q_0\} \cup \phi = \{q_0\}$

$\delta_D(\{q_0,q_2\},1) = \delta_N(\{q_0,q_2\},1) = \delta_N(q_0,1) \cup \delta_N(q_2,1) = \{q_0,q_1\} \cup \{q_2\} = \{q_0,q_1,q_2\}$

for state $\{q_0, q_1, q_2\}$

$\delta_D(\{q_0,q_1,q_2\},0) = \delta_N(\{q_0,q_1,q_2\},0) = \delta_N(q_0,0) \cup \delta_N(q_1,0) \cup \delta_N(q_2,0)$

$= \{q_0\} \cup \{q_2\} \cup \phi$

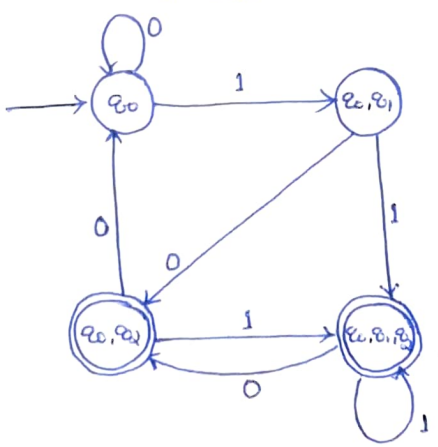
$= \{q_0, q_2\}$

$\delta_D(\{q_0,q_1,q_2\},1) = \delta_N(\{q_0,q_1,q_2\},1) = \delta_N(q_0,1) \cup \delta_N(q_1,1) \cup \delta_N(q_2,1)$

$= \{q_0,q_1\} \cup \{q_2\} \cup \phi$

$= \{q_0, q_1, q_2\}$

δ	0	1
→{q0}	{q0}	{q0, q1}
{q0, q1}	{q0, q2}	{q0, q1, q2}
*{q0, q2}	{q0}	{q0, q1, q2}
*{q0, q1, q2}	{q0, q2}	{q0, q1, q2}



Q8. Draw a NFA to accept the string of a's & b's such that it can accept either the string consisting of one a followed by any number of a's or one b followed by any number of b's. (i.e., aa^*/bb^*) & Obtain the corresponding DFA.

The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$ where,

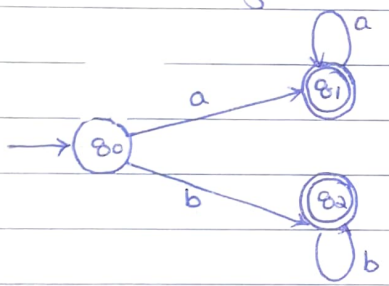
$Q : \{q_0, q_1, q_2\}$

$\Sigma : \{a, b\}$

$q_0 \in Q$ is the start state

$F : \{q_1, q_2\}$ is the final state

Transition diagram



Transition table

δ	a	b
$\rightarrow\{q_0\}$	$\{q_1\}$	$\{q_2\}$
$*\{q_1\}$	$\{q_1\}$	ϕ
$*\{q_2\}$	ϕ	$\{q_2\}$

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

$S_D, \Sigma : \{a, b\}$

STEP 3 : for state $\{q_0\}$

$\delta_D(\{q_0\}, a) = \delta_N(\{q_0\}, a) = \{q_1\}$

$\delta_D(\{q_0\}, b) = \delta_N(\{q_0\}, b) = \{q_2\}$

for state $\{q_1\}$

$\delta_D(\{q_1\}, a) = \delta_N(\{q_1\}, a) = \{q_1\}$

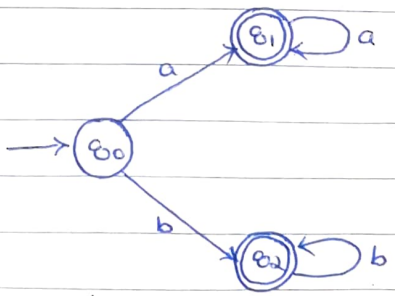
$\delta_D(\{q_1\}, b) = \delta_N(\{q_1\}, b) = \phi$

for state $\{q_2\}$

$\delta_D(\{q_2\}, a) = \delta_N(\{q_2\}, a) = \phi$

$\delta_D(\{q_2\}, b) = \delta_N(\{q_2\}, b) = \{q_2\}$

δ	a	b
$\rightarrow\{q_0\}$	$\{q_1\}$	$\{q_2\}$
$*\{q_1\}$	$\{q_1\}$	ϕ
$*\{q_2\}$	ϕ	$\{q_2\}$



Q9. Give Non-deterministic finite automata to accept the following languages over $\{0,1\}^*$

(a) The set of all strings such that containing either 101 or 110 as substring.

The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$ where

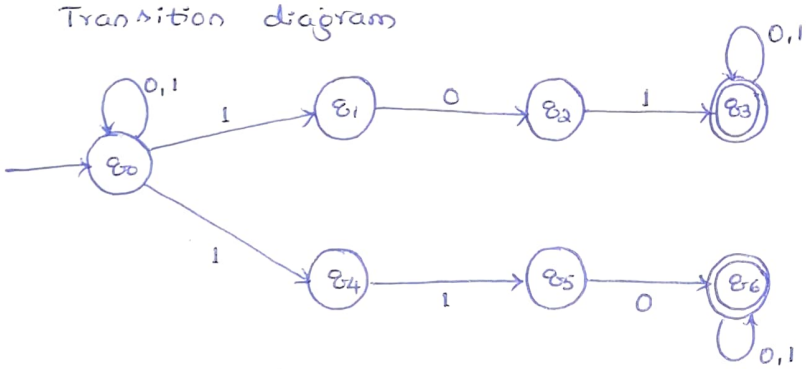
$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

$\Sigma = \{0,1\}$

$q_0 \in Q$ is the start state

$F = \{q_3, q_6\}$ are the final states

Transition diagram



Transition table

δ	0	1
$\rightarrow\{q_0\}$	$\{q_0\}$	$\{q_0, q_1, q_4\}$
$\{q_1\}$	$\{q_2\}$	ϕ
$\{q_2\}$	ϕ	$\{q_3\}$
$*\{q_3\}$	$\{q_3\}$	$\{q_3\}$
$\{q_4\}$	ϕ	$\{q_5\}$
$\{q_5\}$	$\{q_6\}$	ϕ
$*\{q_6\}$	$\{q_6\}$	$\{q_6\}$

(b) The set of all strings such that every 1 is followed immediately by 00.

The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$ where

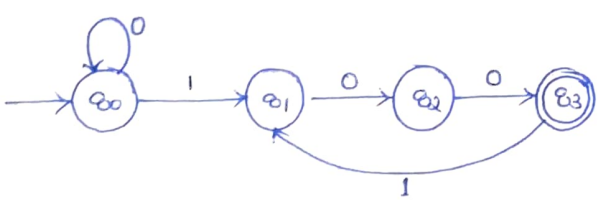
$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0,1\}$

$q_0 \in Q$ is the start state

$F = \{q_3\}$ is the final state

Transition Diagram



Transition table

δ	0	1
$\rightarrow\{q_0\}$	$\{q_0\}$	$\{q_1\}$
$\{q_1\}$	$\{q_2\}$	ϕ
$\{q_2\}$	$\{q_3\}$	ϕ
$*\{q_3\}$	ϕ	$\{q_1\}$

(c) The set of all strings containing exactly two occurrences of 10.

The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$ where

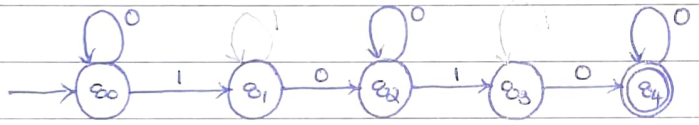
$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_4\}$ is the final state

Transition Diagram



Transition table

δ	0	1
$\rightarrow\{q_0\}$	$\{q_0\}$	$\{q_1\}$
$\{q_1\}$	$\{q_2\}$	ϕ
$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
$\{q_3\}$	$\{q_4\}$	ϕ
$*\{q_4\}$	$\{q_4\}$	ϕ

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Q10. Construct a NFA that accepts the language $aa^*(a+b)$

The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

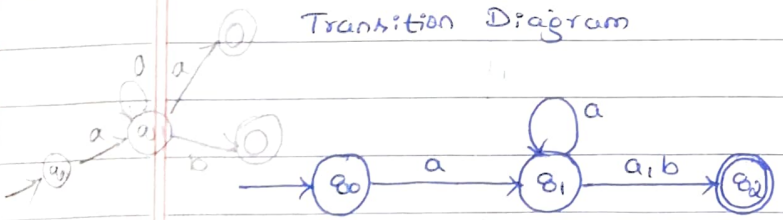
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$q_0 \in Q$ is the start state

$F = \{q_2\}$ is the final state

Transition Diagram



Transition table

δ	a	b
$\rightarrow\{q_0\}$	$\{q_1\}$	ϕ
$\{q_1\}$	$\{q_1, q_2\}$	$\{q_2\}$
$*\{q_2\}$	ϕ	ϕ

Q11. Design NFA's to recognise the following set of strings & convert to DFA's.

(a) lab, cab and dab, Assume the alphabet is $\{a, b, c, d, l\}$

(b) 1101, 101 & 111 Assume the alphabet $\{0, 1\}$

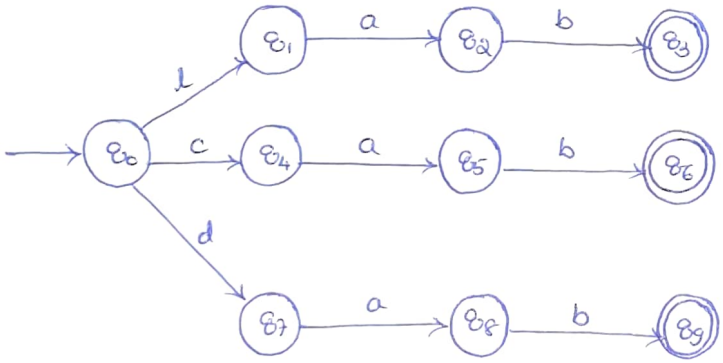
(a) The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9\}$

$\Sigma = \{a, b, c, d, \lambda\}$

$q_0 \in Q$ is the start state

$F = \{q_3, q_6, q_9\}$ are the final states.



δ	a	b	c	d	λ
→{q0}	ϕ	ϕ	{q4}	{q7}	{q1}
{q1}	{q2}	ϕ	ϕ	ϕ	ϕ
{q2}	ϕ	{q3}	ϕ	ϕ	ϕ
*{q3}	ϕ	ϕ	ϕ	ϕ	ϕ
{q4}	{q5}	ϕ	ϕ	ϕ	ϕ
{q5}	ϕ	{q6}	ϕ	ϕ	ϕ
*{q6}	ϕ	ϕ	ϕ	ϕ	ϕ
{q7}	{q8}	ϕ	ϕ	ϕ	ϕ
{q8}	ϕ	{q9}	ϕ	ϕ	ϕ
*{q9}	ϕ	ϕ	ϕ	ϕ	ϕ

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{a, b, c, d, \lambda\}$

STEP 3 : for state $\{q_0\}$

$\delta_D(\{q_0\}, a) = \delta_N(\{q_0\}, a) = \phi$

$\delta_D(\{q_0\}, b) = \delta_N(\{q_0\}, b) = \phi$

$\delta_D(\{q_0\}, c) = \delta_N(\{q_0\}, c) = \{q_4\}$

$\delta_D(\{q_0\}, d) = \delta_N(\{q_0\}, d) = \{q_7\}$

$\delta_D(\{q_0\}, \lambda) = \delta_N(\{q_0\}, \lambda) = \{q_1\}$

for state $\{q_1\}$

$\delta_D(\{q_1\}, a) = \{q_2\}$

$\delta_D(\{q_1\}, b) = \phi$

$\delta_D(\{q_1\}, c) = \phi$

$\delta_D(\{q_1\}, d) = \phi$

$\delta_D(\{q_1\}, \lambda) = \phi$

for state $\{q_2\}$

$\delta_D(\{q_2\}, a) = \phi$

$\delta_D(\{q_2\}, b) = \{q_3\}$

$\delta_D(\{q_2\}, c) = \phi$

$\delta_D(\{q_2\}, d) = \phi$

$\delta_D(\{q_2\}, \lambda) = \phi$

⋮

The DFA is defined as

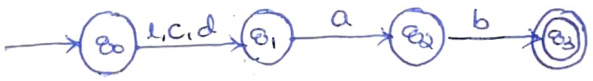
$M = (Q, \Sigma, \delta, q_0, F)$

$Q =$

$\Sigma = \{a, b, c, d, \lambda\}$

$q_0 \in Q$ is the start state

$F = \{q_3\}$ is the final state



δ	a	b	c	d	λ
→{q0}	ϕ	ϕ	{q4}	{q7}	{q1}
{q1}	{q2}	ϕ	ϕ	ϕ	ϕ
{q2}	ϕ	{q3}	ϕ	ϕ	ϕ
*{q3}	ϕ	ϕ	ϕ	ϕ	ϕ

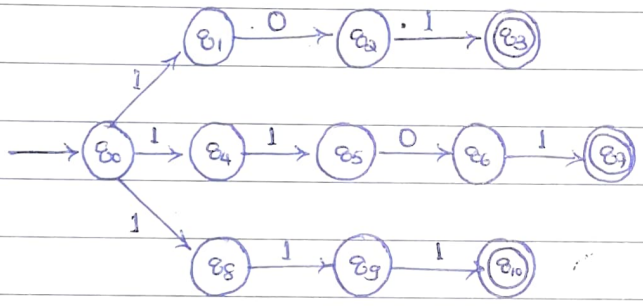
(b) The NFA is defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q =$

$\Sigma = \{0, 1\}$

$q_0 \in Q$ is the start state

$F = \{q_3, q_7, q_{10}\}$ are the final states



δ	0	1
$\rightarrow\{q_0\}$	ϕ	$\{q_1, q_4, q_8\}$
$\{q_1\}$	$\{q_2\}$	ϕ
$\{q_2\}$	ϕ	$\{q_3\}$
$*\{q_3\}$	ϕ	ϕ
$\{q_4\}$	ϕ	$\{q_5\}$
$\{q_5\}$	$\{q_6\}$	ϕ
$\{q_6\}$	ϕ	$\{q_7\}$
$*\{q_7\}$	ϕ	ϕ
$\{q_8\}$	ϕ	$\{q_9\}$
$\{q_9\}$	ϕ	$\{q_{10}\}$
$*\{q_{10}\}$	ϕ	ϕ

STEP 1 : Since q_0 is the start state of NFA, $\{q_0\}$ is the start state of DFA.

STEP 2 : The input alphabets of NFA are the input alphabets of DFA.

So, $\Sigma = \{0, 1\}$

STEP 3 : for state $\{q_0\}$

$\delta_D(\{q_0\}, 0) = \delta_N(\{q_0\}, 0) = \phi$

$\delta_D(\{q_0\}, 1) = \delta_N(\{q_0\}, 1) = \{q_1, q_4, q_8\}$

for state $\{q_1, q_4, q_8\}$

$\delta_D(\{q_1, q_4, q_8\}, 0) = \delta_N(\{q_1, q_4, q_8\}, 0)$

$= \delta_N(q_1, 0) \cup \delta_N(q_4, 0) \cup \delta_N(q_8, 0)$

$= \{q_2\} \cup \phi \cup \phi$

$= \{q_2\}$

$\delta_D(\{q_1, q_4, q_8\}, 1) = \delta_N(\{q_1, q_4, q_8\}, 1)$

$= \delta_N(q_1, 1) \cup \delta_N(q_4, 1) \cup \delta_N(q_8, 1)$

$= \phi \cup \{q_5\} \cup \{q_9\}$

$= \{q_5, q_9\}$

for state $\{q_2\}$

$\delta_D(\{q_2\},0) = \delta_N(\{q_2\},0) = \phi$

$\delta_D(\{q_2\},1) = \delta_N(\{q_2\},1) = \{q_3\}$

for state $\{q_5, q_9\}$

$\delta_D(\{q_5,q_9\},0) = \delta_N(\{q_5,q_9\},0) = \delta_N(\{q_5\},0) \cup \delta_N(\{q_9\},0)$

$= \{q_6\} \cup \phi$

$= \{q_6\}$

$\delta_D(\{q_5,q_9\},1) = \delta_N(\{q_5,q_9\},1) = \delta_N(\{q_5\},1) \cup \delta_N(\{q_9\},1)$

$= \phi \cup \{q_{10}\}$

$= \{q_{10}\}$

for state $\{q_3\}$

$\delta_D(\{q_3\},0) = \delta_N(\{q_3\},0) = \phi$

$\delta_D(\{q_3\},1) = \delta_N(\{q_3\},1) = \phi$

for state $\{q_6\}$

$\delta_D(\{q_6\},0) = \delta_N(\{q_6\},0) = \phi$

$\delta_D(\{q_6\},1) = \delta_D(\{q_6\},1) = \{q_7\}$

for state $\{q_{10}\}$

$\delta_D(\{q_{10}\},0) = \delta_N(\{q_{10}\},0) = \phi$

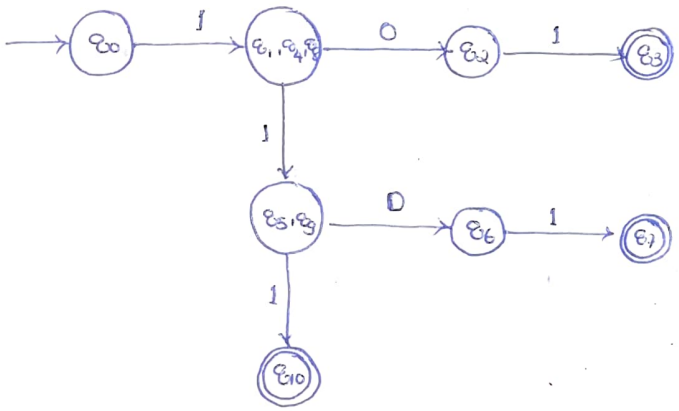
$\delta_D(\{q_{10}\},1) = \delta_N(\{q_{10}\},1) = \phi$

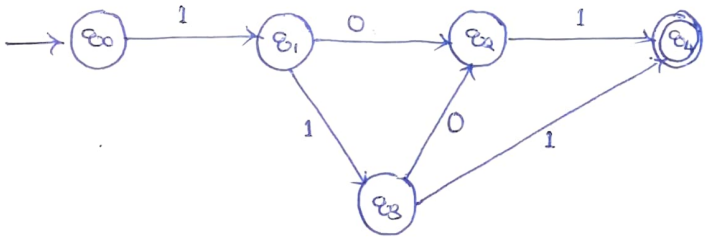
for state $\{q_7\}$

$\delta_D(\{q_7\},0) = \delta_N(\{q_7\},0) = \phi$

$\delta_D(\{q_7\},1) = \delta_N(\{q_7\},1) = \phi$

Transition Diagram





δ	0	1
$\rightarrow\{q_0\}$	ϕ	$\{q_1,q_4,q_8\}$
$\{q_1,q_4,q_8\}$	$\{q_2\}$	$\{q_5,q_9\}$
$\{q_2\}$	ϕ	$\{q_3\}$
$*\{q_3\}$	ϕ	ϕ
$\{q_5,q_9\}$	$\{q_6\}$	$\{q_{10}\}$
$\{q_6\}$	ϕ	$\{q_7\}$
$*\{q_7\}$	ϕ	ϕ
$*\{q_{10}\}$	ϕ	ϕ

δ	0	1
$\rightarrow\{q_0\}$	ϕ	$\{q_1\}$
$\{q_1\}$	$\{q_2\}$	$\{q_3\}$
$\{q_2\}$	ϕ	$\{q_4\}$
$\{q_3\}$	$\{q_2\}$	$\{q_4\}$
$*\{q_4\}$	ϕ	ϕ

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ε - NFA (finite Automata with Epsilon Transitions)

* An NFA with zero or more ε-transitions is called an ε-NFA.
* A transition with an empty input string is called an ε-transition. that is, if there is a transition from one state to another state without any input is called ε-transition.
* The ε-NFA is 5-tuple or quintuple indicating five components:

$M = (Q, \Sigma, \delta, q_0, F)$

where, M is the name of the machine

Q is non-empty, finite set of states.

Σ is non-empty, finite set of input alphabets.

$\delta : Q \times (\Sigma \cup \varepsilon)$ to 2^Q i.e., δ is transition function which is a mapping from $Q \times (\Sigma \cup \varepsilon)$ to 2^Q. Based on the current state there can be a transition to other states with or without any input symbols.

$q_0 \in Q$ is the start state

$F \subseteq Q$ is set of accepting or final states.

Example:

* The ε-NFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2\}$

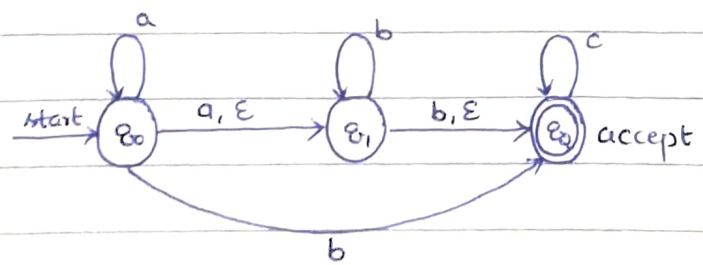
$\Sigma = \{a, b, c\}$

q_0 is the start state

$F = \{q_2\}$ is the final state.

Transition table

δ	ε	a	b	c
→ q_0	q_1	$\{q_0, q_1\}$	$\{q_2\}$	ϕ
q_1	q_2	ϕ	$\{q_1, q_2\}$	ϕ
* q_2	ϕ	ϕ	ϕ	$\{q_2\}$



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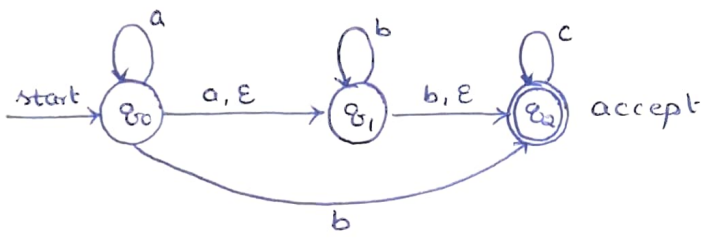
NOTE:

* The ε-CLOSURE of q denoted by ECLOSE(q) is the set of all states which are reachable from q on ε-transitions only.

Example :



$ECLOSE(q_0) = \{q_0, q_1, q_2\}$

$ECLOSE(q_1) = \{q_1, q_2\}$

$ECLOSE(q_2) = \{q_2\}$

Extended Transition function of ε-NFA

Let $M = (Q, \Sigma, \delta, q_0, F)$ be an ε-NFA. The extended transition function $\delta^* : Q \times (\Sigma \cup \varepsilon)^* \to 2^Q$ is defined as :

Basis : $\delta^*(q, \varepsilon) = ECLOSE(q)$

Induction : Let $w = xa$ where a is the last symbol of w and x is the remaining string of w.

Let q is the current state and x is the string to be processed and after consuming the string x, let the states of the machine is $\{p_1, p_2, p_3, \ldots p_m\}$

i.e., $\delta^*(q, x) = \{p_1, p_2, p_3, \ldots p_m\}$

Let the transition from $\{p_1, p_2, p_3, \ldots, p_m\}$ on input symbol a is

$$\delta(\{p_1, p_2, p_3, \ldots p_m\}, a) = \delta(p_1, a) \cup \delta(p_2, a) \cup \delta(p_3, a) \ldots \cup \delta(p_m, a)$$
$$= \{r_1, r_2, r_3, \ldots, r_m\}$$

So, $\delta^*(q, w) = ECLOSE(r_1, r_2, r_3, \ldots, r_m)$

$$= ECLOSE(r_1) \cup ECLOSE(r_2) \cup \ldots\ldots ECLOSE(r_m)$$

NOTE :

- $\delta^*(q, \varepsilon) = ECLOSE(q)$
- $\delta^*(q, w) = \delta^*(q, xa) = ECLOSE(\delta(\delta^*(q, x), a))$ where $w = xa$
- $\delta^*(q, w) = \delta^*(q, ax) = ECLOSE(\delta^*(\delta(q, a), x))$ where $w = ax$

Q1. Obtain an ε-NFA which accepts strings consisting of zero or more a's followed by zero or more b's followed by zero or more c's.

An ε-NFA is given by $M = (Q, \Sigma, \delta, q_0, F)$

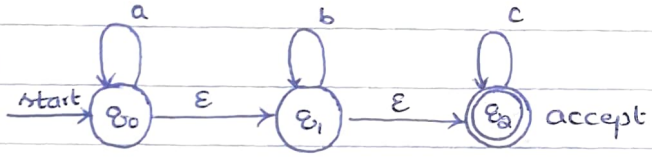
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b, c\}$

q_0 is the start state

$F = \{q_2\}$ is the final state

Transition Diagram



Transition table

δ	a	b	c	ε
→ q_0	$\{q_0\}$	ϕ	ϕ	$\{q_1\}$
q_1	ϕ	$\{q_1\}$	ϕ	$\{q_2\}$
* q_2	ϕ	ϕ	$\{q_2\}$	ϕ

June 11 Conversion from ε-NFA to DFA

Let $M_E = (Q_E, \Sigma, \delta_E, q_{0E}, F_E)$ be an ε-NFA where Q_E is set of finite states, Σ is set of input alphabets, δ_E is transition from $Q_E \times \{\Sigma \cup \varepsilon\}$ to 2^Q, q_{0E} is the start state and F_E is the set of final states.

The equivalent DFA $M_D = (Q_D, \Sigma, \delta_D, q_{0D}, F_D)$ can be obtained as shown below:

STEP 1 : If q_0 is the start state of NFA, then ECLOSE(q_0) is the start state of DFA i.e., q_{0D} = ECLOSE(q_0)

STEP 2 : Compute the transitions for DFA.

Let $\{q_i, q_j, \ldots, q_k\}$ is a state in DFA. Then, the transitions from this state on a i.e., $\delta_D(\{q_i, q_j, \ldots, q_k\}, a)$ is computed as shown below:

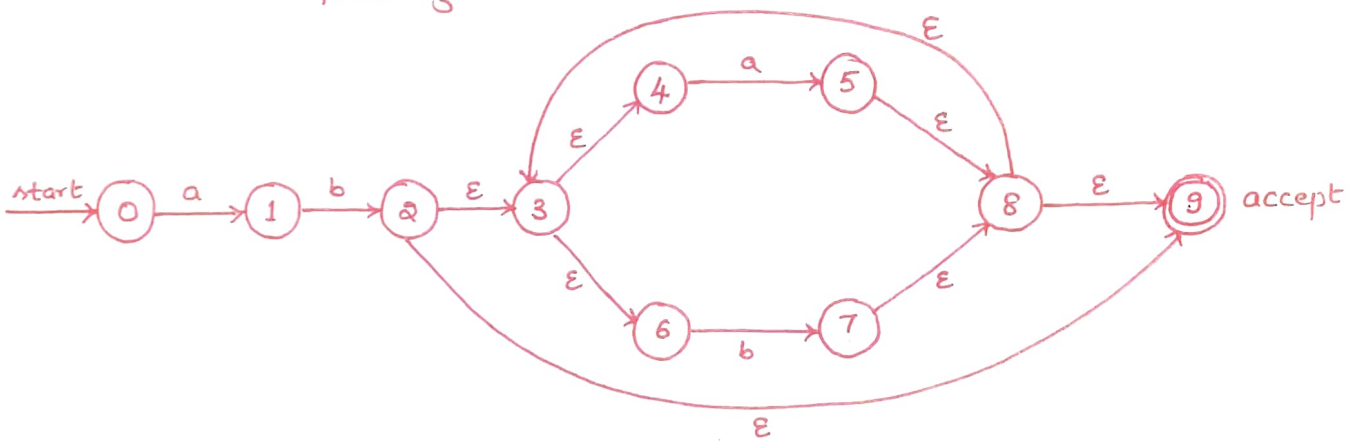
Let $\delta_E(\{q_i, q_j, \ldots, q_k\}, a) = \{p_1, p_2, \ldots, p_m\}$

Then take ECLOSE($\{p_1, p_2, \ldots, p_m\}$)

Thus, $\delta_D(\{q_i, q_j, \ldots, q_k\}, a)$ = ECLOSE($\{p_1, p_2, \ldots, p_m\}$)

STEP 3 : If $\{q_i, q_j, \ldots, q_k\}$ is a state in DFA and if this set contains at least one final state of ε-NFA, then $\{q_i, q_j, \ldots q_k\}$ is a final state of DFA.

Q1. Convert the following NFA to its equivalent DFA.



STEP 1 : Identify the start state of DFA.

Since 0 is the start state of ε-NFA, ECLOSE(0) is the start state of DFA. i.e., ECLOSE(0) = $\{0\}$ $\rightarrow$ (A)

STEP 2 : Consider the state A

$\delta(A,a)$ = ECLOSE ($\delta_E(A,a)$)
= ECLOSE ($\delta_E(0,a)$)
= $\{1\}$ $\rightarrow$ (B)

$\delta(A,b)$ = ECLOSE ($\delta_E(A,b)$)
= ECLOSE ($\delta_E(0,b)$)
= ϕ

Consider the state B

$\delta(B,a)$ = ECLOSE ($\delta_E(B,a)$)
= ECLOSE ($\delta_E(1,a)$)
= ϕ

$\delta(B,b)$ = ECLOSE ($\delta_E(B,b)$)
= ECLOSE ($\delta_E(1,b)$)
= ECLOSE ($\{2\}$)
= $\{2,3,4,6,9\}$ $\rightarrow$ (C)

Consider the state C

$\delta(C,a)$ = ECLOSE ($\delta_E(C,a)$)
= ECLOSE ($\delta_E(\{2,3,4,6,9\},a)$
= ECLOSE ($\{5\}$)
= $\{5,8,9,3,4,6\}$
= $\{3,4,5,6,8,9\}$ $\rightarrow$ (D)

$\delta(C,b)$ = ECLOSE ($\delta_E(C,b)$)
= ECLOSE ($\delta_E(\{2,3,4,6,9\},b)$)
= ECLOSE ($\{7\}$)
= $\{7,8,9,3,4,6\}$
= $\{3,4,6,7,8,9\}$ $\rightarrow$ (E)

Consider the state D

$\delta(D,a) = \text{ECLOSE}(\delta_E(D,a))$
$= \text{ECLOSE}(\delta_E(\{3,4,5,6,8,9\},a))$
$= \text{ECLOSE}(\{5\})$
$= \{5,8,9,3,4,6\}$
$= \{3,4,5,6,8,9\}$

$\delta(D,b) = \text{ECLOSE}(\delta_E(D,b))$
$= \text{ECLOSE}(\delta_E(\{3,4,5,6,8,9\},b))$
$= \text{ECLOSE}(\{7\})$
$= \{7,8,9,3,4,6\}$
$= \{3,4,6,7,8,9\}$

Consider the state E

$\delta(E,a) = \text{ECLOSE}(\delta_E(E,a))$
$= \text{ECLOSE}(\delta_E(\{3,4,6,7,8,9\},a))$
$= \text{ECLOSE}(\{5\})$
$= \{5,8,9,3,4,6\}$
$= \{3,4,5,6,8,9\}$

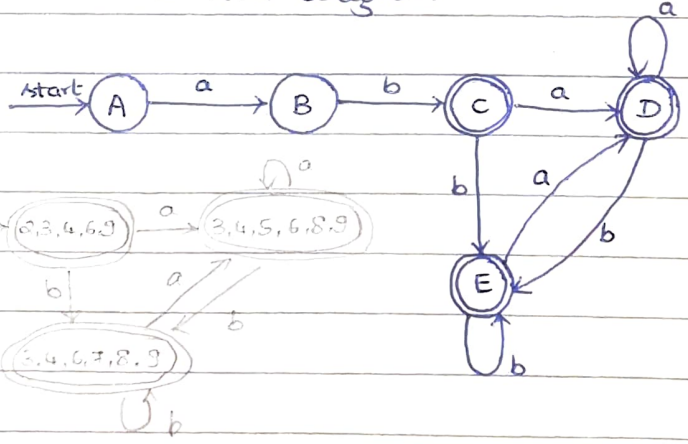
$\delta(E,b) = \text{ECLOSE}(\delta_E(E,b))$
$= \text{ECLOSE}(\delta_E(\{3,4,6,7,8,9\},b))$
$= \text{ECLOSE}(\{7\})$
$= \{7,8,9,3,4,6\}$
$= \{3,4,6,7,8,9\}$

STEP 3 : The states C, D and E are the final states.
Since 9 which is the final state of ε-NFA is present in C, D and E.

Transition table

	δ	a	b
0	→A	B	ϕ
1	B	ϕ	C
2	*C	D	E
3	*D	D	E
4	*E	D	E

Transition diagram



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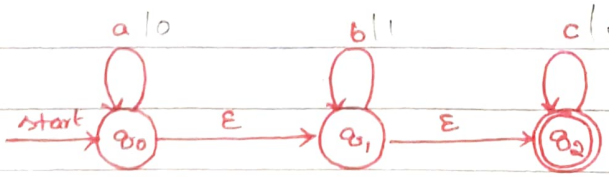
Q2. Convert the following NFA to its equivalent DFA.



STEP 1 : Identify the start state of DFA.

Since q_0 is the start state of ε-NFA, ECLOSE(q_0) is the start state of DFA i.e., ECLOSE(q_0) = $\{q_0, q_1, q_2\}$ → (A)

STEP 2 : Consider the state $\{q_0, q_1, q_2\}$

$\delta(\{q_0, q_1, q_2\}, a)$ = ECLOSE($\{q_0\}$) = $\{q_0, q_1, q_2\}$ → (A)

$\delta(\{q_0, q_1, q_2\}, b)$ = ECLOSE($\{q_1\}$) = $\{q_1, q_2\}$ → (B)

$\delta(\{q_0, q_1, q_2\}, c)$ = ECLOSE($\{q_2\}$) = $\{q_2\}$ → (C)

Consider the state $\{q_1, q_2\}$

$\delta(\{q_1, q_2\}, a) = \phi$

$\delta(\{q_1, q_2\}, b)$ = ECLOSE($\{q_1\}$) = $\{q_1, q_2\}$

$\delta(\{q_1, q_2\}, c)$ = ECLOSE($\{q_2\}$) = $\{q_2\}$

Consider the state $\{q_2\}$

$\delta(\{q_2\}, a) = \phi$

$\delta(\{q_2\}, b) = \phi$

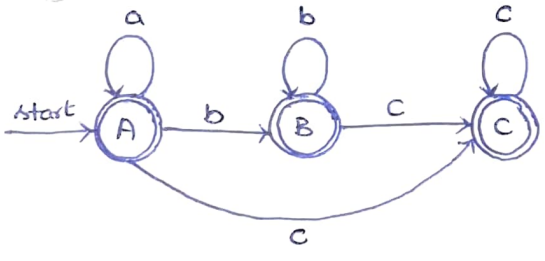
$\delta(\{q_2\}, c)$ = ECLOSE($\{q_2\}$) = $\{q_2\}$

STEP 3 : The states A, B and C are the final states, since q_2 is the final state of ε-NFA is present in A, B and C.

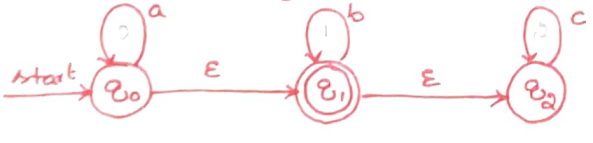
Transition table

δ	a	b	c
→*A	A	B	C
*B	ϕ	B	C
*C	ϕ	ϕ	C

Transition diagram

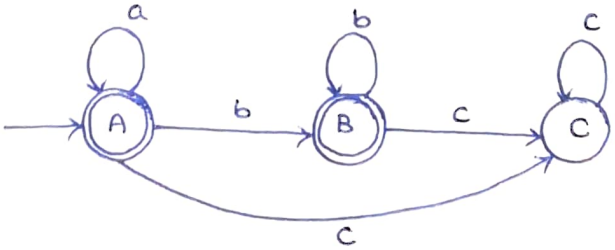


Q3. Convert the following NFA to its equivalent DFA.



Transition table

δ	a	b	c
→* A	A	B	C
* B	ϕ	B	C
C	ϕ	ϕ	C



Transition diagram

Q4. Obtain an NFA with ε-transitions to accept decimal numbers and obtain $\delta^*(q_0, 4.7)$.

The ε-NFA can be defined as $M = (Q, \Sigma, \delta, q_0, F)$

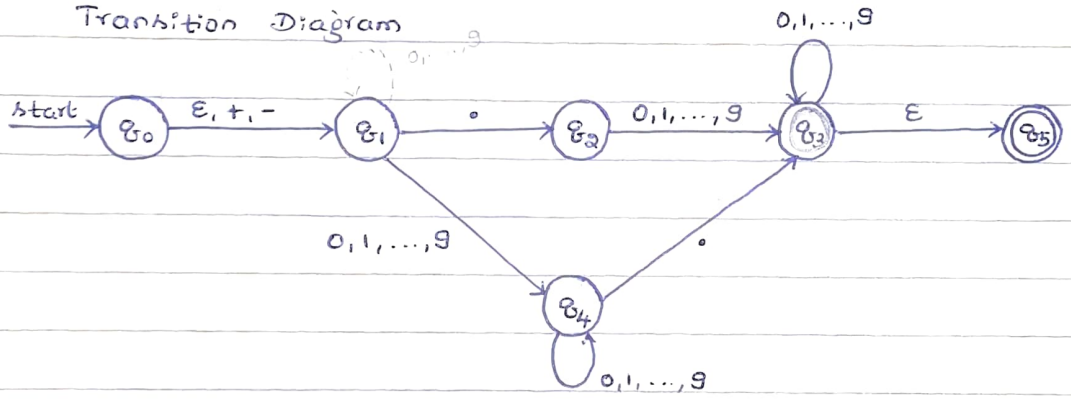
$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5 \}$

$\Sigma = \{ +, -, ., 0, 1, 2, \ldots, 9 \}$

q_0 is the start state

$F = \{q_5\}$ is the final state

Transition Diagram



Transition table

δ	ε	+	−	.	0 to 9
→q_0	q_1	q_1	q_1	ϕ	ϕ
q_1	ϕ	ϕ	ϕ	q_2	q_4
q_2	ϕ	ϕ	ϕ	ϕ	q_3
q_3	q_5	ϕ	ϕ	ϕ	q_3
q_4	ϕ	ϕ	ϕ	q_3	q_4
*q_5	ϕ	ϕ	ϕ	ϕ	ϕ

To obtain $\delta^*(q_0, 4.7)$

$\delta^*(q_0, 4) = \text{ECLOSE}(\delta_E\{q_0, q_1\}, 4))$

$= \text{ECLOSE}(\delta_E(q_0, 4) \cup \delta_E(q_1, 4))$

$= \text{ECLOSE}(\phi \cup q_4)$

$= \{q_4\}$

$\delta^*(q_4, \cdot) = \text{ECLOSE}(\delta_E(\{q_4\}, \cdot))$

$= \text{ECLOSE}(q_3)$

$= \{q_3, q_5\} \Rightarrow \{q_3\}$

$\delta^*(\{q_3, q_5\}, 7) = \text{ECLOSE}(\delta_E(\{q_3, q_5\}, 7)$

$= \text{ECLOSE}(\delta_E(q_3, 7) \cup \delta_E(q_5, 7))$

$= \text{ECLOSE}(q_3 \cup \phi)$

$= \text{ECLOSE}(q_3)$

$= \{q_3, q_5\}$

$\delta^*(q_3, 7) = \text{ECLOSE}(\delta_E(\{q_3\}, 7))$

$= \text{ECLOSE}(q_3)$

$= \{q_3\}$

Thus, $\delta^*(q_0, 4 \cdot 7) = \{q_3, q_5\}$

Dec08 Q5. Obtain an equivalent DFA for the above ε-NFA.

STEP 1 : Identify the start state of DFA.

Since q_0 is the start state of ε-NFA, ECLOSE(q_0) is the start state of DFA. i.e., $\text{ECLOSE}(q_0) = \{q_0, q_1\} \rightarrow (A)$

STEP 2 : Consider the state A

$\delta(A, \pm) = \text{ECLOSE}(\delta_E(A, \pm))$

$= \text{ECLOSE}(\delta_E(\{q_0, q_1\}, \pm))$

$= \text{ECLOSE}(\delta_E(q_0, \pm) \cup \delta_E(q_1, \pm))$

$= \text{ECLOSE}(q_1 \cup \phi)$

$= \text{ECLOSE}(q_1)$

$= \{q_1\} \rightarrow (B)$

$\delta(A, \cdot) = \text{ECLOSE}(\delta_E(A, \cdot))$

$= \text{ECLOSE}(\delta_E(\{q_0, q_1\}, \cdot)$

$= \text{ECLOSE}(\delta_E(q_0, \cdot) \cup \delta_E(q_1, \cdot))$

$= \text{ECLOSE}(\phi \cup q_2)$

$= \text{ECLOSE}(q_2)$

$= \{q_2\} \rightarrow (C)$

$\delta(A, \{0, 1, \ldots, 9\}) = \text{ECLOSE}(\delta_E(A, \{0, 1, \ldots, 9\}))$

$= \text{ECLOSE}(\delta_E(\{q_0, q_1\}, \{0, 1, \ldots, 9\}))$

$= \text{ECLOSE}(\delta_E(q_0, \{0, 1, \ldots, 9\}) \cup (q_1, \{0, 1, \ldots, 9\}))$

$= \text{ECLOSE}(\phi \cup q_4)$ $\quad \{q_1, q_4\}$

$= \text{ECLOSE}(q_4)$

$= \{q_4\} \rightarrow (D)$ $\quad \{q_1, q_4\} \rightarrow (D)$

Consider the state B

$\delta(B, \pm) = \text{ECLOSE}(\delta_E(B, \pm))$

$= \text{ECLOSE}(\delta_E(q_1, \pm))$

$= \text{ECLOSE}(\phi)$

$= \phi$

$\delta(B, \cdot) = \text{ECLOSE}(\delta_E(B, \cdot))$

$= \text{ECLOSE}(\delta_E(q_1, \cdot))$

$= \text{ECLOSE}(q_2)$

$= \{q_2\}$

$\delta(B, \{0,1,\ldots,9\}) = \text{ECLOSE}(\delta_E(B, \{0,1,\cdots,9\}))$

$= \text{ECLOSE}(\delta_E(q_1, \{0,1,\cdots,9\}))$

$= \text{ECLOSE}(q_4)$

$= \{q_4\}$ $\{q_1, q_4\} \rightarrow D$

Consider the state C

$\delta(C, \pm) = \text{ECLOSE}(\delta_E(C, \pm))$

$= \text{ECLOSE}(\delta_E(q_2, \pm))$

$= \text{ECLOSE}(\phi)$

$= \phi$

$\delta(C, \cdot) = \text{ECLOSE}(\delta_E(C, \cdot))$

$= \text{ECLOSE}(\delta_E(q_2, \cdot))$

$= \text{ECLOSE}(\phi)$

$= \phi$

$\delta(C, \{0,1,\ldots,9\}) = \text{ECLOSE}(\delta_E(C, \{0,1,\cdots,9\}))$

$= \text{ECLOSE}(\delta_E(q_2, \{0,1,\cdots,9\}))$

$= \text{ECLOSE}(q_3)$

$= \{q_3, q_5\} \rightarrow (E)$ $\{q_3\}$ $\{q_3, q_5\} \rightarrow E$

Consider the state D

$\delta(D, \pm) = \text{ECLOSE}(\delta_E(D, \pm))$

$= \text{ECLOSE}(\delta_E(q_4, \pm))$

$= \text{ECLOSE}(\phi)$

$= \phi$

$\delta(D, \cdot) = \text{ECLOSE}(\delta_E(D, \cdot))$

$= \text{ECLOSE}(\delta_E(q_4, \cdot))$ q_2

$= \text{ECLOSE}(q_3)$

$= \{q_3, q_5\} \rightarrow E$ $\{q_3\}$

$(q_1, \cdot) \cup (q_4, \cdot)$

$q_2 \cup \{q_3, q_5\}$

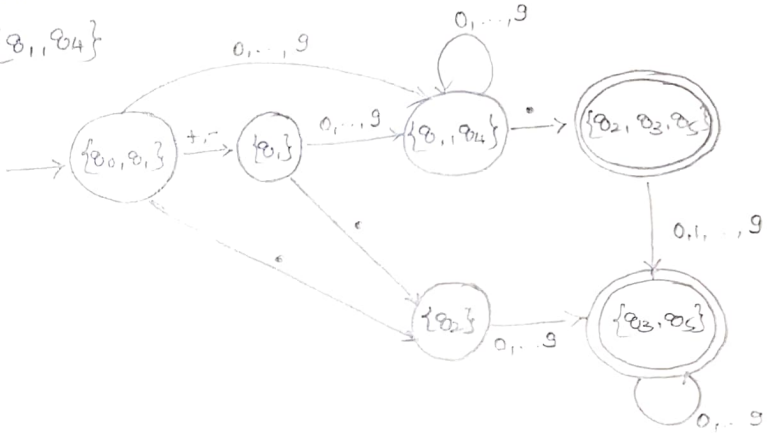
$\{q_2, q_3, q_5\}$

$\delta(D, \{0,1,\ldots,9\}) = \text{ECLOSE}(\delta_E(D, \{0,1,\ldots,9\}))$

$= \text{ECLOSE}(\delta_E(q_4, \{0,1,\ldots,9\}))$

$= \text{ECLOSE}(q_4)$ $\{q_1, q_4\}$

$= \{q_4\}$



Consider the state E

$\delta(E, \pm) = \text{ECLOSE}(\delta_E(E, \pm))$

$= \text{ECLOSE}(\delta_E(\{q_3, q_5\}, \pm))$

$= \text{ECLOSE}(\phi)$

$= \phi$

$\delta(E, \cdot) = \text{ECLOSE}(\delta_E(E, \cdot))$

$= \text{ECLOSE}(\delta_E(\{q_3, q_5\}, \cdot))$

$= \text{ECLOSE}(\phi)$

$= \phi$

$\delta(E, \{0,1,\ldots,9\}) = \text{ECLOSE}(\delta_E(E, \{0,1,\ldots,9\}))$

$= \text{ECLOSE}(\delta_E(\{q_3, q_5\}, \{0,1,\ldots,9\}))$

$= \text{ECLOSE}(q_3)$

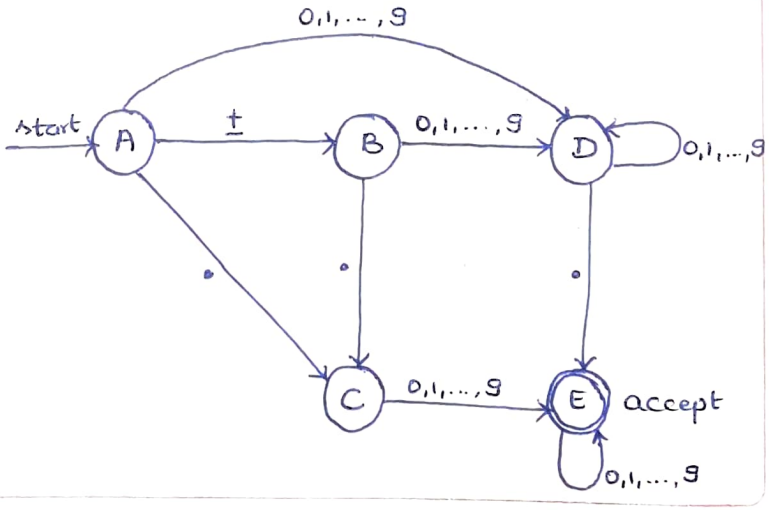
$= \{q_3, q_5\}$ $\{q_3\}$ $\{q_2, q_5\}$

STEP 3 : The state E is the final state, since q_5 is the final state of ϵ-NFA is present in E.

Transition table

δ	+	−	.	0 to 9
→A	B	B	C	D
B	ϕ	ϕ	C	D
C	ϕ	ϕ	ϕ	E
D	ϕ	ϕ	E	D
*E	ϕ	ϕ	ϕ	E

Transition Diagram



Q6. Prove that the language L is accepted by DFA M iff L is accepted by an (ε-NFA) N.

Let $N = (Q_E, \Sigma, \delta_E, q_0, F_E)$ be an ε-NFA.

Let the DFA obtained is $M = (Q_D, \Sigma, \delta_D, q_{0D}, F_D)$

Since all the transitions of DFA are obtained from the transitions of ε-NFA, the language accepted by ε-NFA is accepted by DFA.

Basis : if $\omega = \varepsilon$

$$\delta_E^*(q_0, \varepsilon) = \text{ECLOSE}(q_0) \rightarrow (1)$$

We also know that

$q_{0D} = \text{ECLOSE}(q_0)$

is correct since that is how the start state of DFA is obtained.

So, if $\omega = \varepsilon$, then

$$\delta_D^*(q_{0D}, \varepsilon) = \text{ECLOSE}(q_0) \rightarrow (2)$$

By comparing (1) & (2)

$$\delta_E^*(q_0, \varepsilon) = \delta_D^*(q_{0D}, \varepsilon)$$

Thus, the language accepted by ε-NFA is same as the language accepted by DFA when $\omega = \varepsilon$.

Induction hypotheses : Let $\omega = x$ and assume that the result is true for the string x.

i.e., $\delta_E^*(q_0, x) = \delta_D^*(q_{0D}, x) = \{q_i, q_j, \ldots, q_k\}$

Inductive Proof : Let $\omega = xa$, where a is the last symbol of ω.

We know that

$$\delta_E^*(q_0, x) = \delta_D^*(q_{0D}, x) = \{q_i, q_j, \ldots, q_k\}$$

is true for the string x.

Now, we have to prove that the result is true for $w = xa$.

By definition of δ^* for ε-NFA,

let $\delta_E(\{q_i, q_j, \ldots, q_k\}, a) = \{p_1, p_2, \ldots, p_m\}$ → ③

Then, $\delta_E^*(q_0, w) = \text{ECLOSE}(\{p_1, p_2, \ldots, p_m\})$ → ④

If we examine the construction of DFA from ε-NFA, the above two steps are used. Thus, $\delta_D^*(q_0, w) = \delta_E^*(q_0, w)$

June09 Dec09 Dec11 June14 Q7. Consider the following ε-NFA

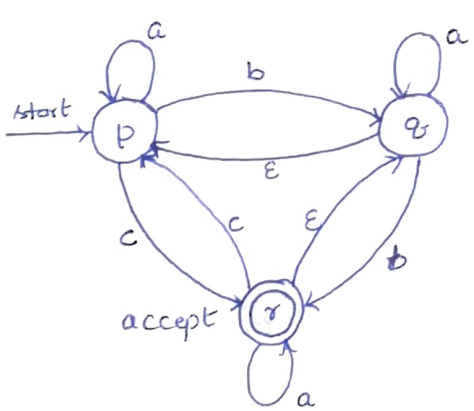
δ	ε	a	b	c
→ p	ϕ	{p}	{q}	{r}
q	{p}	{q}	{r}	ϕ
*r	{q}	{r}	ϕ	{p}

(i) Compute ε-CLOSURE of each state

(ii) Give all the strings of length three or less accepted by the automata.

(iii) Convert the automata to DFA.

Transition diagram



(i) $\text{ECLOSE}(p) = \{p\}$

$\text{ECLOSE}(q) = \{p, q\}$

$\text{ECLOSE}(r) = \{p, q, r\}$

(ii) Strings of length 1 - c

Strings of length 2 - bb, cc, (6)
ac, ca,
bc, cb

Strings of length 3 - ~~bbb~~, ~~ccc~~, (21)
~~abb~~, bab, bba, ~~acb~~
~~acc~~, ~~cac~~, ~~cca~~
~~aac~~, ~~aca~~, caa
~~bcc~~, ~~cbc~~, ~~ccb~~
~~bbc~~, ~~bcb~~, ~~cbb~~
~~abc~~, ~~bac~~, ~~bca~~

6 + 6 + 6 + 3 ⇒ (21)

~~acb~~	~~bbb~~	~~cbb~~	bab
~~acc~~	~~bcc~~	~~ccc~~	bba
~~aac~~	~~bac~~	~~cac~~	caa
~~aca~~	~~bca~~	~~cca~~	~~cba~~
~~abc~~	~~abc~~	~~cbc~~	~~cab~~
~~acb~~	~~bcb~~	~~ccb~~	

(iii) STEP 1 : Since p is the start state of ε-NFA, ECLOSE(p) is the start state of DFA.

i.e., ECLOSE(p) = {p} → (A)

STEP 2 : Consider the state A

$\delta(A,a)$ = ECLOSE($\delta(\{p\},a)$)
= {p}

$\delta(A,b)$ = ECLOSE($\delta(\{p\},b)$)
= ECLOSE(q)
= {p, q} → (B)

$\delta(A,c)$ = ECLOSE($\delta(\{p\},c)$)
= ECLOSE(r)
= {p, q, r} → (C)

Consider the state B

$\delta(B,a)$ = ECLOSE($\delta(\{p,q\},a)$)
= ECLOSE($\delta(p,a) \cup \delta(q,a)$)
= ECLOSE(p, q)
= {p, q}

$\delta(B,b)$ = ECLOSE($\delta(\{p,q\},b)$)
= ECLOSE($\delta(p,b) \cup \delta(q,b)$)
= ECLOSE(q, r)
= ECLOSE(q) ∪ ECLOSE(r)
= {p, q} ∪ {p, q, r}
= {p, q, r}

$\delta(B,c)$ = ECLOSE($\delta(\{p,q\},c)$)
= ECLOSE($\delta(p,c) \cup \delta(q,c)$)
= ECLOSE($r \cup \phi$)
= ECLOSE(r)
= {p, q, r}

Consider the state C

$\delta(C,a)$ = ECLOSE($\delta(\{p,q,r\},a)$)

= ECLOSE($\delta(p,a) \cup \delta(q,a) \cup \delta(r,a)$)

= ECLOSE(p,q,r)

= $\{p,q,r\}$

$\delta(C,b)$ = ECLOSE($\delta(\{p,q,r\},b)$)

= ECLOSE($\delta(p,b) \cup \delta(q,b) \cup \delta(r,b)$)

= ECLOSE($q \cup r \cup \phi$)

= ECLOSE(q,r)

= $\{p,q,r\}$

$\delta(C,c)$ = ECLOSE($\delta(\{p,q,r\},c)$)

= ECLOSE($\delta(p,c) \cup \delta(q,c) \cup \delta(r,c)$)

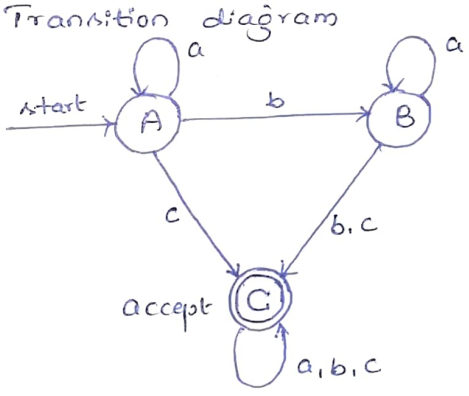
= ECLOSE($r \cup \phi \cup p$)

= $\{p,q,r\}$

Transition table

δ	a	b	c
→A	A	B	C
B	B	C	C
*C	C	C	C

Transition diagram



May 10
Dec 11

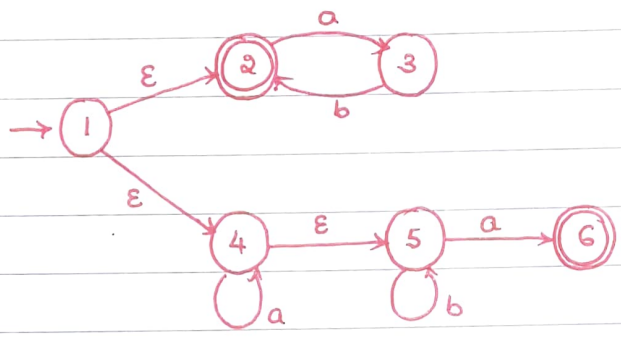
Difference between DFA, NFA and ε-NFA

DFA	NFA	ε-NFA
The DFA is a 5-tuple $M=(Q,\Sigma,\delta,q_0,F)$ where Q is set of finite states; Σ is set of input alphabets; $\delta: Q\times\Sigma$ to Q; q_0 is the start state; $F\subseteq Q$ is set of final states	An NFA is a 5-tuple $M=(Q,\Sigma,\delta,q_0,F)$ where Q is set of finite states; Σ is set of input alphabets; $\delta: Q\times\Sigma$ to 2^Q; q_0 is the start state; $F\subseteq Q$ is set of final states	An ε-NFA is a 5-tuple $M=(Q,\Sigma,\delta,q_0,F)$ where Q is set of finite states; Σ is set of input alphabets; $\delta: Q\times(\Sigma\cup\varepsilon)$ to 2^Q; q_0 is the start state; $F\subseteq Q$ is set of final states

*	There can be zero or one transition from a state on an input symbol.	There can be zero, one or more transitions from a state on an input symbol.	There can be zero, one or more transitions from a state with or without giving any input.
*	More number of transitions.	Less number of transitions	Relatively more transitions when compared with NFA
*	Difficult to construct	Easy to construct	Easy to construct using regular expressions
*	Less powerful since at any point of time it will be in only one state.	More powerful than DFA since at any point of time it will be in more than one state.	More powerful than NFA since at any point of time it will be in more than one state with or without giving any input.

Exercises

Q1. Convert the following ε-NFA to DFA



STEP 1 : Identify the start state of DFA

Since 1 is the start state of ε-NFA, ECLOSE(1) is the start state of DFA.

ECLOSE(1) = {1, 2, 4, 5} → (A)

STEP 2 : Consider the state A

$\delta(A,a)$ = ECLOSE($\delta_E(A,a)$)

= ECLOSE(δ_E({1, 2, 4, 5}, a)

= ECLOSE(3, 4, 5, 6)

= {3, 4, 5, 6} → (B)

$\delta(A,b)$ = ECLOSE($\delta_E(A,b)$)

= ECLOSE(δ_E({1, 2, 4, 5}, b)

= ECLOSE(5)

= {5} → (C)

Consider the state B

$\delta(B,a)$ = ECLOSE($\delta_E(B,a)$)

= ECLOSE(δ_E({3, 4, 5, 6}, a)

= ECLOSE(4, 6)

= {4, 5, 6} → (D)

$\delta(B,b)$ = ECLOSE($\delta_E(B,b)$)

= ECLOSE(δ_E({3, 4, 5, 6}, b)

= ECLOSE(2, 5)

= {2, 5} → (E)

Consider state C

$\delta(C,a) = \text{ECLOSE}(\delta_E(C,a))$

$= \text{ECLOSE}(\delta_E(\{5\},a))$

$= \text{ECLOSE}\ \{6\}$

$= \{6\} \rightarrow (F)$

$\delta(C,b) = \text{ECLOSE}(\delta_E(C,b))$

$= \text{ECLOSE}(\delta_E(\{5\},b))$

$= \text{ECLOSE}\ \{5\}$

$= \{5\} \rightarrow (C)$

Consider state D

$\delta(D,a) = \text{ECLOSE}(\delta_E(D,a))$

$= \text{ECLOSE}(\delta_E(\{4,5,6\},a))$

$= \text{ECLOSE}(4,6)$

$= \{4,5,6\} \rightarrow (D)$

$\delta(D,b) = \text{ECLOSE}(\delta_E(D,b))$

$= \text{ECLOSE}(\delta_E(\{4,5,6\},b))$

$= \text{ECLOSE}(5)$

$= \{5\} \rightarrow (C)$

Consider state E

$\delta(E,a) = \text{ECLOSE}(\delta_E(E,a))$

$= \text{ECLOSE}(\delta_E(\{2,5\},a))$

$= \text{ECLOSE}(3,6)$

$= \{3,6\} \rightarrow (G)$

$\delta(E,b) = \text{ECLOSE}(\delta_E(E,b))$

$= \text{ECLOSE}(\delta_E(\{2,5\},b))$

$= \text{ECLOSE}(5)$

$= \{5\} \rightarrow (C)$

Consider state F

$\delta(F,a) = \text{ECLOSE}(\delta_E(F,a))$

$= \text{ECLOSE}(\delta_E(G,a))$

$= \phi$

$\delta(F,b) = \phi$

Consider state G

$\delta(G,a) = \text{ECLOSE}(\delta_E(G,a))$

$= \text{ECLOSE}(\delta_E(\{3,6\},a))$

$= \text{ECLOSE}(\phi)$

$= \phi$

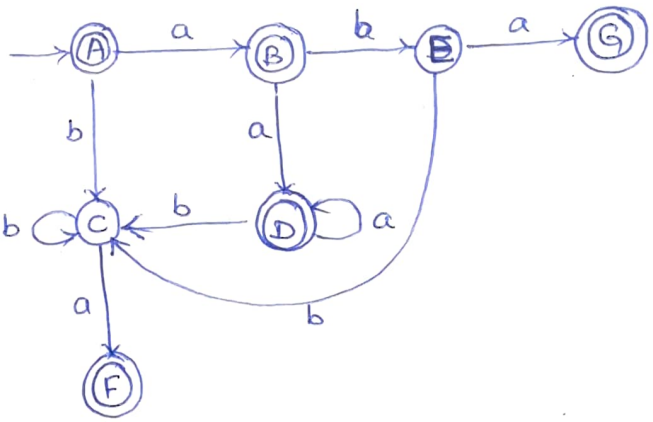
$\delta(G,b) = \text{ECLOSE}(\delta_E(G,b))$

$= \text{ECLOSE}(\delta_E(\{3,6\},b))$

$= \text{ECLOSE}(\phi)$

$= \phi$

δ	a	b
→*{1,2,4,5}	{3,4,5,6}	{5}
*{3,4,5,6}	{4,5,6}	{2,5}
{5}	{6}	{5}
*{4,5,6}	{4,5,6}	{5}
{2,5}	{3,6}	{5}
*{6}	ϕ	ϕ
*{3,6}	ϕ	ϕ



δ	a	b
→*A	B	C
*B	D	E
C	F	C
*D	D	C
E	G	C
*F	ϕ	ϕ
*G	ϕ	ϕ